Business Process Management

Poornima Jayathunga

General Information

- Course Code: IIT 448-2
- Course Name: Business Process Management
- Credit Value: 2 credits
- Compulsory Course – Attendance 80% is must for every students
- Hourly Breakdown
    - Theory : 30 hours
    - Independent Learning : 70 hours
- Evaluation
    - Continuous Assessment 40%
    - Final Examinations 60%

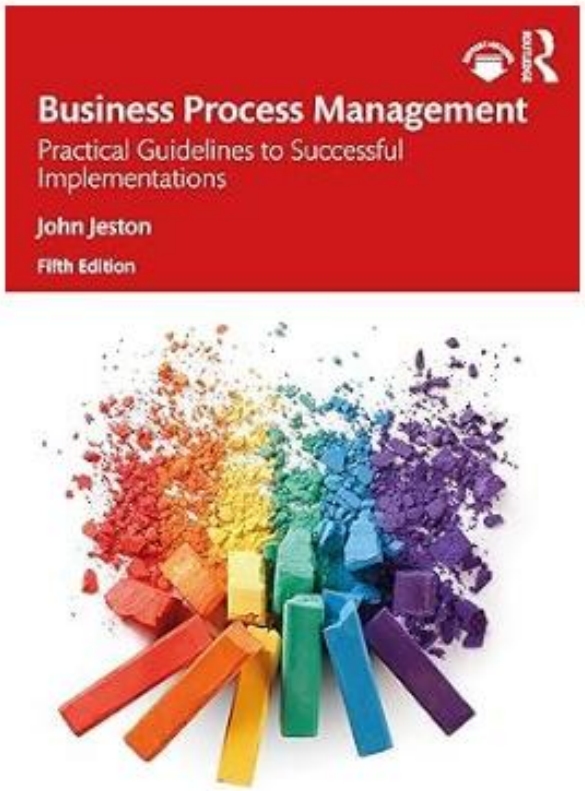
Evaluation Process

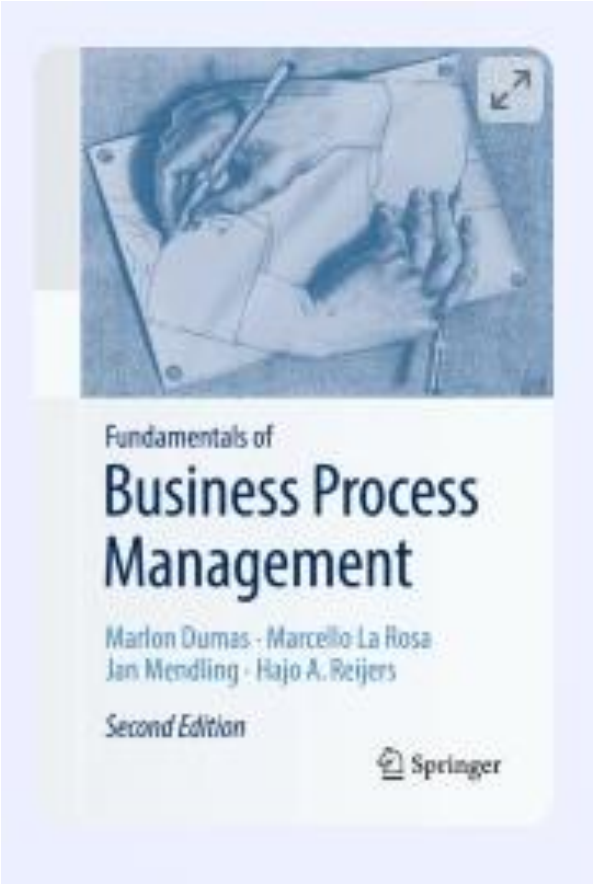
- **Continuous assessments (40%) :**
  - Quizzes (02) –

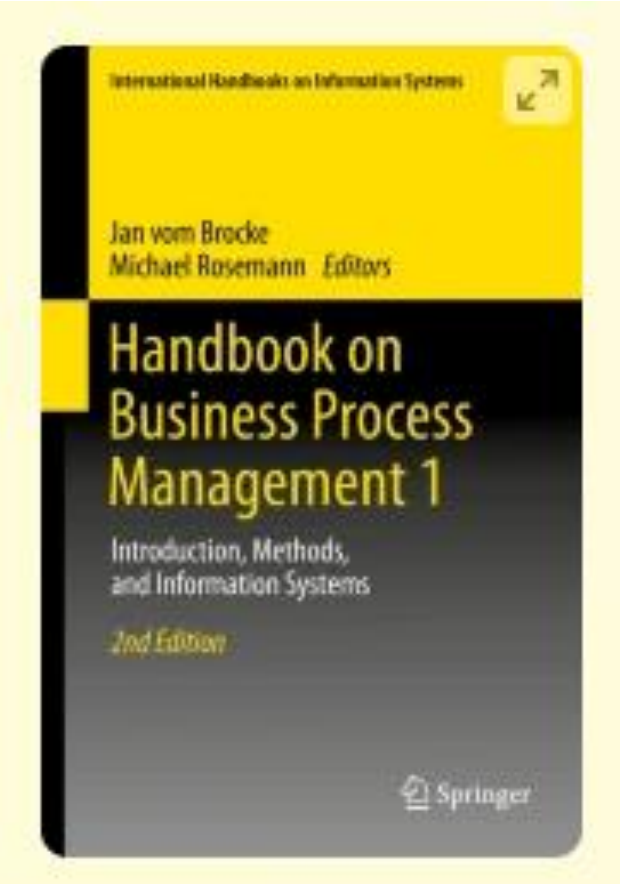
    week 3 – 20%

    First week after mid break – 40%
  - Assignment (01) (submit on last week) – 40%
- **End Semester Examination (60%)**
  - 4 Essay questions – 2 hours

References


Business Process Management
Practical Guidelines to Successful Implementations
John Jeston
Fifth Edition


Fundamentals of
Business Process Management
Marlon Dumas · Marcello La Rosa
Jan Mendling · Hajo A. Reijers
Second Edition
Springer


Jan vom Brocke
Michael Rosemann Editors
Handbook on Business Process Management 1
Introduction, Methods, and Information Systems
2nd Edition
Springer

Intended Learning Outcome

- After successful completion of the course students should be able to;
    - synthesize the principles of organizational strategy and process design and model simple business processes.
    - explain the role of Information Technology and other resources in BPM and document processes using a process mapping tool complying with Business Process Management Notation.
    - assess the documented business processes using their key operations characteristics, e.g., efficiency, intended service quality, process flexibility and costs associated with delays, material low volume and level of service or product customization.
    - propose business solutions in written and verbal forms for process innovation and redesign
    - create a BPM implementation strategy and implementation plan for an organization.

Lesson 1: Introduction to Business Process Management

Why you should learn?

- Can Become a
    - Business Process Analyst- Analyze and optimize business processes using BPM tools and methodologies.
    - BPM Developer- Design, implement, and deploy BPM solutions using IT platforms.
    - IT Business Analyst- ridge the gap between business needs and IT solutions.
    - Process Automation Specialist - Focus on automating repetitive tasks and workflows using tools like Robotic Process Automation (RPA).
    - BPM Consultant - Advise organizations on implementing BPM frameworks and technologies.

What is Business Process Management(BPM) ?

- Definitions
  - "*Business process management (BPM) is a discipline involving any combination of modeling, automation, execution, control, measurement and optimization of business activity flows, in support of enterprise goals, spanning systems, employees, customers and partners within and beyond the enterprise boundaries*."

    - Workflow Management Coalition, BPM.com

  - *"the discipline of managing processes (rather than tasks) as the means for improving business performance outcomes and operational agility. Processes span organizational boundaries, linking together people, information flows, systems, and other assets to create and deliver value to customers and constituents.*"

    - Gartner

What is BPM?

- A management science (art and science) that focus on improving the corporate performance by managing and optimizing a company's business processes.

or

- Business Process Management (BPM) is the **practice of managing and optimizing how tasks are carried out within an organization** to ensure **consistent results** and capitalize on **opportunities for improvement**

What is it and what it is NOT……..

- A method for managing processes
- A method of identify which processes are executed within an organization
- A business led methodology or project
- The foundation of facilities a continues improvement culture within a team or organization

- A pure improvement methodology (but it provides the means for process improvement)
- An IT led solution
- A tactical level tool (BPM is strategic led, operationally deployed)

Why BPM is needed

- Improve Efficiency
- Enhance Customer Satisfaction
- Support Continuous Improvement
- Increase Agility
- Cost Reduction
- Enhance Compliance and Risk Management
- Enable Better Decision-Making
- Drive Innovation
- Empower Employees

Key components - Processes Everywhere

- Each organization—be it a governmental agency, a non-profit organization, or an enterprise has to manage a number of processes.
- **process** is a **structured series of related activities or tasks** that are carried out — either manually or automatically — to achieve a specific business goal or outcome.

- Ingredients of a Business Process - events and activities, decision points, actors, physical objects, information objects, final outcome, customer

Exercise 01: Exercise to Understand the Process

1. What you can see as a process/s in the following image


MENU

Components of process

- Ingredients of a Business Process –
    - Events
    - Activities
    - Decision points
    - Actors
    - Physical objects
    - Information objects
    - Final outcome
    - Customer

Ingredients of a Business Process - Events

- Events: correspond to things that happen atomically, which means that they have no duration.

- Events are triggers that start, change, or end a process. These can be external or internal.

- Example : The arrival of a piece of equipment at a construction site is an event. This event may trigger the execution of a series of activities

Ingredients of a Business Process - Activities

- These are specific tasks or actions taken during a process, either automated, manual, or hybrid.

- For example, when a piece of equipment arrives, the site engineer inspects it. This inspection is an activity, in the sense that it takes time.

- When an activity is rather simple and can be seen as one single unit of work, we call it a task.

Ingredients of a Business Process – Decision Points

- These are points in a process where choices or conditions determine the next step or path or next process.
- In addition to events and activities, a typical process includes decision points, that is, points in time when a decision is made that affects the way the process is executed.

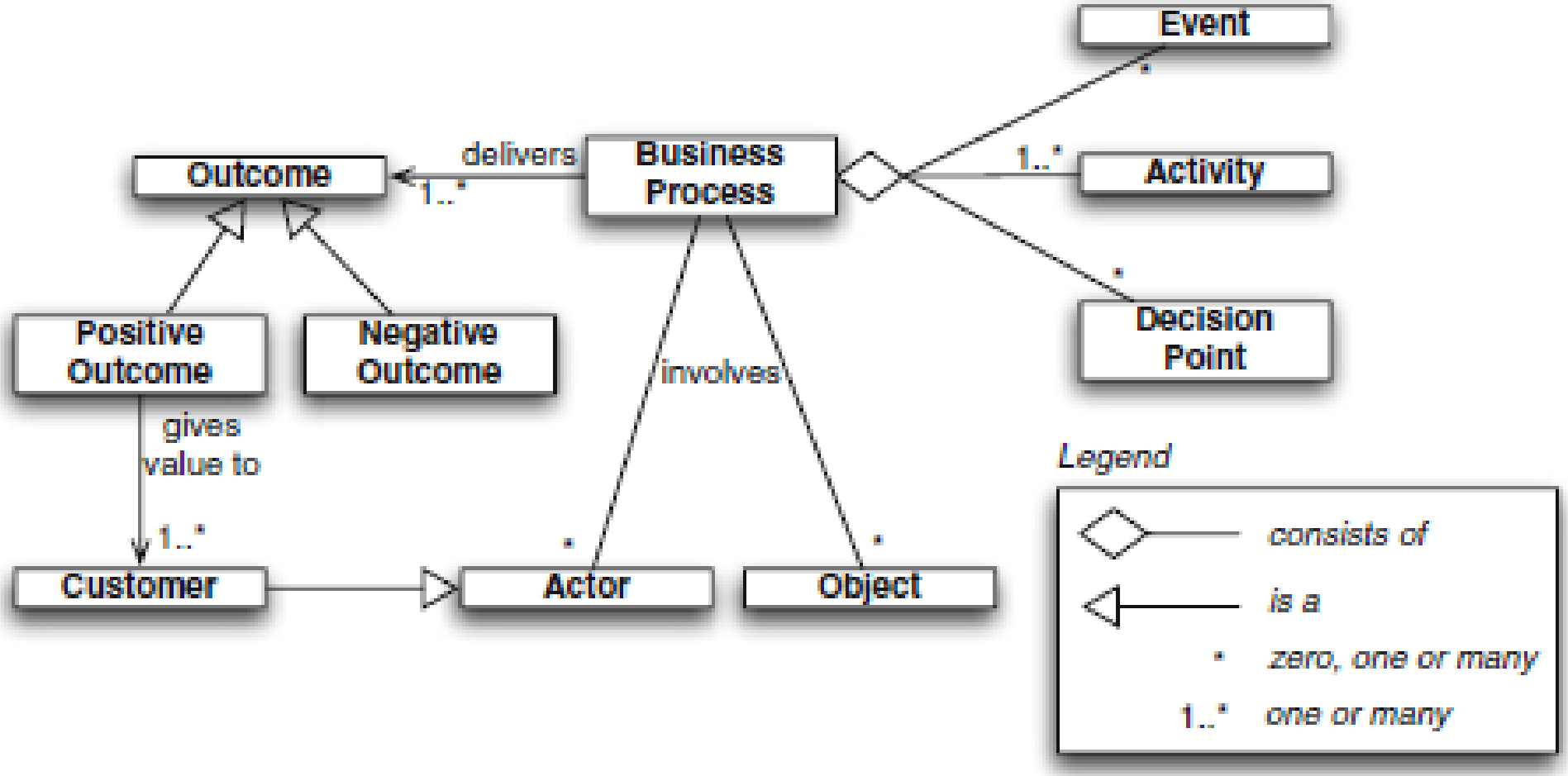
- For example, as a result of the inspection, the site engineer may decide that the equipment should be returned or that the equipment should be accepted.
- This decision affects what happens later in the process.

Ingredients of a Business Process - A process also involves

- Actors : human actors, organizations, or software systems acting on behalf of human actors or organizations.
- Physical objects : equipment, materials, products, paper documents.
- Informational objects : electronic documents and electronic records.

Ingredients of a Business Process – Final Outcome

- The execution of a process leads to one or several outcomes
- Ideally, an outcome should deliver value to the actors involved in the process
- In some cases, this value is not achieved or is only partially achieved. This corresponds to a *negative outcome*, as opposed to a *positive outcome* that delivers value to the actors involved.
- Among the actors involved in a process, the one who consumes the output is called the *customer*.

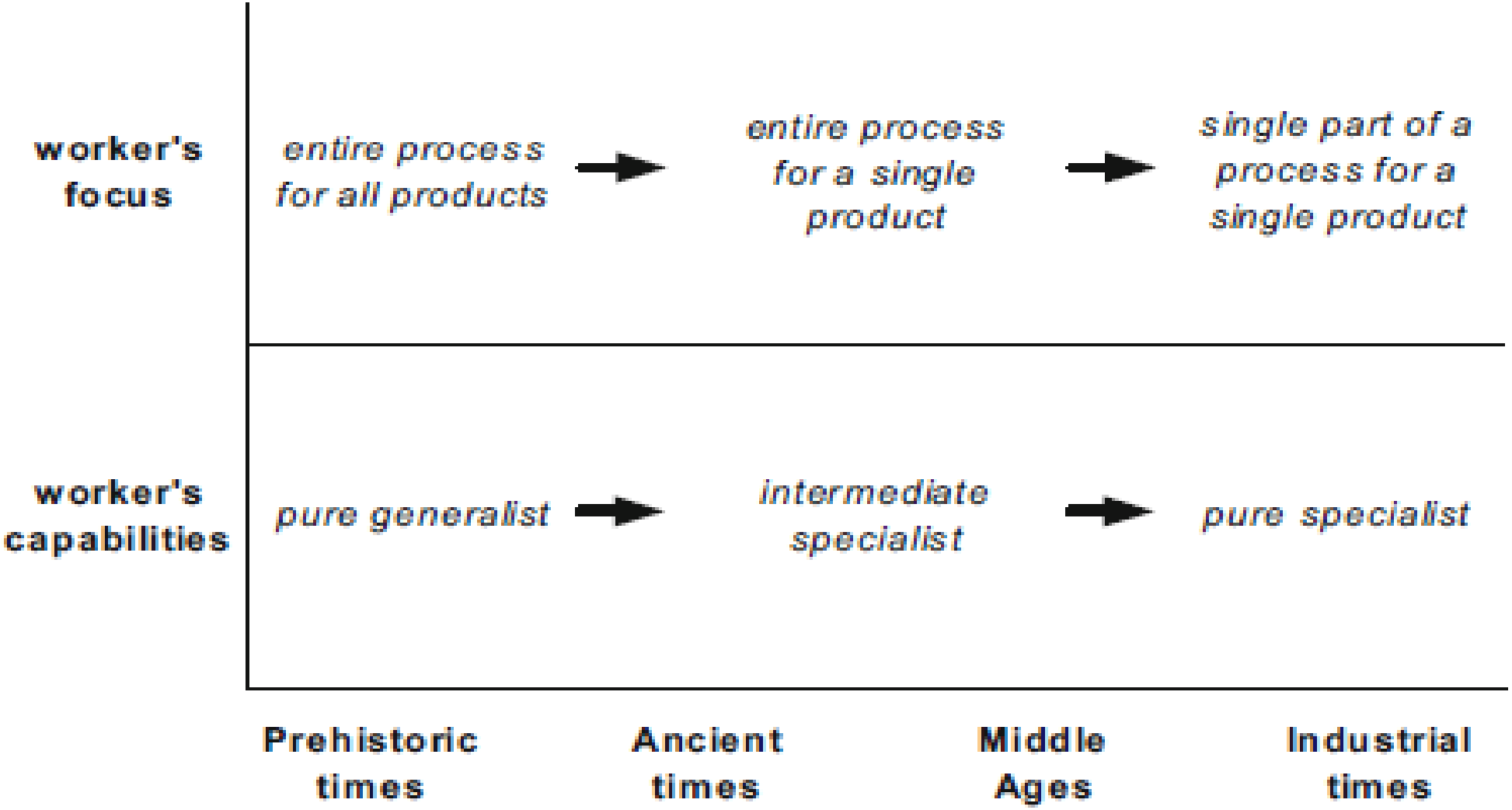

Event
Outcome
delivers
1..*
Business Process
1..*
Activity
Positive Outcome
Negative Outcome
involves
Decision Point
gives value to
1..*
Customer
Actor
Object
Legend
consists of
is a
* zero, one or many
1..* one or many

Exercise 02: Identify the event, activities, decision points, actors, physical objects, Information Objects and final outcome from the following scenario.

Online Food Delivery of a restaurant

- In this scenario, the process begins when a customer opens a food delivery mobile application and places an order from a chosen restaurant. Once the order is submitted, the restaurant receives and confirms it before the kitchen staff begins preparing the meal. After the food is ready, a delivery rider is assigned through the app to collect and deliver the order to the customer's location. During this process, several decision points are encountered, such as verifying whether the payment has been successfully processed, confirming the restaurant's acceptance of the order, and checking the availability of a delivery rider. The key actors involved include the customer, restaurant staff, delivery rider, and the food delivery application system. The main physical objects handled in the process are the food, delivery box, and delivery bike, while essential information objects include the order details, payment receipt, delivery tracking information, and customer feedback data. The process concludes when the customer receives the meal in good condition and provides feedback, marking the successful completion of the transaction.

History of BPM



Historical milestone

- BPR: One of the breakthrough events for the development of BPM was Ford's acquisition of a big financial stake in Mazda during the 1980s.

- Technology: Different types of IT systems emerged, most notably Enterprise Resource Planning (ERP) systems and Workflow Management Systems (WfMSs)

Stakeholders in BPM Lifecycle

- **Management Team.**
- **Process Owners**
- **Process Participants.**
- **Process Analysts**
- **Process Methodologist**
- **System Engineers**
- **BPM Group (***BPM Center of Excellence***)**

Management Team

- CEO (Chief Executive Officer), responsible for overall business performance
- COO (Chief Operations Officer), responsible for the way operations are setup
- CPO (Chief Process Officer), responsible for process performance
- CIO (Chief Information Officer), responsible for efficient and effective operation of information system infrastructure
- CFO (Chief Financial Officer), responsible for the overall financial performance of the company
- HRD (Human Resources Director), responsible when big numbers of process participants are involved

- **Process Owners**
  - Responsible for efficient and effective operations of a given process
  - Planning, organizing, controlling, monitoring, securing resources, guidance (in resolving errors)
  - One person could be responsible for multiple processes

- **Process Participants**
  - Human actors who perform the activities of a business process on a day-to-day basis
  - Conducting of routine work according to standards and guidelines
  - Support redesign activities and implementation efforts

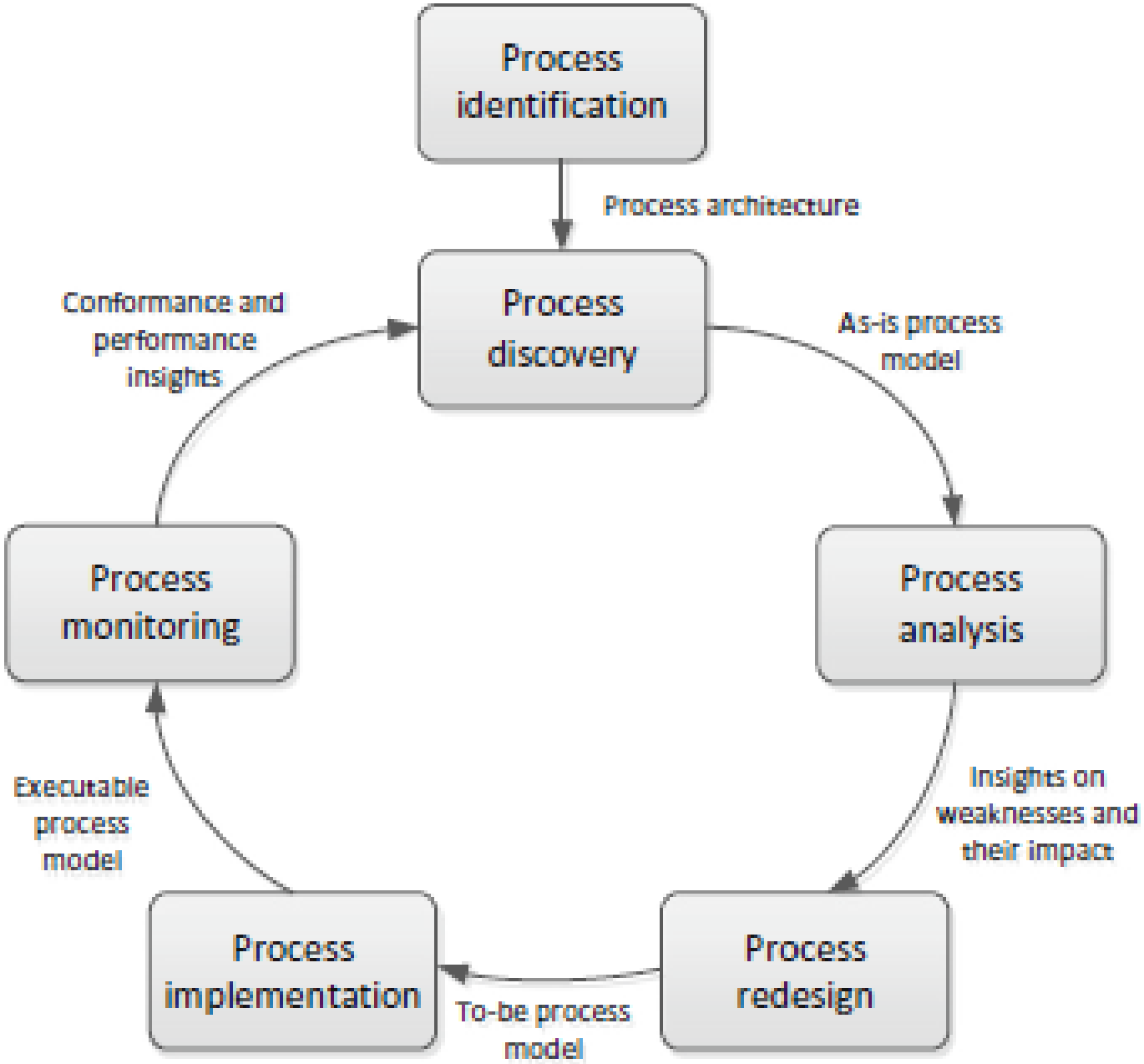
- **Process Analysts/supervisors**
  - Conduct process identification, discovery, analysis and redesign activities
  - Coordinate process implementation, process monitoring and process controlling
  - Report to management
  - Closely interact with process participants
  - Have a business or IT background depending on which change facet they focus

- System Engineers
    - Involved in process redesign and implementation
    - Interact with process analysts to capture system requirements
    - Can be an external contractor

BPM team (*BPM Center of Excellence*)

- analyzing, designing, implementing, and optimizing business processes to align them with organizational goals. Their responsibilities span multiple areas, ensuring processes are efficient, effective, and adaptable to changing business needs.
- Not all organizations have a dedicated BPM group
- Preserving of process documentation
- Knowledge of how to plan and execute an BPM project
- Responsible for maintaining process architecture, prioritizing of redesign projects and giving support to other stakeholders of a BPM project

BPM life cycle



1. Process Identification

- Recognize and prioritize processes to be managed or improved.
- Activities to be done
  - Map existing processes at a high level.
  - Identify key processes that are critical to achieving business objectives.
  - Define processes including boundaries, inputs, outputs, and stakeholders.

- Intended outcome: A list of prioritized processes for management or improvement/optimization –process architecture

2. Process Discovery (As-Is Analysis)

- Understand the current state of the process.
- Activities to be done:
  - Document existing processes using flowcharts, **BPMN diagrams**, or other tools.
  - Collect data from stakeholders, process owners, and employees.
  - Identify inefficiencies, bottlenecks, and pain points.
- Intended outcome: clear and detailed map of the current process (as-is process)-process models.

3. Process Analysis

- Analyze the as-is process to identify opportunities for improvement.
- Activities to be done:
  - Use root cause analysis, value stream mapping, or other techniques to assess inefficiencies.
  - Evaluate the process against performance metrics and business objectives.
  - Identify areas for automation, streamlining, or redesign.
- Intended outcome: Insights into problems and opportunities within the current process.

4. Process Design (To-Be Design)/redesign

- Create an optimized version of the process.
- Activities to be done:
  - Redesign the process to address identified inefficiencies and align it with business goals.
  - Incorporate technology, automation, and best practices where applicable.
  - Define roles, responsibilities, and workflows for the new process.
- Intended Outcome: A blueprint for the improved/optimized process (to-be process).

5. Process Implementation

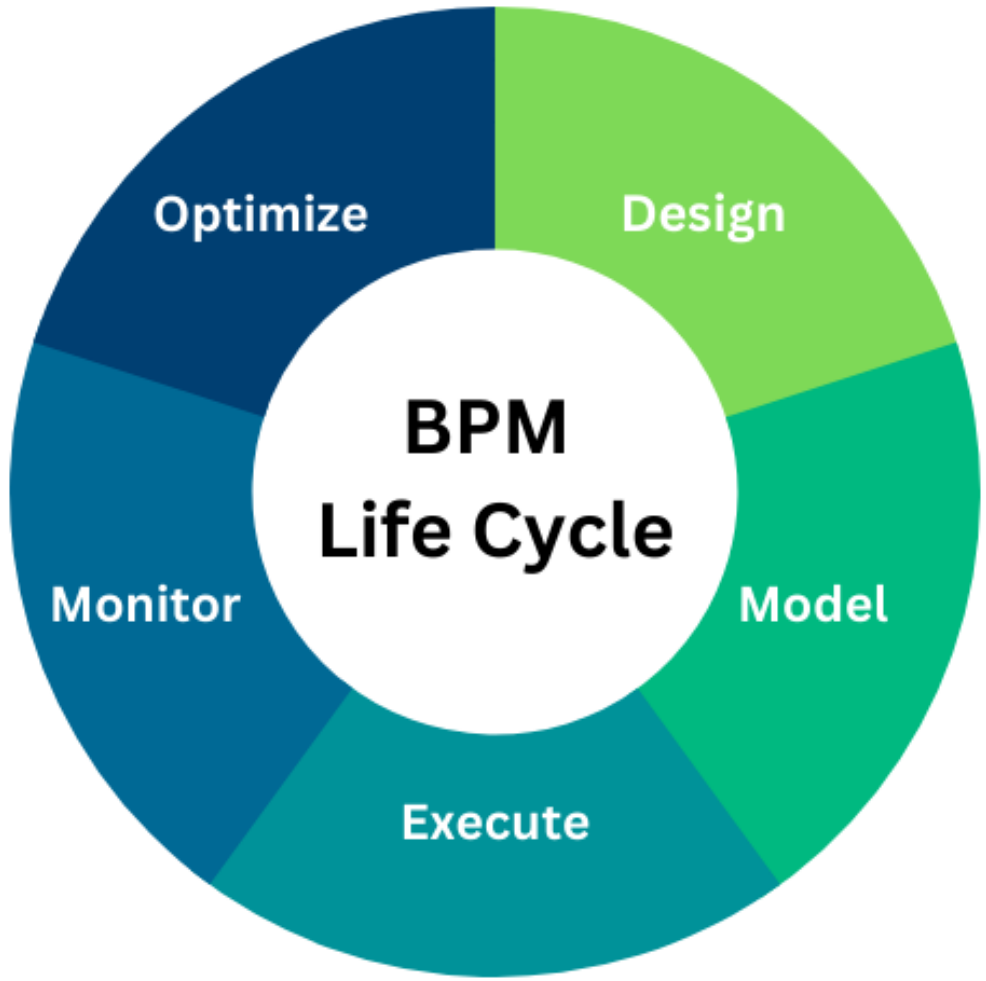
- Deploy the redesigned process into the organization.
- Activities to be done:
  - Implement process changes, including new tools, technologies, or workflows.
  - Train employees and stakeholders on the new process.
  - Conduct pilot testing to identify issues before full-scale deployment.
- Intended outcome: The new process is operational within the organization.

6. Process Monitoring and Controlling

- Track process performance and ensure it operates as intended.
- Activities to be done:
  - Use KPIs and dashboards to monitor process outcomes.
  - Collect real-time feedback from stakeholders and systems.
  - Compare performance data against defined goals and benchmarks.
- Intended outcome: Continuous insights into process performance.

7. Process Optimization (Continuous Improvement)

- Enhance the process further based on performance data and feedback.
- Activities to be done :
    - Identify areas for refinement or additional improvement.
    - Use methodologies like Six Sigma, Lean, or PDCA (Plan-Do-Check-Act) for optimization.
    - Adapt processes to changing business needs or technological advancements.
- Intended Outcome: Incremental or breakthrough improvements in process performance.


Optimize
Design
BPM
Life Cycle
Monitor
Model
Execute

Recap

- What is BPM?
- Why BPM is needed?
- What is a process?
- What are the components in a process?
- What are the steps in the BPM lifecycle?

Thank You

Business Process Management (Lecture 02) Process Identification

Poornima Jayathunga

Process Identification

- management activities that aim to systematically define the set of business processes of an organization and establish clear criteria for selecting specific processes for improvement

Focusing on Key Processes

- Some processes need to receive priority because they are of strategic importance to an organization's survival.
- The processes that an organization should focus on are found in areas where there is either great value created or significant trouble present (or both)
- Need to answer two questions
    1. what processes are executed in the organization? and
    2. which ones should the organization focus on?

Focusing on Key Processes

- Process identification is concerned with two successive phases:
  - Designation : Objective is to gain an understanding of the processes an organization is involved in as well as their interrelationships.
  - Evaluation: based on the understanding that is established in the previous phase, intends to develop a prioritization among these for process management activities (modeling, redesign, automation, monitoring, etc.).

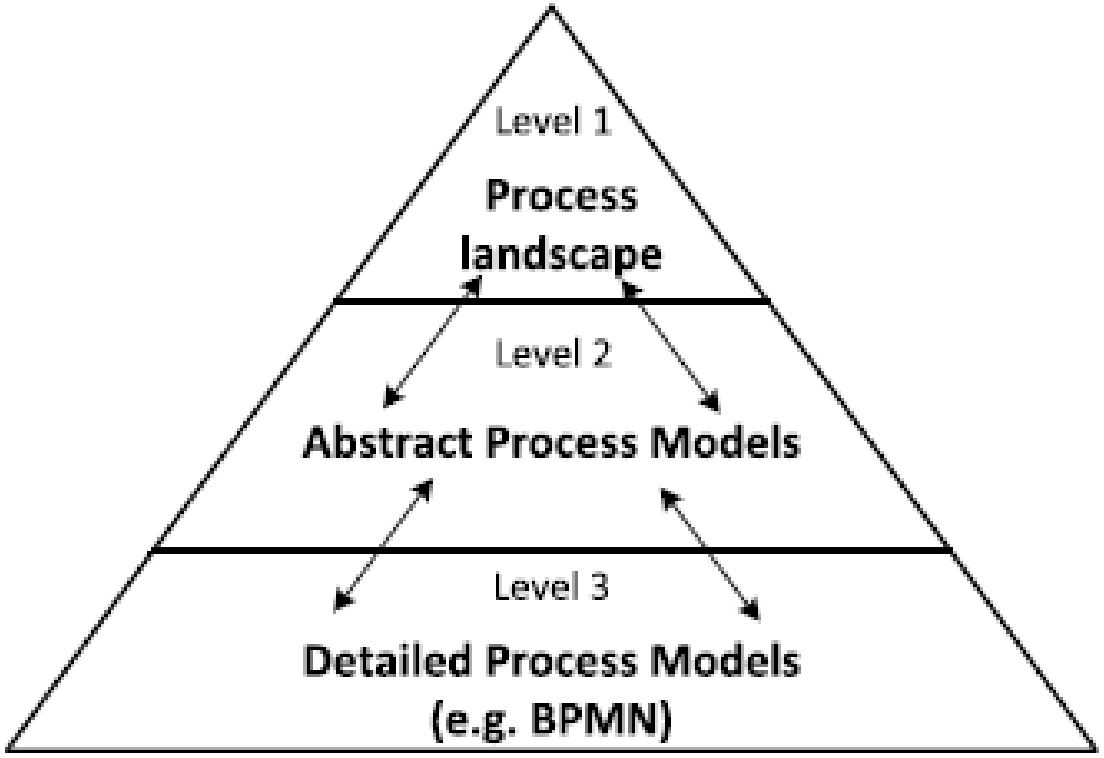
The Designation Phase

- Identify Processes
- Map Interrelationships
- Document Process Details
- Establish Process Boundaries
- Analyze Existing Performance

Evaluation Phrase

- Assess Process Performance
- Prioritize Process Improvements
- Choose Process Management Activities
- Evaluate Feasibility and Resources

Fig. 2.1 The different levels of detail in a process architecture



PROCESS CHECKLIST

- **Is it a process at all?**
- **Can the process be controlled?**
- **Is the process important enough to manage?**
- **Is the scope of the process not too big?**
- **Is the scope of the process not too small?**

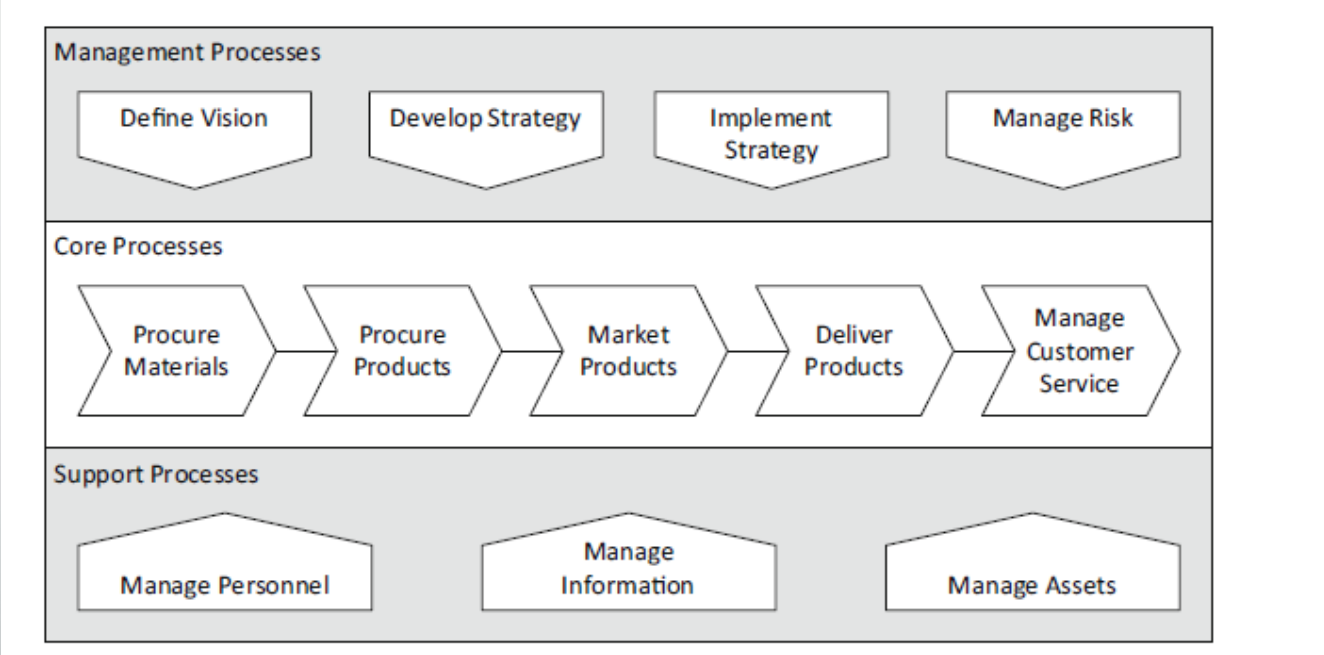
Why Process Architecture

- aim of a process architecture is to provide a representation of the processes that exist in an organization
- The definition of a process architecture has to face the complexity of the whole organization.
- First - differentiate categories of processes , Second- describe different relationships between processes

Level 01: Process Categories for Landscaping

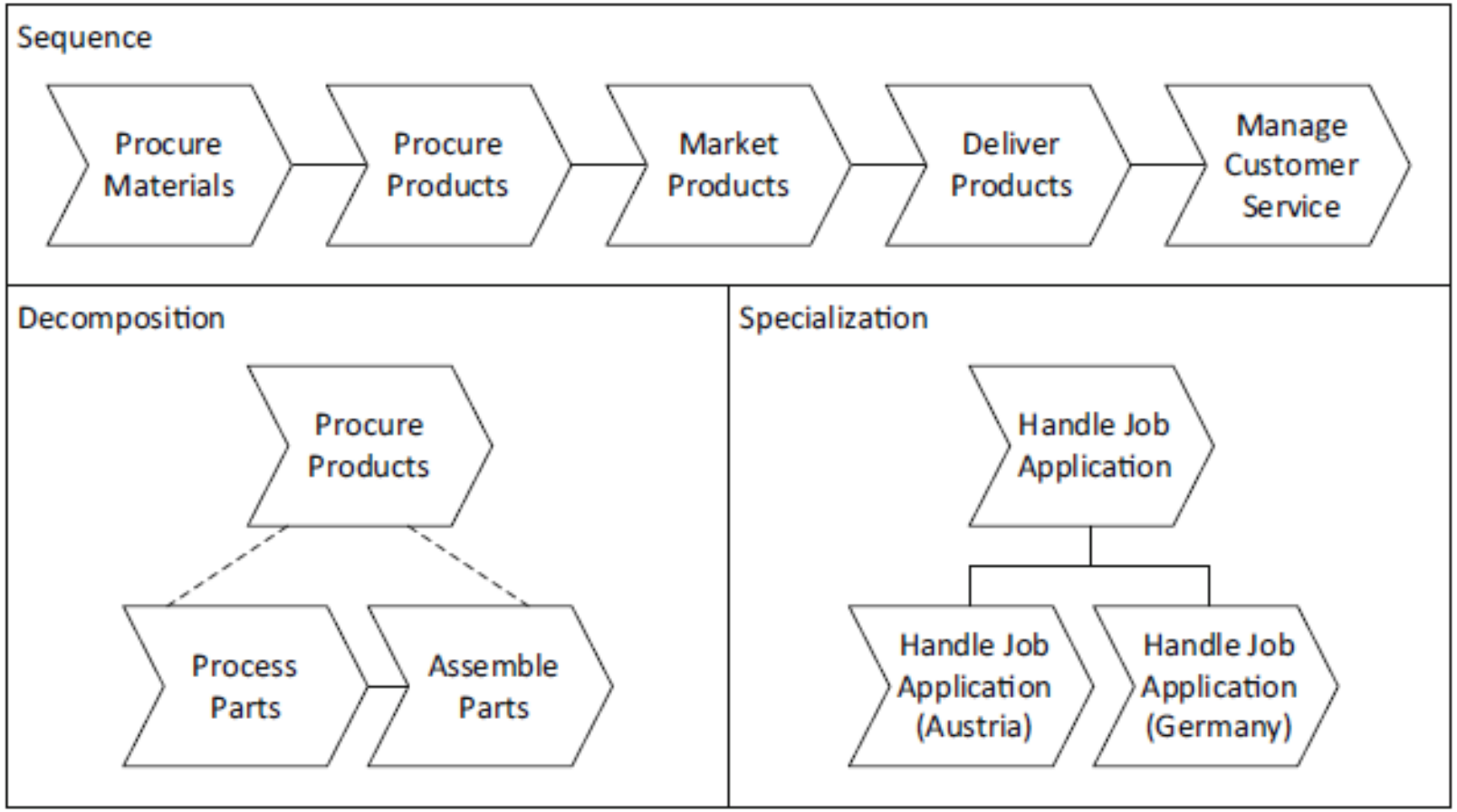
- **Porter's Value Chain model.**
  - **Core processes: include design and development, manufacturing, marketing and sales, delivery, after-sales, and direct procurement**
  - **Support processes: include indirect procurement (i.e., sourcing of hardware, furniture, stationery, etc.), human resource management, information technology management, accounting, financial management, and legal services.**
  - **Management processes: include strategic planning, budgeting, compliance and risk management, as well as investors, suppliers, and partners management.**

Process Landscaping



Relationships Between Processes

- Sequence
- Decomposition
- Specialization


Sequence
Procure Materials
Procure Products
Market Products
Deliver Products
Manage Customer Service
Decomposition
Procure Products
Process Parts
Assemble Parts
Specialization
Handle Job Application
Handle Job Application (Austria)
Handle Job Application (Germany)

Reuse of Reference Models

- process analysts find it difficult to identify processes of an organization and the levels of a process architecture
- It might be helpful to use reference models as an aid.
- These reference models are developed by a range of industry consortia, non-profit associations, government research programs, and academia.

Examples of some frameworks

- Information Technology Infrastructure Library (ITIL)
- Supply Chain Operations Reference Model (SCOR)
- Process Classification Framework
- Performance Framework

Process landscape model of Vienna's public transport operator

Example

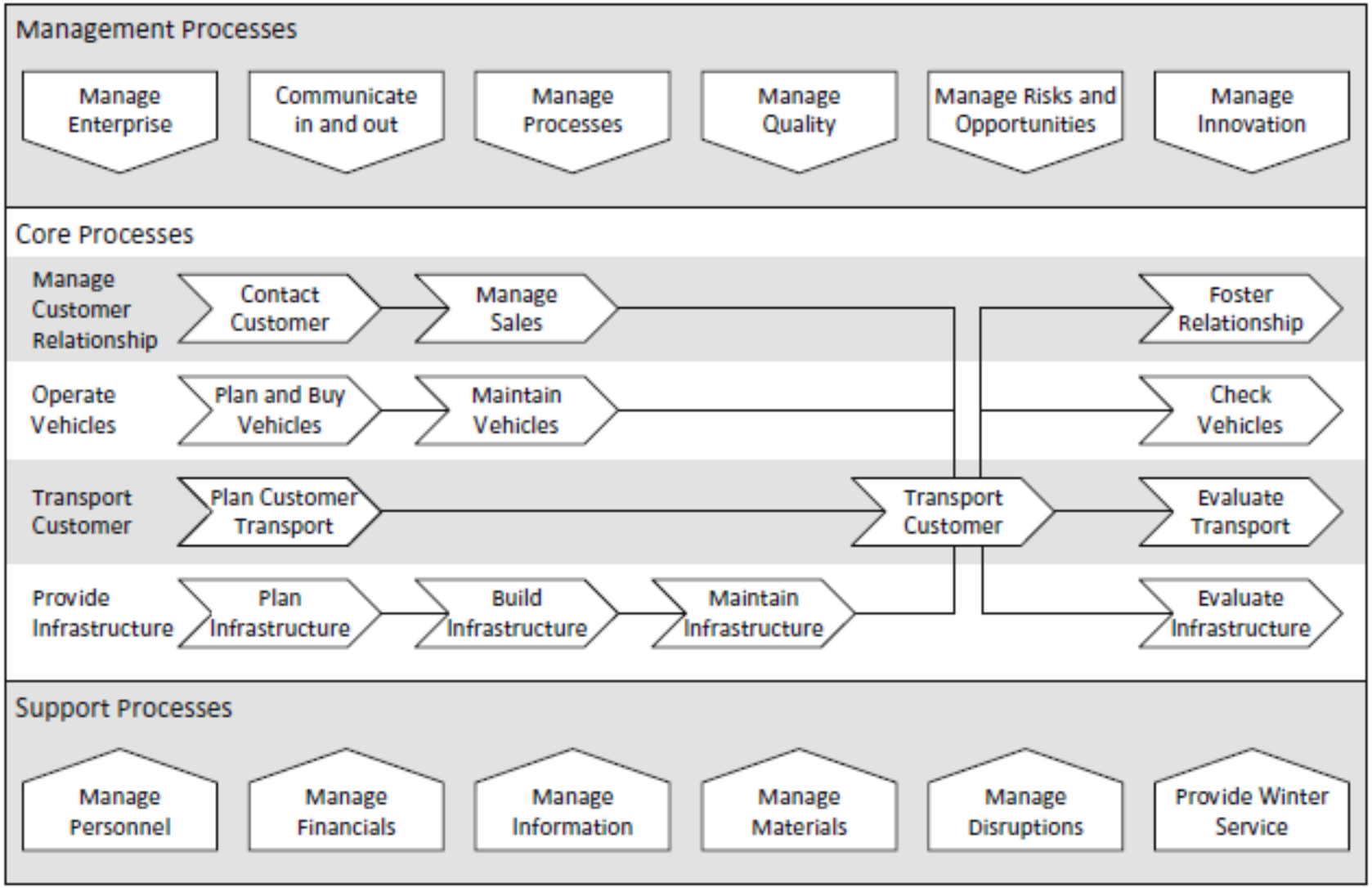

Management Processes
Manage Enterprise
Communicate in and out
Manage Processes
Manage Quality
Manage Risks and Opportunities
Manage Innovation
Core Processes
Manage Customer Relationship
Contact Customer
Manage Sales
Foster Relationship
Operate Vehicles
Plan and Buy Vehicles
Maintain Vehicles
Check Vehicles
Transport Customer
Plan Customer Transport
Transport Customer
Evaluate Transport
Provide Infrastructure
Plan Infrastructure
Build Infrastructure
Maintain Infrastructure
Evaluate Infrastructure
Support Processes
Manage Personnel
Manage Financials
Manage Information
Manage Materials
Manage Disruptions
Provide Winter Service

Table 2.1 Level 1 and Level 2 of the APQC Process Classification Framework

1.0 Develop Vision and Strategy
1.1 Define the business concept and long-term vision
1.2 Develop business strategy
1.3 Execute and measure strategic initiatives

2.0 Develop and Manage Products and Services
2.1 Govern and manage product and service development program
2.2 Generate and define new product and service ideas
2.3 Develop products and services

3.0 Market and Sell Products and Services
3.1 Understand markets, customers, and capabilities
3.2 Develop marketing strategy
3.3 Develop and manage marketing plans
3.4 Develop sales strategy
3.5 Develop and manage sales plans

4.0 Deliver Physical Products
4.1 Plan for and align supply chain resources
4.2 Procure materials and services
4.3 Produce, manufacture, and deliver product
4.4 Manage logistics and warehousing

5.0 Deliver Services
5.1 Establish service delivery governance and strategies
5.2 Manage service delivery resources
5.3 Deliver service to customer

6.0 Manage Customer Service
6.1 Develop customer care and customer service strategy
6.2 Plan and manage customer service contacts
6.3 Service products after sales
6.4 Manage product recalls and regulatory audits
6.5 Evaluate customer service operations and customer satisfaction

7.0 Develop and Manage Human Capital
7.1 Develop and manage human resources planning, policies, and strategies
7.2 Recruit, source, and select employees
7.3 Develop and counsel employees
7.4 Manage employee relations
7.5 Reward and retain employees
7.6 Redeploy and retire employees
7.7 Manage employee information and analytics
7.8 Manage employee communication
7.9 Deliver employee communications

8.0 Manage Information Technology (IT)
8.1 Manage the business of information technology
8.2 Develop and manage IT customer relationships
8.3 Develop and implement security, privacy, and data protection controls
8.4 Manage enterprise information
8.5 Develop and maintain information technology solutions
8.6 Deploy information technology solutions
8.7 Deliver and support information technology services

9.0 Manage Financial Resources
9.1 Perform planning and management accounting
9.2 Perform revenue accounting
9.3 Perform general accounting and reporting
9.4 Manage fixed-asset project accounting
9.5 Process payroll
9.6 Process accounts payable and expense reimbursements
9.7 Manage treasury operations
9.8 Manage internal controls
9.9 Manage taxes
9.10 Manage international funds/consolidation
9.11 Perform global trade services

10.0 Acquire, Construct, and Manage Assets
10.1 Plan and acquire assets
10.2 Design and construct productive assets
10.3 Maintain productive assets
10.4 Dispose of assets

11.0 Manage Enterprise Risk, Compliance, Remediation and Resiliency
11.1 Manage enterprise risk
11.2 Manage compliance
11.3 Manage remediation efforts
11.4 Manage business resiliency

12.0 Manage External Relationships
12.1 Build investor relationships
12.2 Manage government and industry relationships
12.3 Manage relations with board of directors
12.4 Manage legal and ethical issues
12.5 Manage public relations program

13.0 Develop and Manage Business Capabilities
13.1 Manage business processes
13.2 Manage portfolio, program, and project
13.3 Manage quality
13.4 Manage change
13.5 Develop and manage enterprise-wide knowledge management (KM) capability
13.6 Measure and benchmark
13.7 Manage environmental health and safety (EHS)

Steps for model landscaping

- Clarify terminology
- Identify end-to-end processes
- For each end-to-end process, identify its sequential processes
- For each business process, identify its major management and support processes
- Decompose and specialize business processes
- Compile process profile
- Check completeness and consistency

Name of Process: Procure-to-Pay	
Vision: The objective of the procurement process is to secure that the entire range of external products and services becomes available on time and is at the required level of quality.	
Process Owner: Chief Financial Officer (CFO)	
Customer of process: • Requesting unit	**Expectation of customer:** • Timely, economic and complete provision
Outcome: Delivered products or provided services for the requested unit	
Trigger: Need is identified	
First activity: Submit Request ..... **Last activity:** Create Purchase Order	
Interfaces inbound: Plan-to-Procure **Interfaces outbound:** Construct-to-Complete	
Required resources: • Human resources: Site Engineer, Clerk, Works Engineer • Information, documents, know-how: procurement guidelines, supplier rating, framework contract • Work environment, materials, infrastructure: Procurement information system	
Process Performance Measures: • Cycle Time • Operational Costs • Error Rate	

Selecting and Prioritizing

- Selection Criteria
  - **Strategic Importance**
  - **Health**
  - **Feasibility**

Process Performance Measures

- key performance indicators
  - Four dimension
    - Time
    - Cost
    - Quality
    - Flexibility:

Prioritizing - Process Portfolio

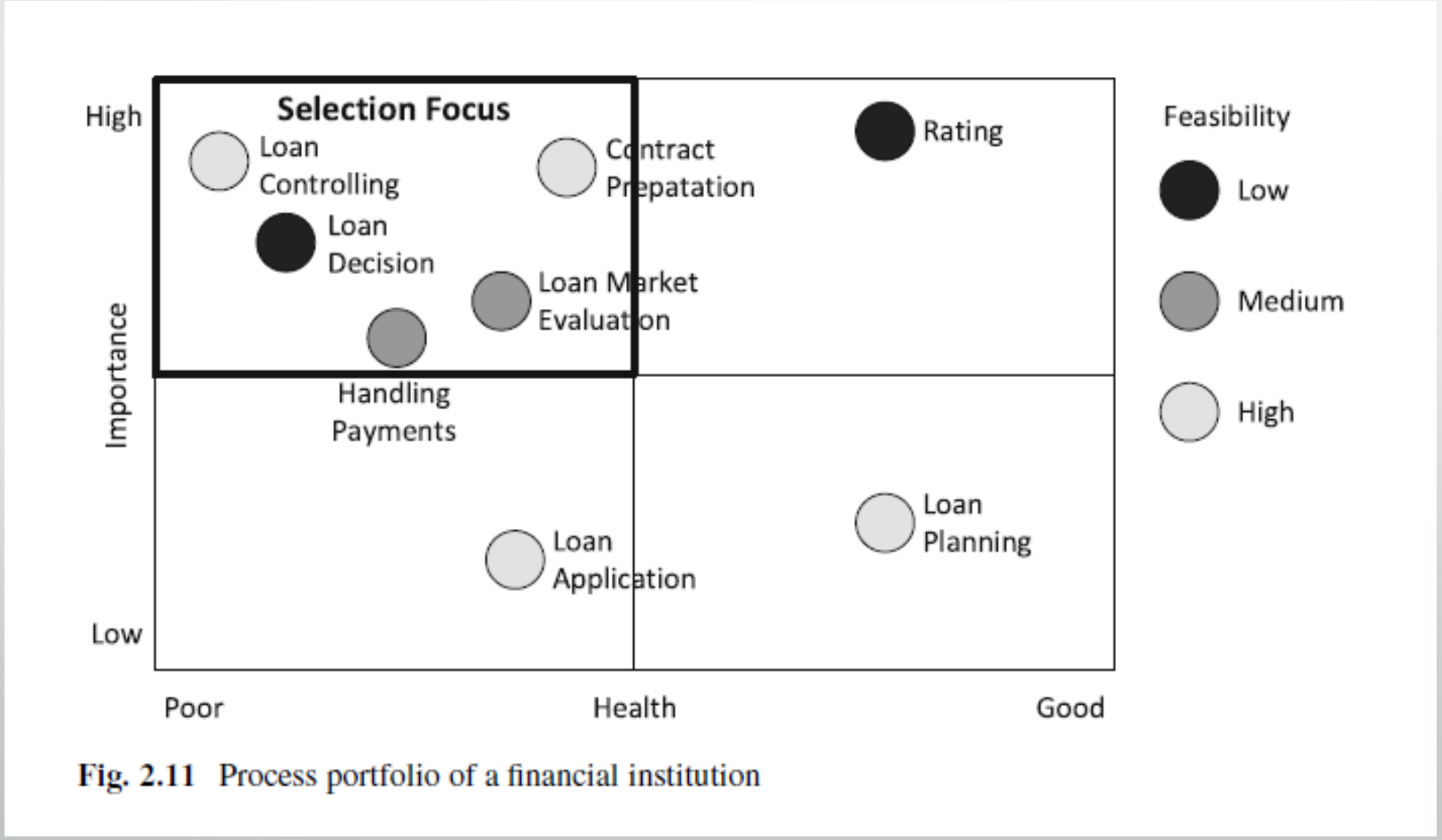
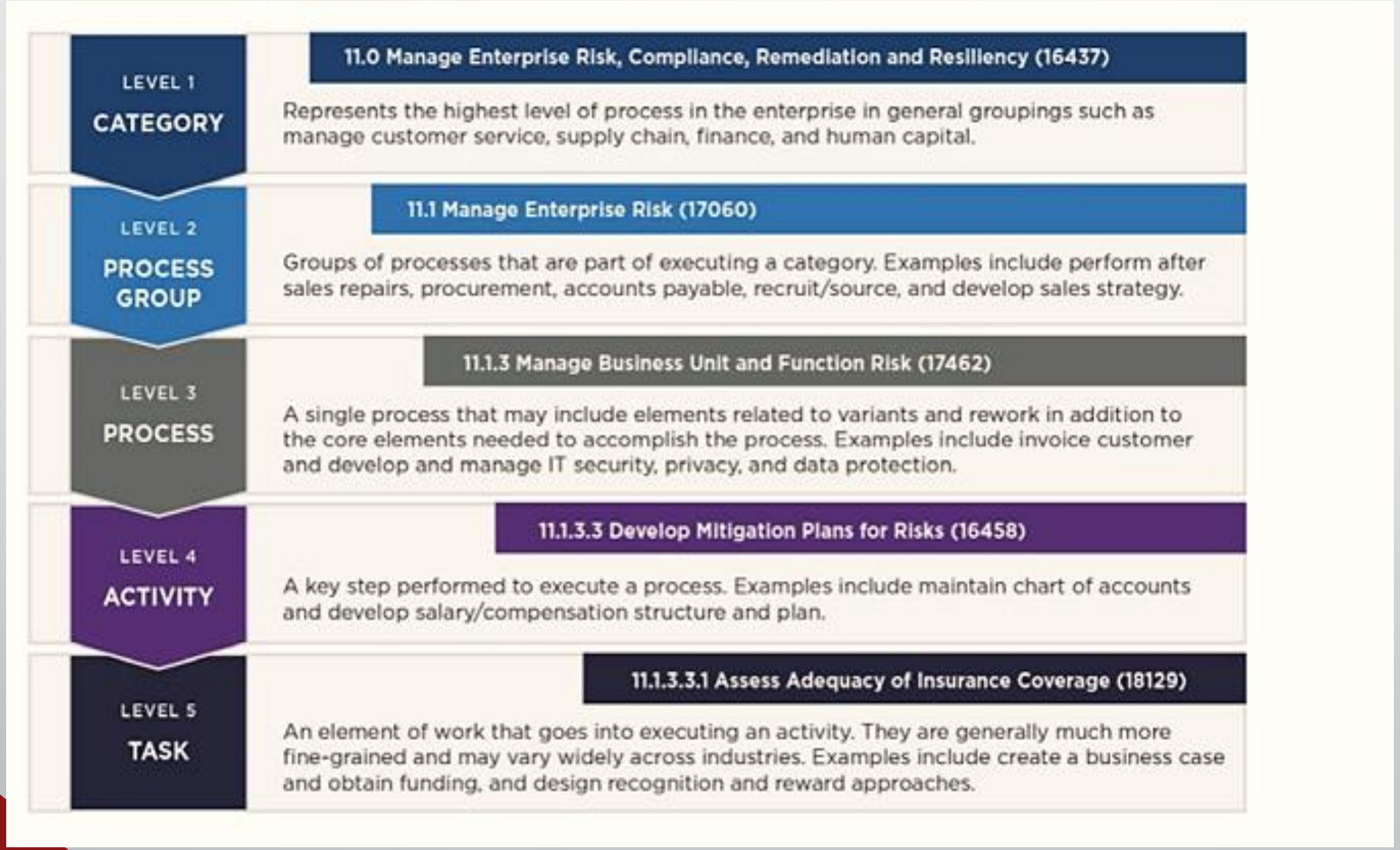
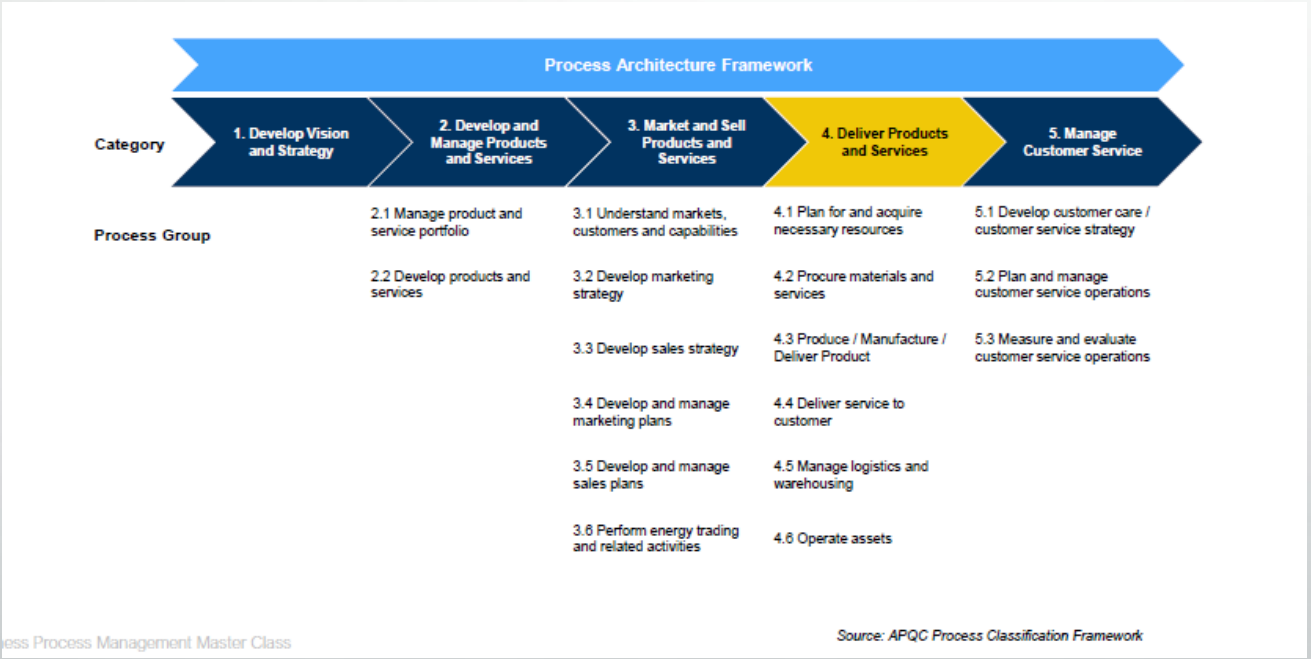


Fig. 2.11 Process portfolio of a financial institution

APQC framework



Example

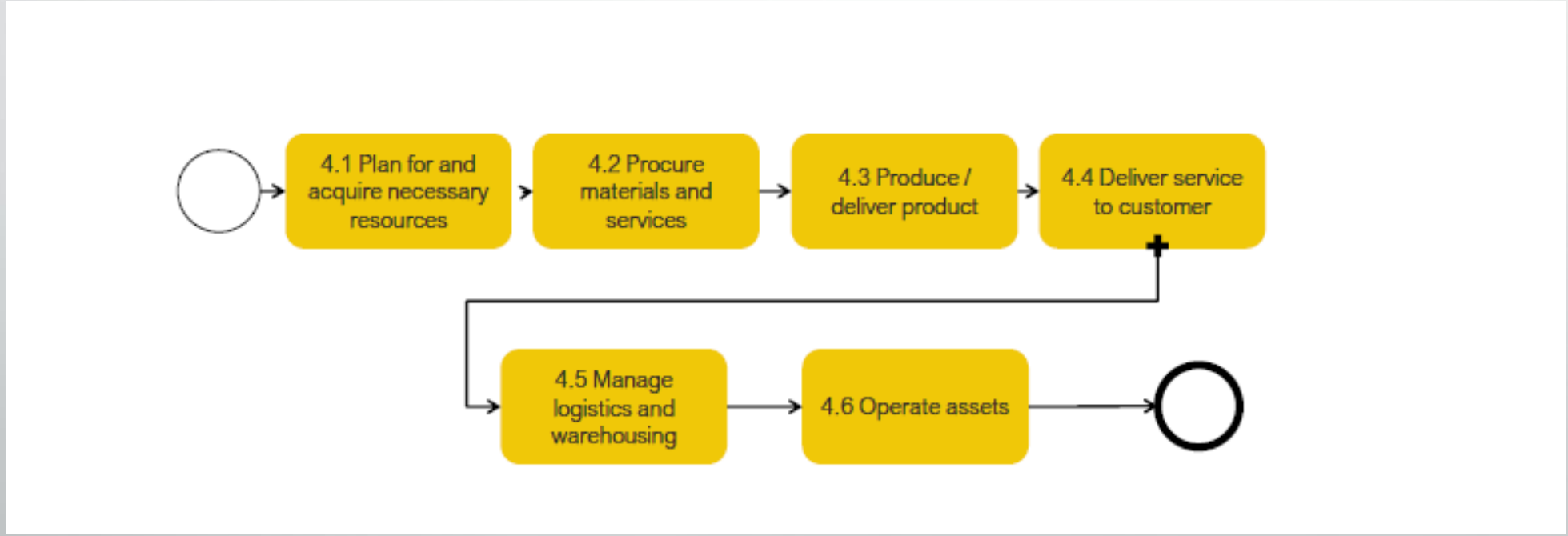

Process Architecture Framework
Category
1. Develop Vision and Strategy
2. Develop and Manage Products and Services
3. Market and Sell Products and Services
4. Deliver Products and Services
5. Manage Customer Service
Process Group
2.1 Manage product and service portfolio
2.2 Develop products and services
3.1 Understand markets, customers and capabilities
3.2 Develop marketing strategy
3.3 Develop sales strategy
3.4 Develop and manage marketing plans
3.5 Develop and manage sales plans
3.6 Perform energy trading and related activities
4.1 Plan for and acquire necessary resources
4.2 Procure materials and services
4.3 Produce / Manufacture / Deliver Product
4.4 Deliver service to customer
4.5 Manage logistics and warehousing
4.6 Operate assets
5.1 Develop customer care / customer service strategy
5.2 Plan and manage customer service operations
5.3 Measure and evaluate customer service operations
Source: APQC Process Classification Framework

Categories



The highest level of the hierarchy and is indicated by whole numbers (e.g., 8.0 and 9.0). These are processes that are grouped by a common subject

Process Grouping



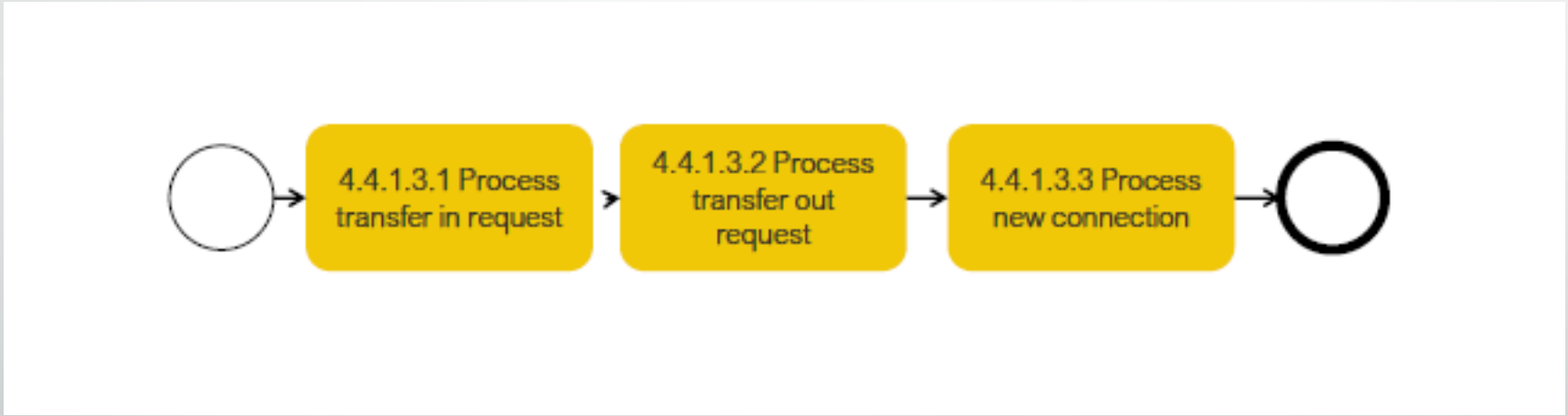
Process


4.4.1 Confirm specific service requirements
4.4.2 Identify and schedule resources to meet service
4.4.3 Provide the specific service to customers
4.4.4 Ensure quality of service

Activities


4.4.1. Process customer request
4.4.1.2 Create customer profile
4.4.1.3 Generate service order

Tasks


4.4.1.3.1 Process transfer in request
4.4.1.3.2 Process transfer out request
4.4.1.3.3 Process new connection

Exercise 2.1

- Develop a University Process Architecture for UWU undergraduate (From admission to graduation (convocation))

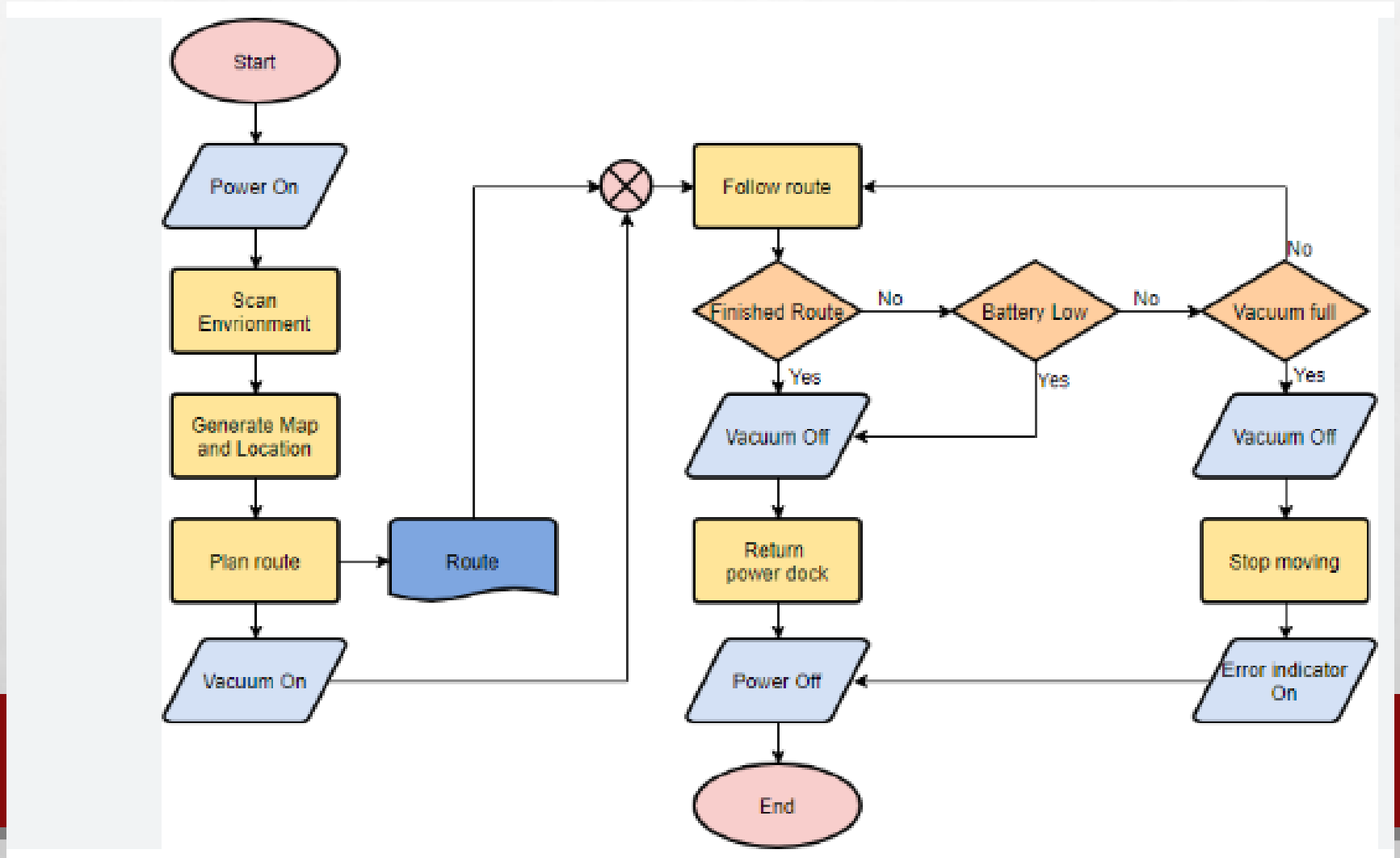
BUSINESS PROCESS MANAGEMENT- PROCESS MODELING

POORNIMA JAYATHUNGA

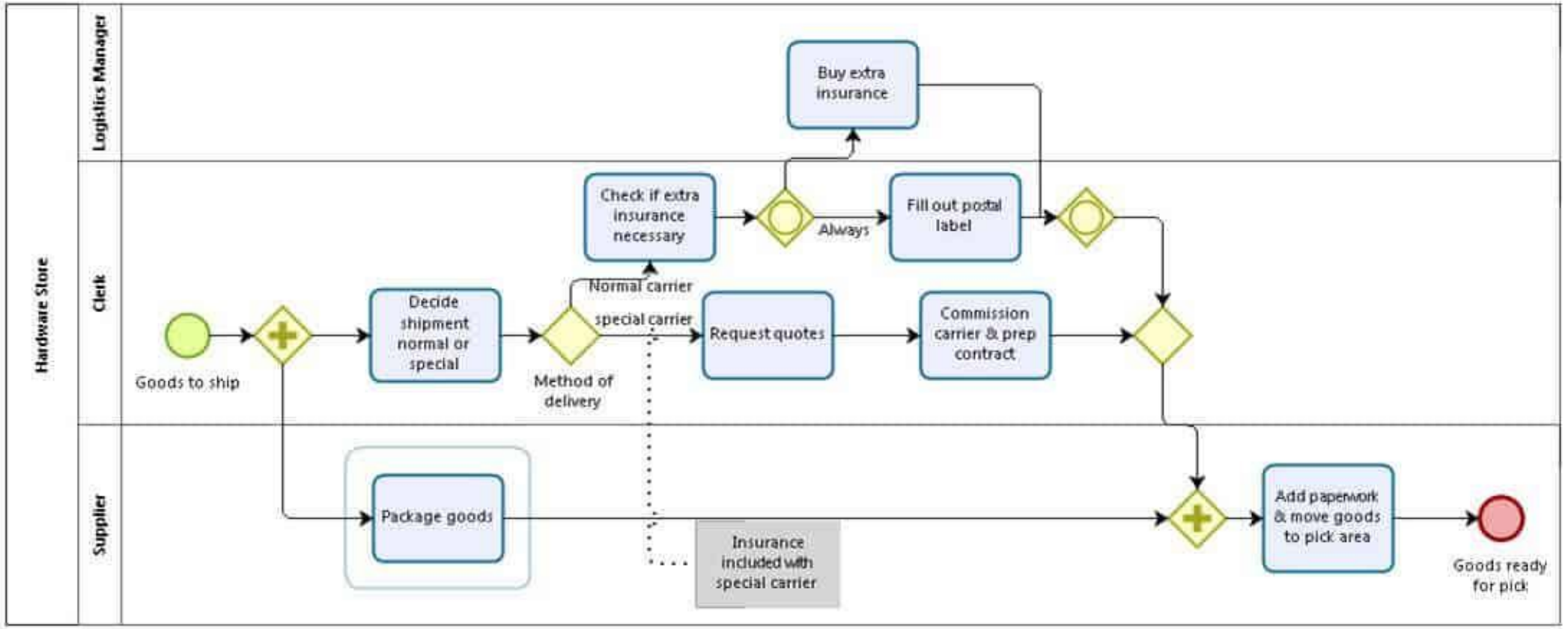
WHY PROCESS MODELING?

- Understand the process and to share our understanding of the process with the people who are involved with it on a daily basis
- Better understand the process and to identify and prevent issues
- Thorough understanding of business processes is the prerequisite to conduct process analysis, redesign, or automation.

MAPPING- STATIC TOOLS



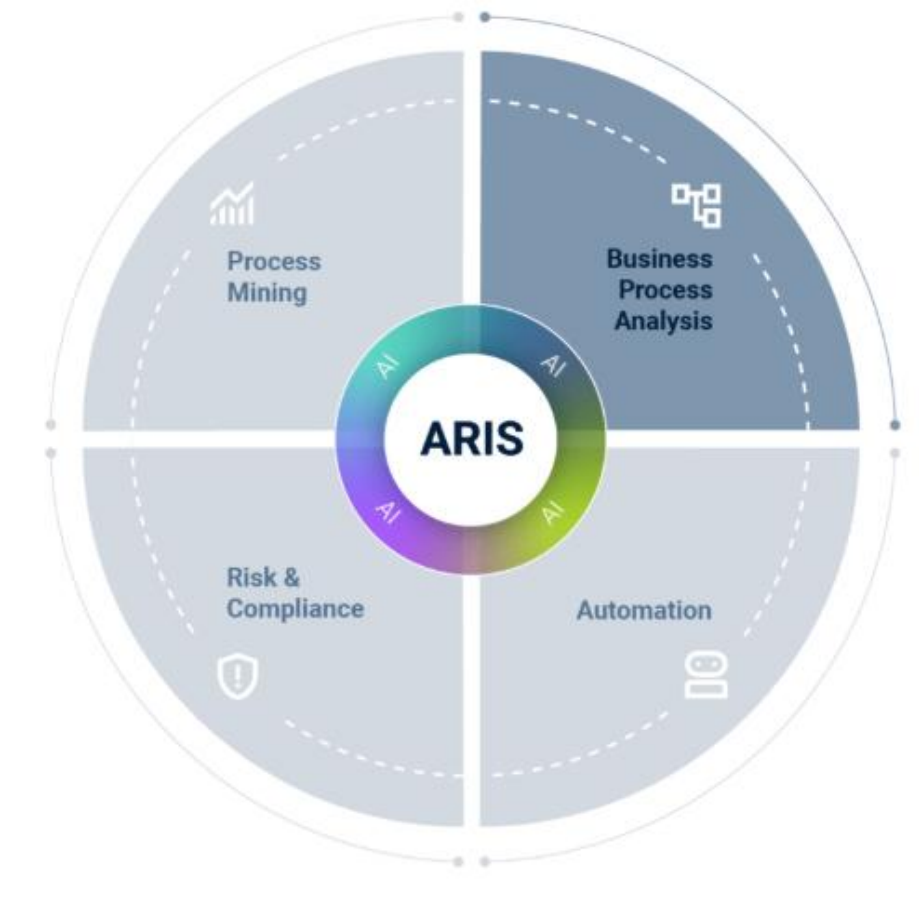
MODELING – DYNAMIC TOOLS



BUSINESS PROCESS MANAGEMENT PLATFORMS REVIEWS AND RATINGS – GARTNER BPM MAGIC QUADRANT

- https://www.gartner.com/reviews/market/business-process-management-platforms

ARIS SOFTWARE TOOL



DIFFERENT MODELING METHODS

- Business Process Model And Notation (BPMN)
- Unified Modeling Language (UML)
- Flowchart

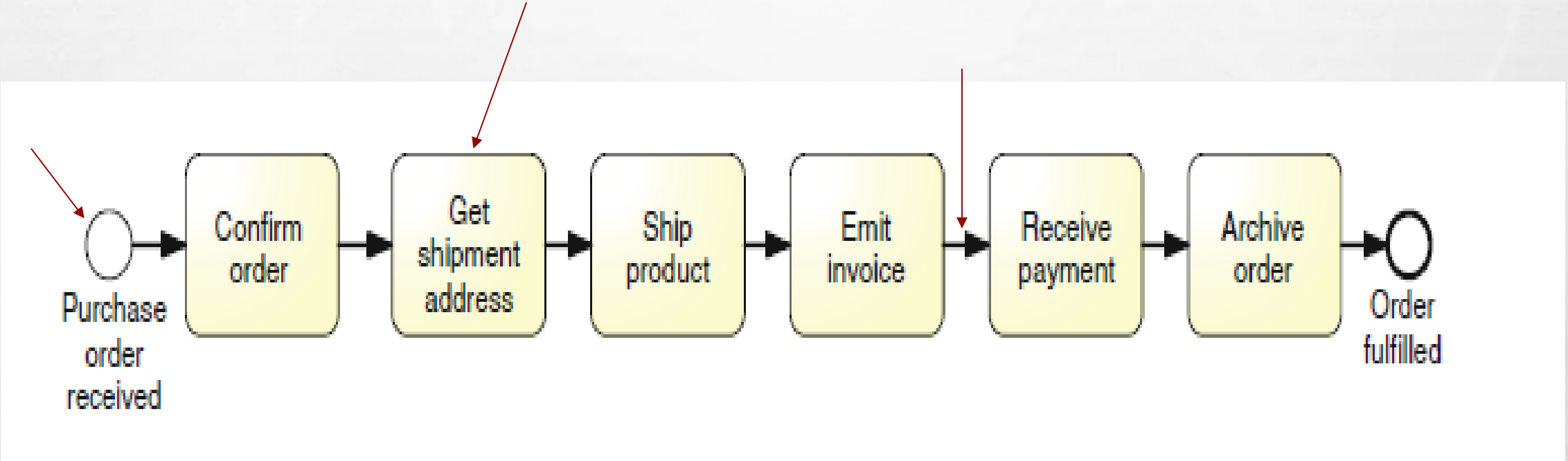
COMPONENTS OF A MODELING LANGUAGE

- includes
  - activities,
  - events,
  - gateways,
  - and sequence flows

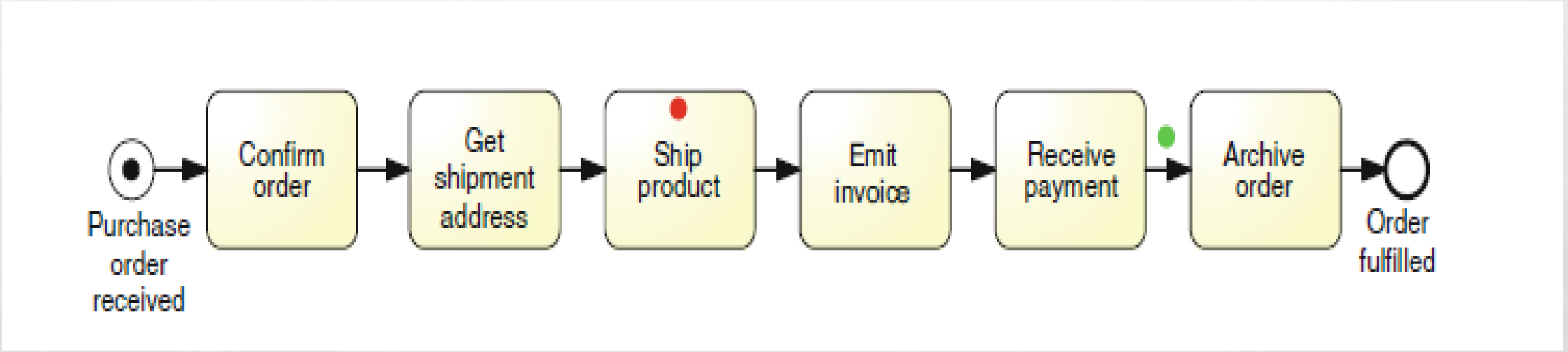
THREE MOST BASIC CONCEPTS - BPMN

- EVENTS - 
- ACTIVITIES - 
- ARCS (SEQUENCE FLOWS) - 

BPMN


Purchase order received
Confirm order
Get shipment address
Ship product
Emit invoice
Receive payment
Archive order
Order fulfilled

TOKEN


Purchase order received
Confirm order
Get shipment address
Ship product
Emit invoice
Receive payment
Archive order
Order fulfilled

NAMING CONVENTIONS

- Activities: label should begin with a verb. Capitalized the first letter
  - E.G. Approve order
- Events
  - label should begin with a noun . Capitalized the first letter
- To name a process model
  - A noun, potentially preceded by an adjective – e.g: “loan origination”, or “order-to-cash”

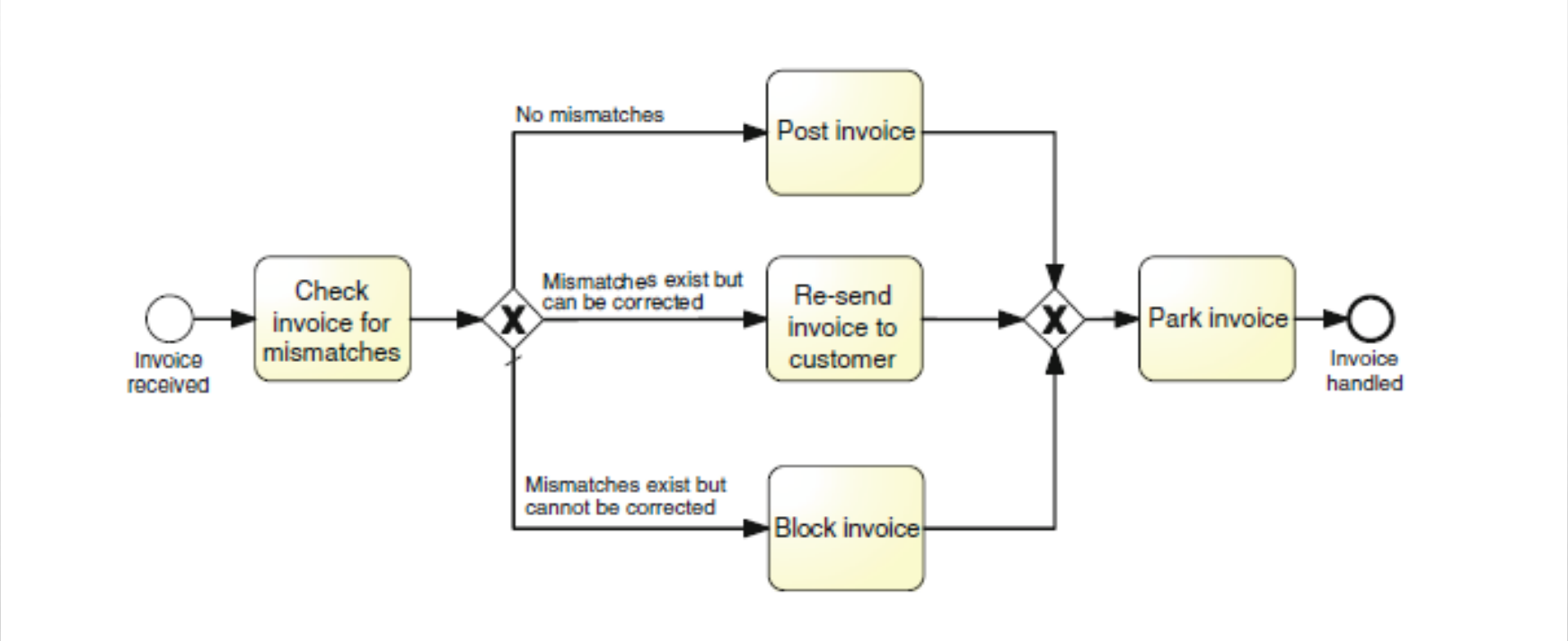
BRANCHING AND MERGING

- Sometimes processes can be mutually exclusive or and *concurrent*
- Introduce the notion of a gateway
- Gateway: A split gateway represents a point where the process flow diverges while a join gateway represents a point where the process flow converges.

BRANCHING AND MERGING - EXCLUSIVE DECISIONS (XOR)

- As soon as an invoice is received from a customer, it needs to be checked for mismatches. The check may result in any of the following three options: (i) there are no mismatches, in which case the invoice is posted; (ii) there are mismatches but these can be corrected, in which case the invoice is resent to the customer; and (iii) there are mismatches but these cannot be corrected, in which case the invoice is blocked. Once one of these three activities is performed the invoice is parked and the process completes.

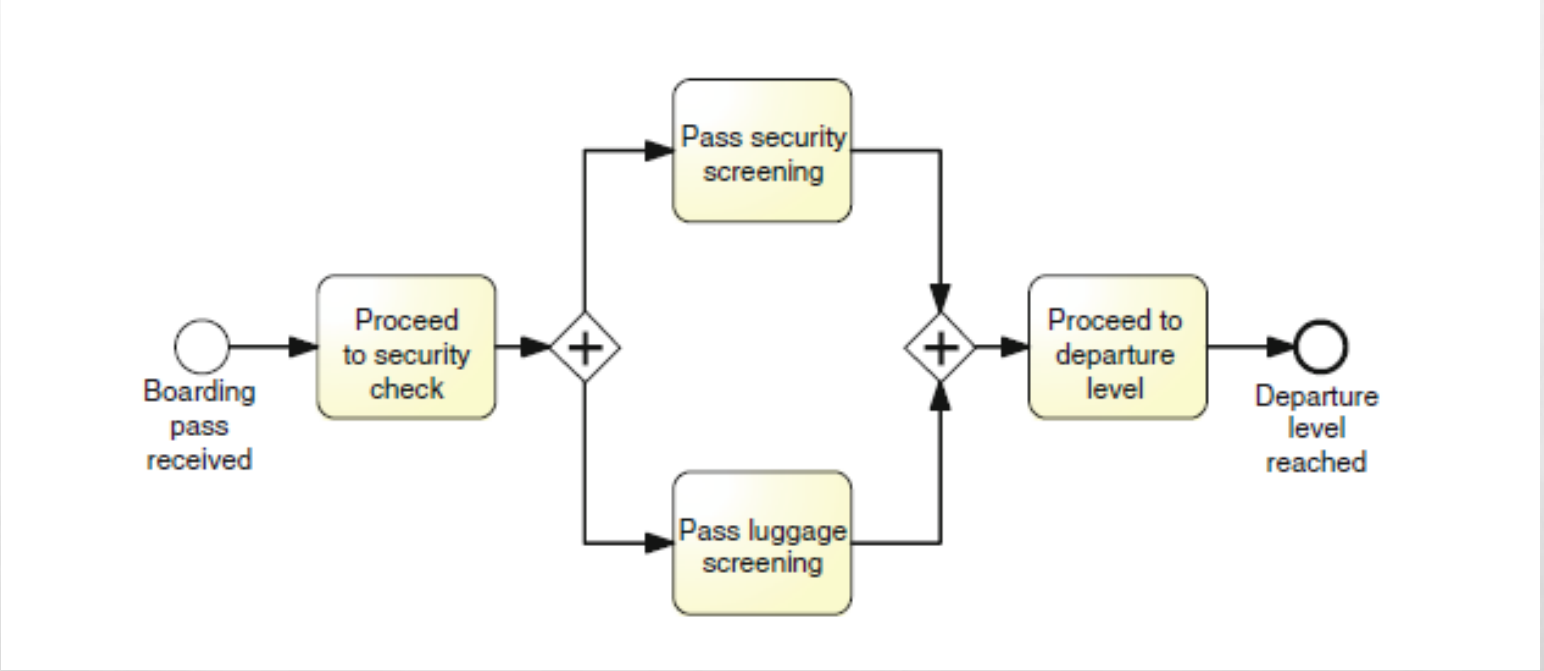
BRANCHING AND MERGING - EXCLUSIVE DECISIONS (XOR)

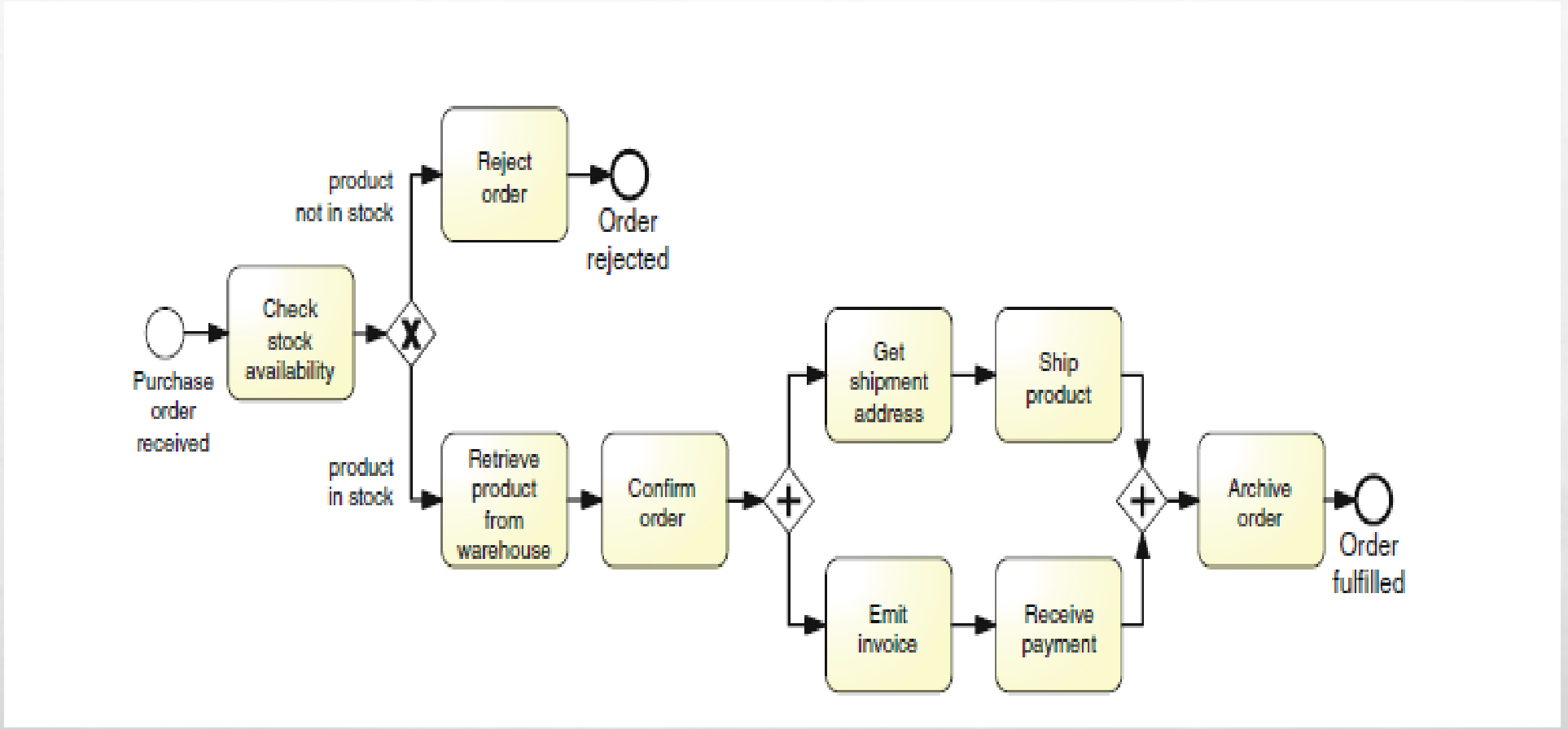


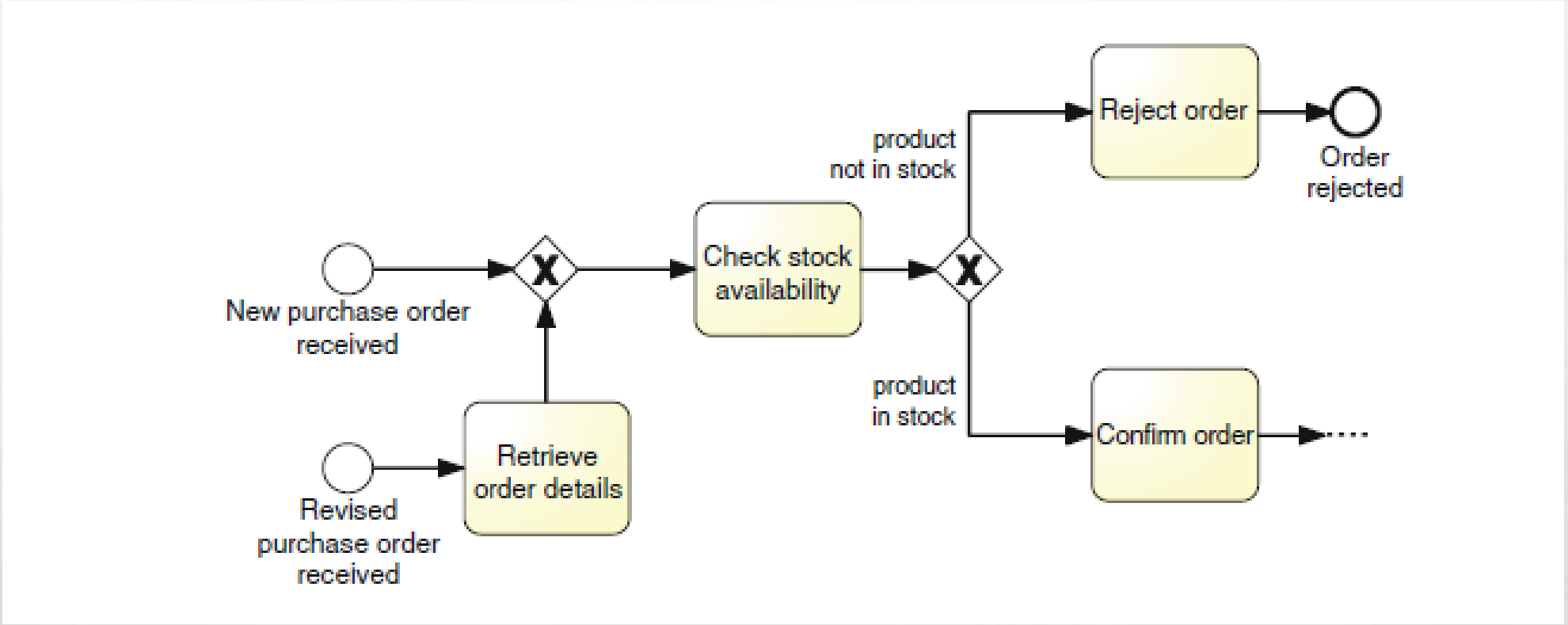
BRANCHING AND MERGING - PARALLEL EXECUTION (AND)

- Once the boarding pass has been received, passengers proceed to the security check. Here they need to pass the personal security screening and the luggage screening. Afterwards, they can proceed to the departure level.

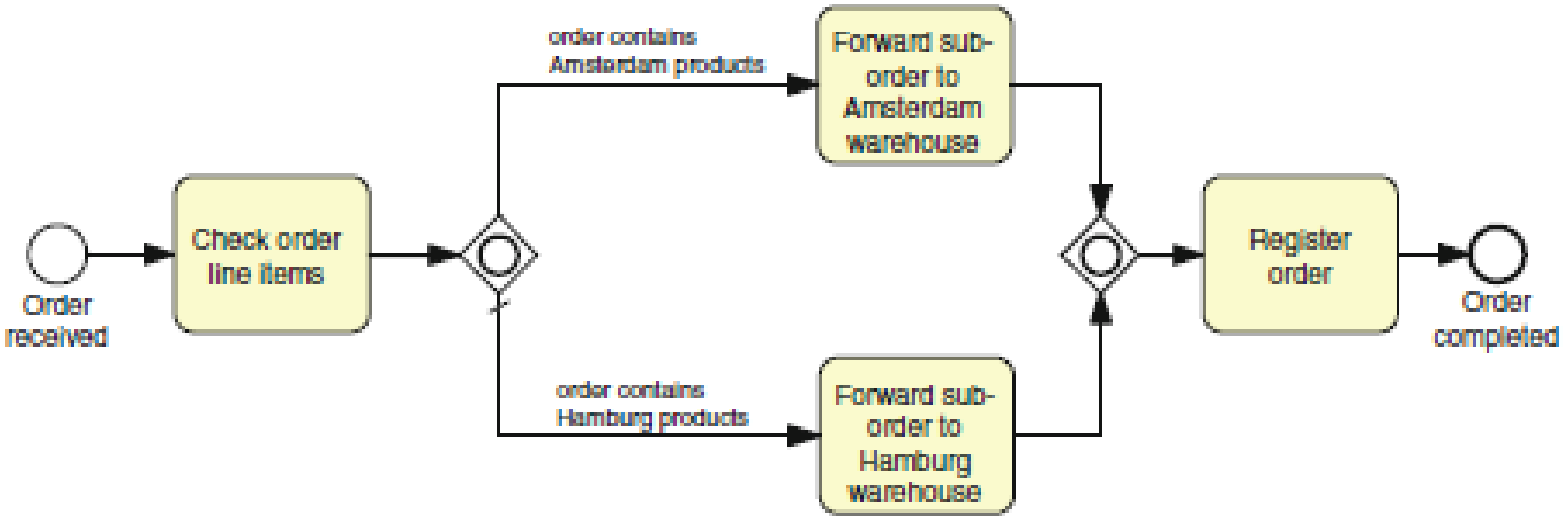
PARALLEL EXECUTION (AND)

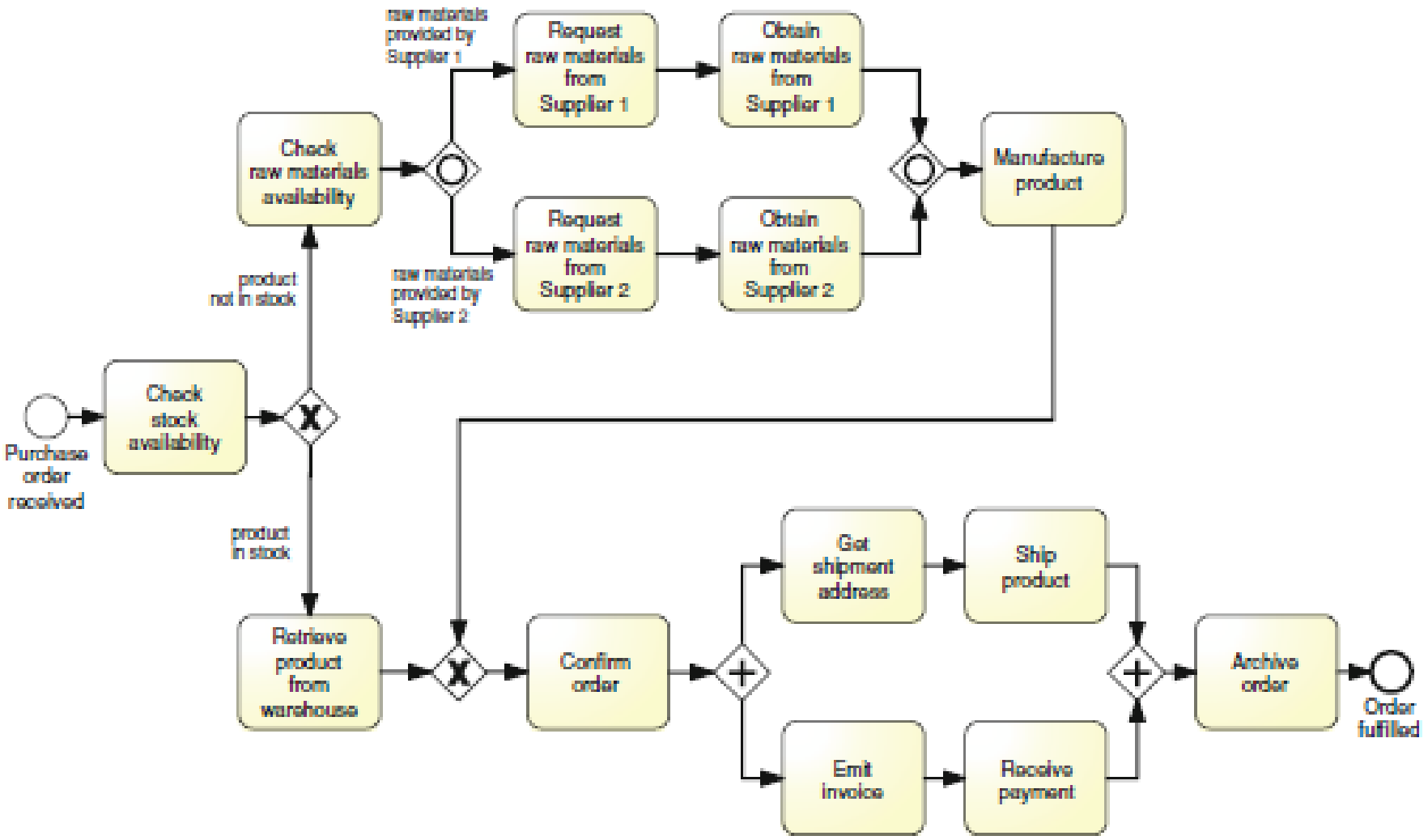



Purchase order received
Check stock availability
product not in stock
Reject order
Order rejected
product in stock
Retrieve product from warehouse
Confirm order
Get shipment address
Ship product
Emit invoice
Receive payment
Archive order
Order fulfilled


New purchase order received
Revised purchase order received
Retrieve order details
Check stock availability
product not in stock
product in stock
Reject order
Order rejected
Confirm order

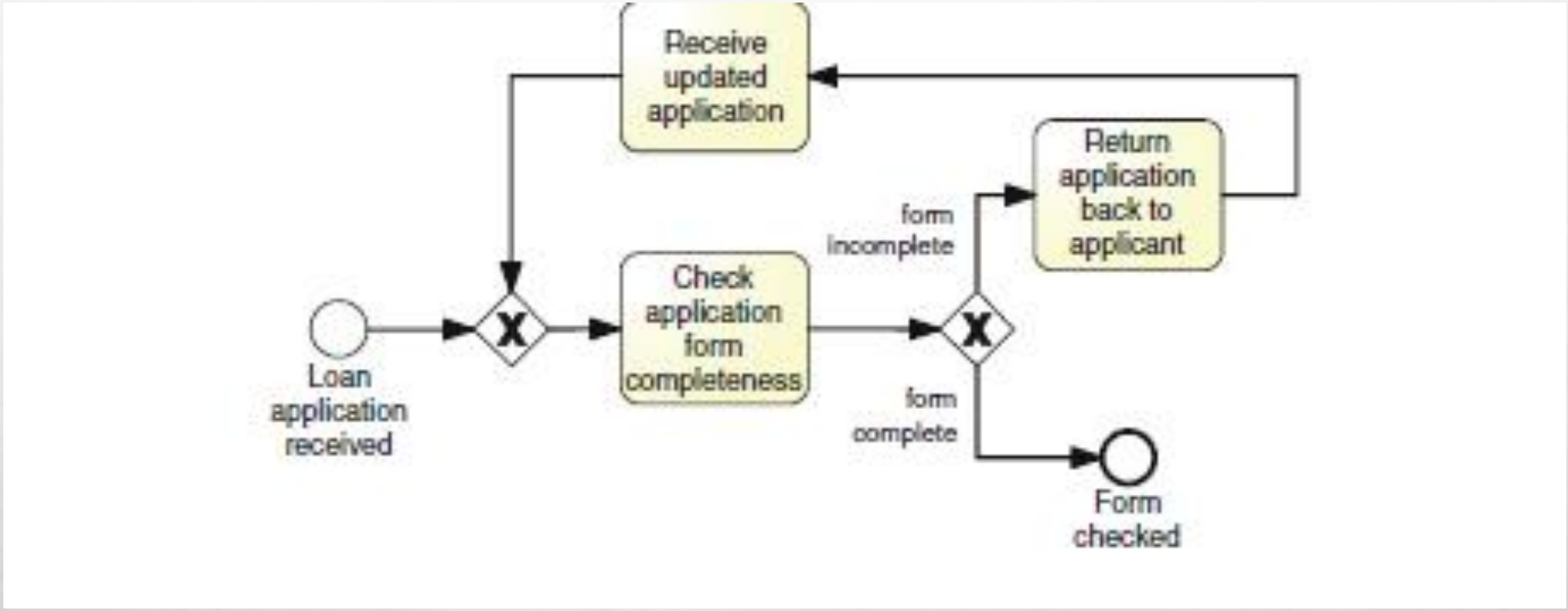
BRANCHING AND MERGING - INCLUSIVE DECISIONS (OR)




Purchase order received
Check stock availability
product not in stock
product in stock
Check raw materials availability
raw materials provided by Supplier 1
Request raw materials from Supplier 1
Obtain raw materials from Supplier 1
raw materials provided by Supplier 2
Request raw materials from Supplier 2
Obtain raw materials from Supplier 2
Manufacture product
Retrieve product from warehouse
Confirm order
Get shipment address
Ship product
Emit invoice
Receive payment
Archive order
Order fulfilled

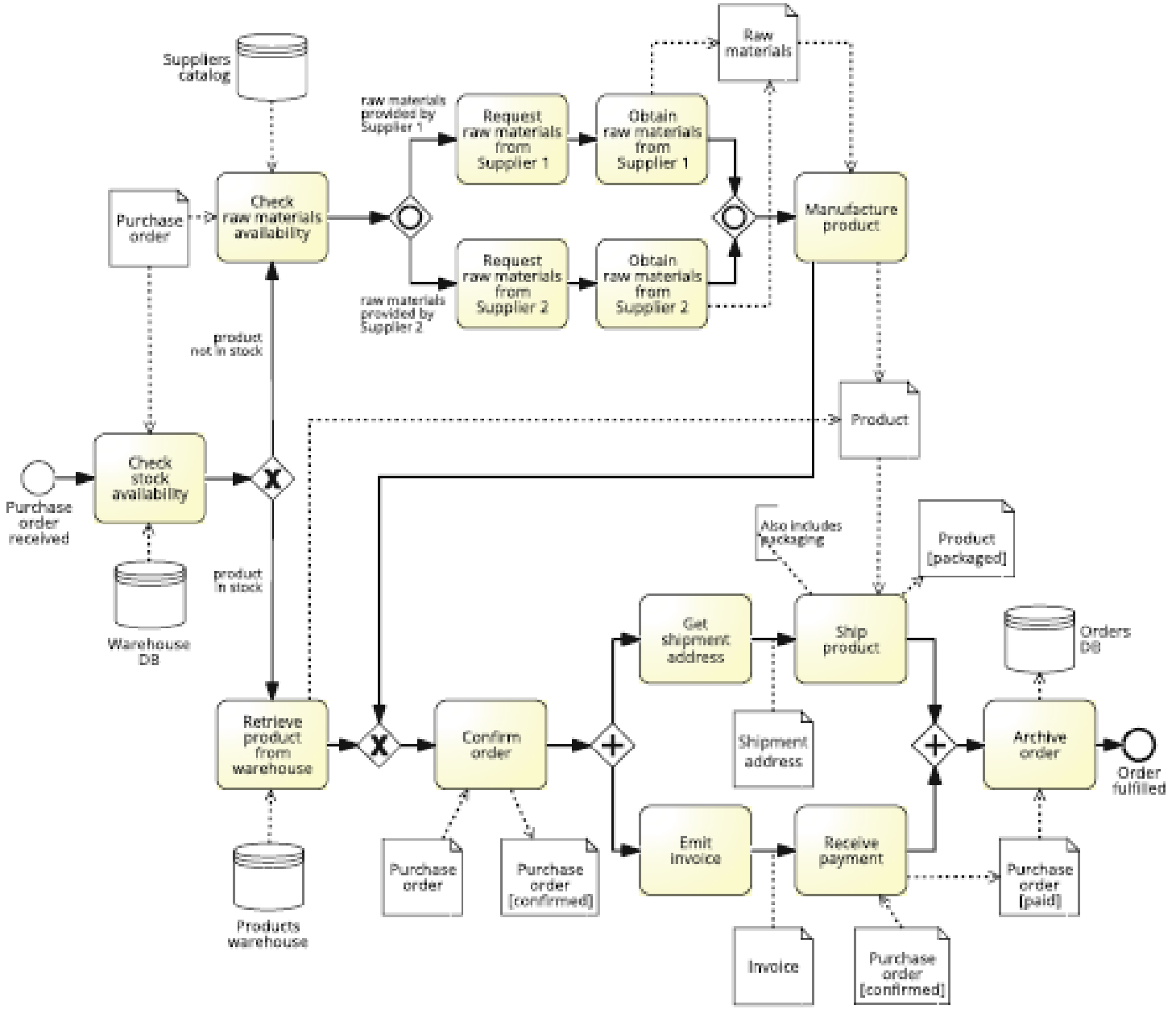
BRANCHING AND MERGING - REWORK AND REPETITION

- Once a loan application is received by the loan provider, and before proceeding with its assessment, the application itself needs to be checked for completeness. If the application is incomplete, it is returned to the applicant, so that they can fill out the missing information and send it back to the loan provider. This process is repeated until the application is found complete.


Receive updated application
Return application back to applicant
form incomplete
Check application form completeness
Loan application received
form complete
Form checked

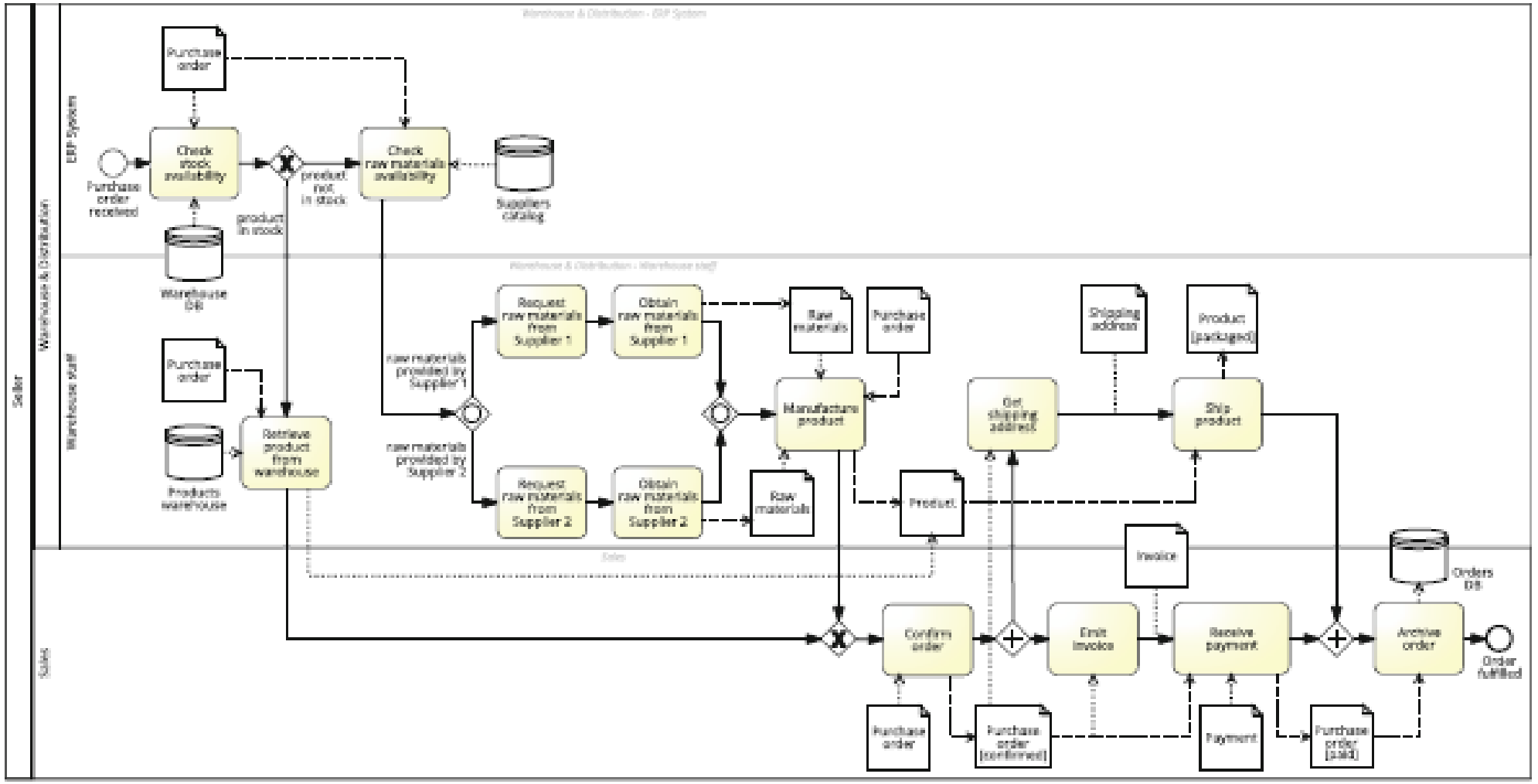
BUSINESS OBJECTS

- Another important perspective that we ought to consider when modeling business processes is the object perspective, also called the data perspective.
- The object perspective indicates which business objects, also known as artifacts (e.g. documents, files, material) are required to perform an activity, and which ones are produced as a result of performing an activity.
- We use the direction of the data association to establish whether a data object is an input or output for a given activity


Suppliers catalog
Raw materials
raw materials provided by Supplier 1
Request raw materials from Supplier 1
Obtain raw materials from Supplier 1
Purchase order
Check raw materials availability
Manufacture product
raw materials provided by Supplier 2
Request raw materials from Supplier 2
Obtain raw materials from Supplier 2
product not in stock
Product
Purchase order received
Check stock availability
Also includes packaging
Product [packaged]
product in stock
Get shipment address
Ship product
Orders DB
Warehouse DB
Shipment address
Retrieve product from warehouse
Confirm order
Archive order
Order fulfilled
Purchase order
Purchase order [confirmed]
Emit invoice
Receive payment
Purchase order [paid]
Products warehouse
Invoice
Purchase order [confirmed]

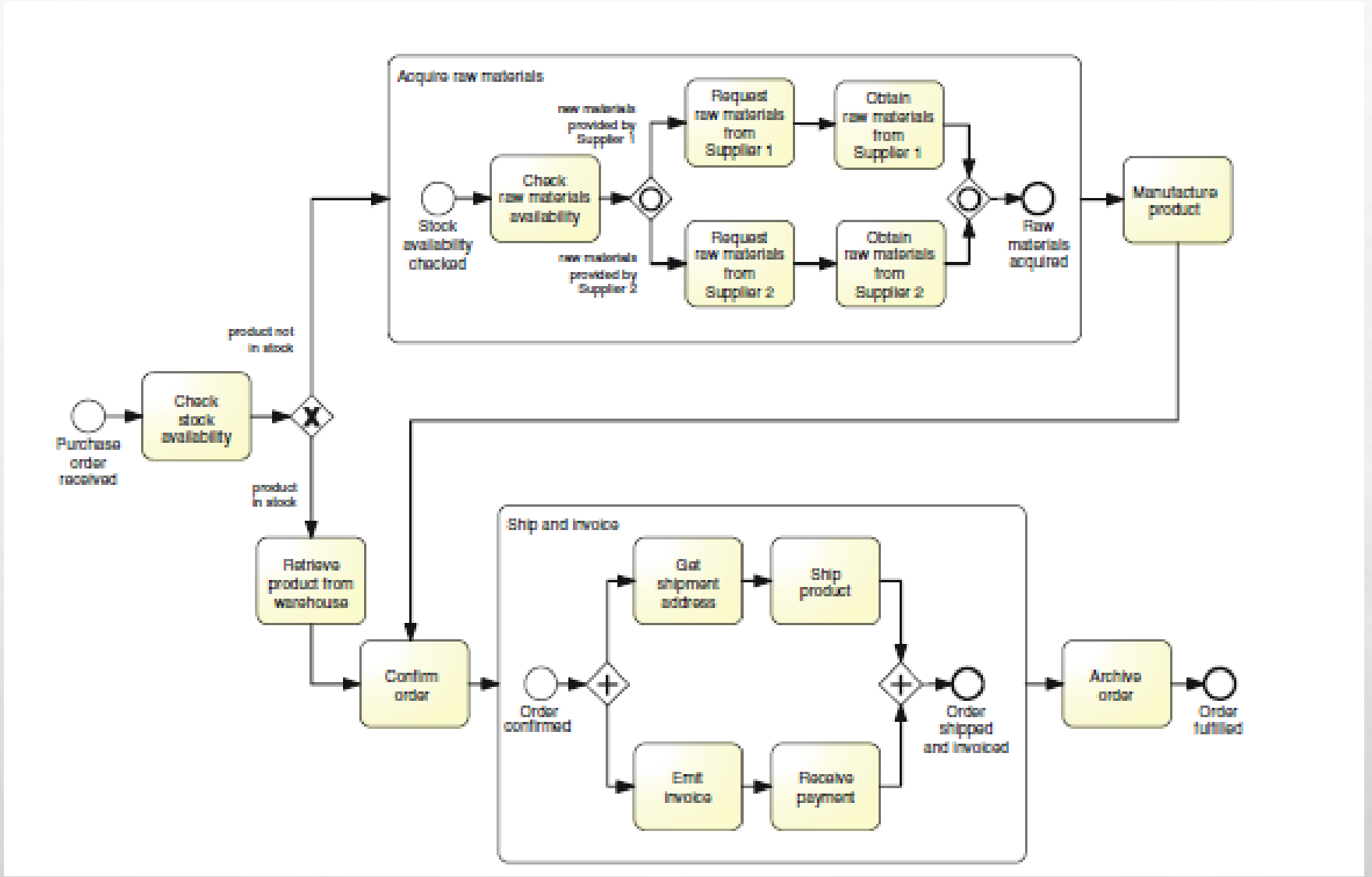
RESOURCE

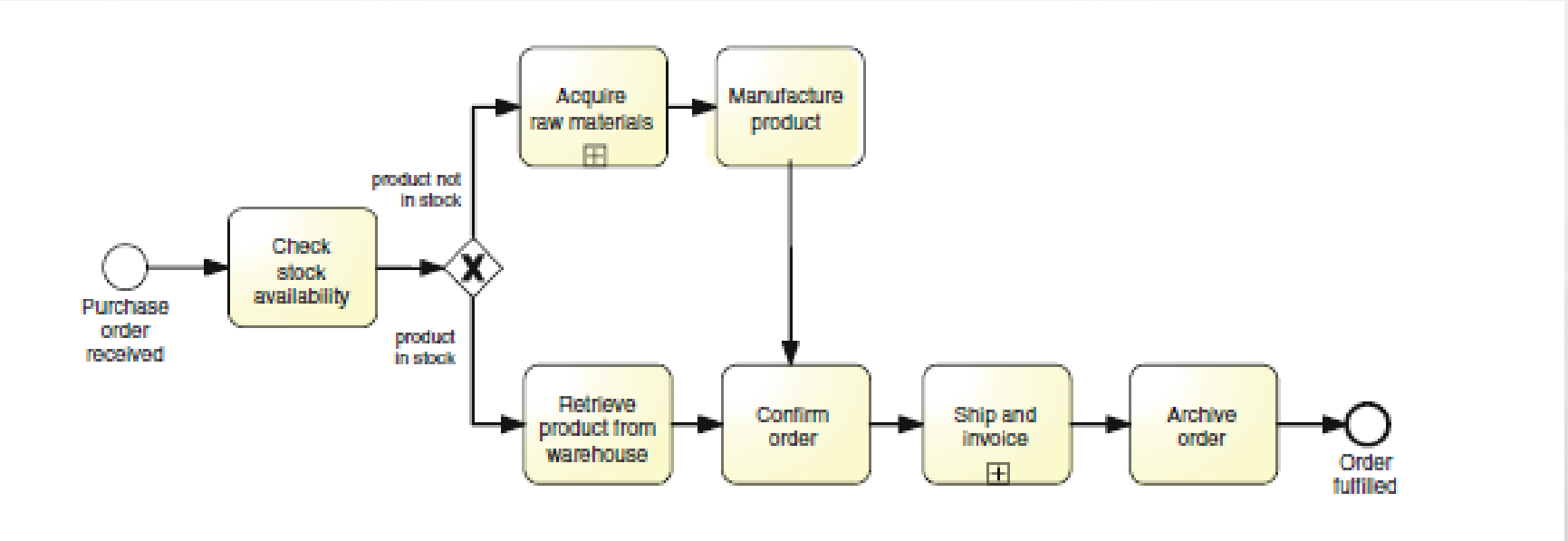
- A process participant, i.E. An individual person like the employee john smith,
- A software system, for example a server or a software application,
- A piece of equipment, such as a printer or a manufacturing plant.

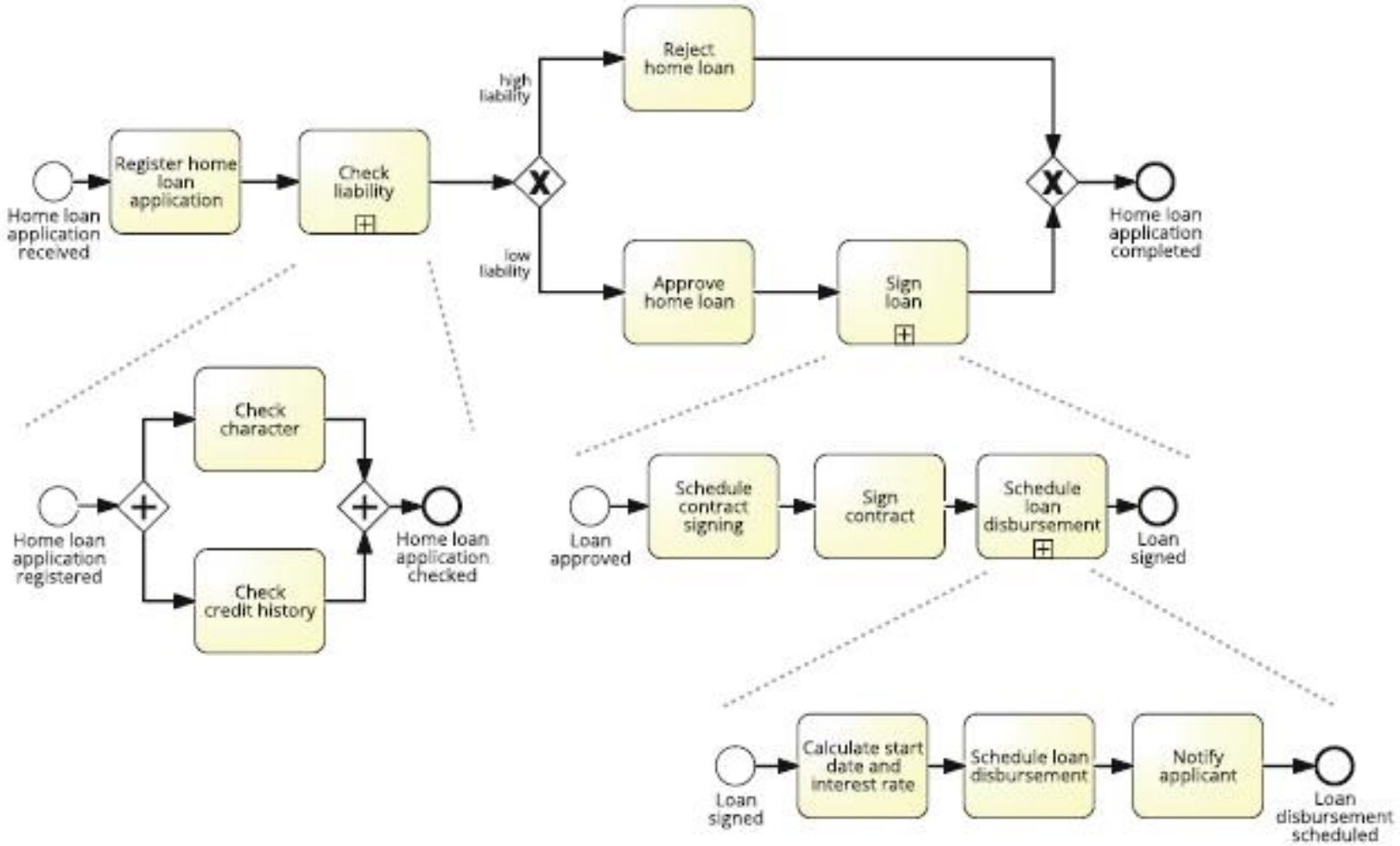

Seller
Warehouse & Distribution
ERP System
Warehouse & Distribution - ERP System
Purchase order
Check stock availability
Purchase order received
product in stock
product not in stock
Check raw materials availability
Supplier's catalog
Warehouse DB
Warehouse & Distribution - Warehouse staff
Warehouse staff
Purchase order
Products warehouse
Retrieve product from warehouse
raw materials provided by Supplier 1
raw materials provided by Supplier 2
Request raw materials from Supplier 1
Obtain raw materials from Supplier 1
Request raw materials from Supplier 2
Obtain raw materials from Supplier 2
Raw materials
Purchase order
Raw materials
Manufacture product
Product
Get shipping address
Shipping address
Ships product
Product (packaged)
Sales
Sales
Invoice
Orders DB
Confirm order
Emit invoice
Receive payment
Archive order
Order fulfilled
Purchase order
Purchase order (confirmed)
Payment
Purchase order (paid)

PROCESS DECOMPOSITION

- To improve understandability, we can simplify the model by hiding certain parts within a sub-process (Process Groups).
- Conversely, an atomic activity, also called *task*, is an activity capturing a unit of work that cannot be further broken down.
- To use a sub-process, we first need to identify groups of related activities.


Acquire raw materials
raw materials provided by Supplier 1
Request raw materials from Supplier 1
Obtain raw materials from Supplier 1
Check raw materials availability
Stock availability checked
raw materials provided by Supplier 2
Request raw materials from Supplier 2
Obtain raw materials from Supplier 2
Raw materials acquired
Manufacture product
product not in stock
Check stock availability
Purchase order received
product in stock
Retrieve product from warehouse
Confirm order
Ship and invoice
Get shipment address
Ship product
Order confirmed
Emit invoice
Receive payment
Order shipped and invoiced
Archive order
Order fulfilled

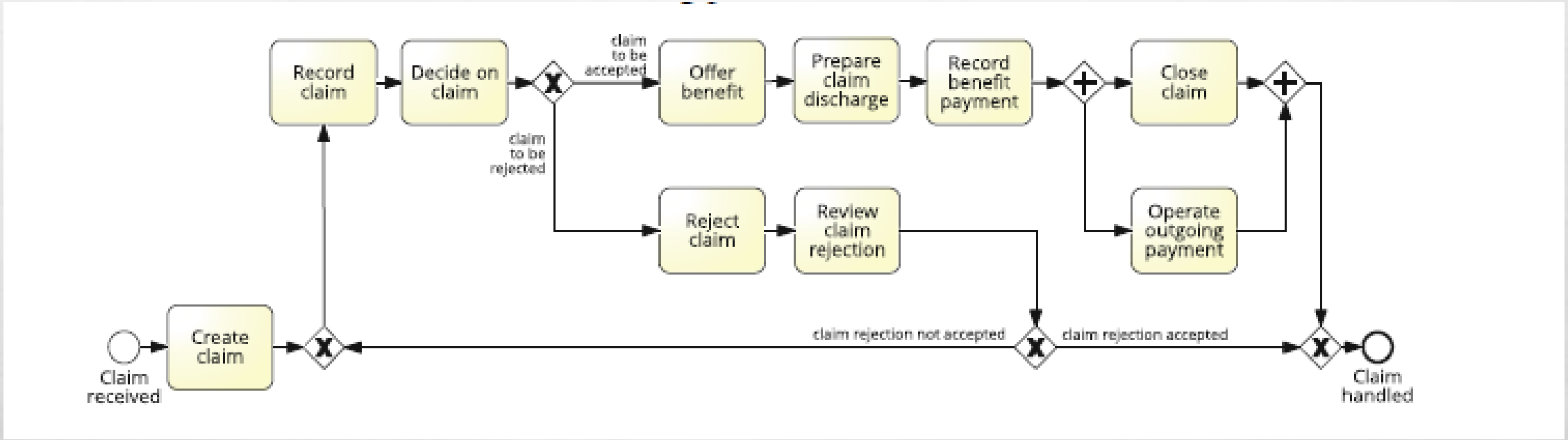

Purchase order received
Check stock availability
product not in stock
product in stock
Acquire raw materials
Manufacture product
Retrieve product from warehouse
Confirm order
Ship and invoice
Archive order
Order fulfilled


Home loan application received
Register home loan application
Check liability
high liability
low liability
Reject home loan
Approve home loan
Sign loan
Home loan application completed
Home loan application registered
Check character
Check credit history
Home loan application checked
Loan approved
Schedule contract signing
Sign contract
Schedule loan disbursement
Loan signed
Loan signed
Calculate start date and interest rate
Schedule loan disbursement
Notify applicant
Loan disbursement scheduled

INTERNATION STANDARD (BPMN 2.0 – OBJECT MANAGEMENT GROUP)

- See the PDF in the VLE.

Exercise 3.1 : describe the following model using one paragraph



EXERCISE 3.2 : MODEL THE FOLLOWING BUSINESS PROCESS FOR HANDLING DOWNPAYMENTS

- The process for handling downpayments starts when a request for payment has been approved. It involves entering the downpayment request into the system, the automatic subsequent payment, the emission of the direct invoice and the clearance of the vendor line items. The clearing of the vendor line items can result in a debit or credit balance. In case of debit balance, the arrears are processed, otherwise the remaining balance is paid.

EXERCISE 3.3 : MODEL THE FOLLOWING BUSINESS PROCESS FOR UWU STUDENT ENROLMENT FOR CERTIFICATE COURSE

- University course enrollment: create a BPMN diagram for a university course enrollment process. Start with a student selecting courses and include tasks like verifying prerequisites, enrolling in courses, and issuing a confirmation. Incorporate decision gateways for scenarios where a course is full or prerequisites are not met.

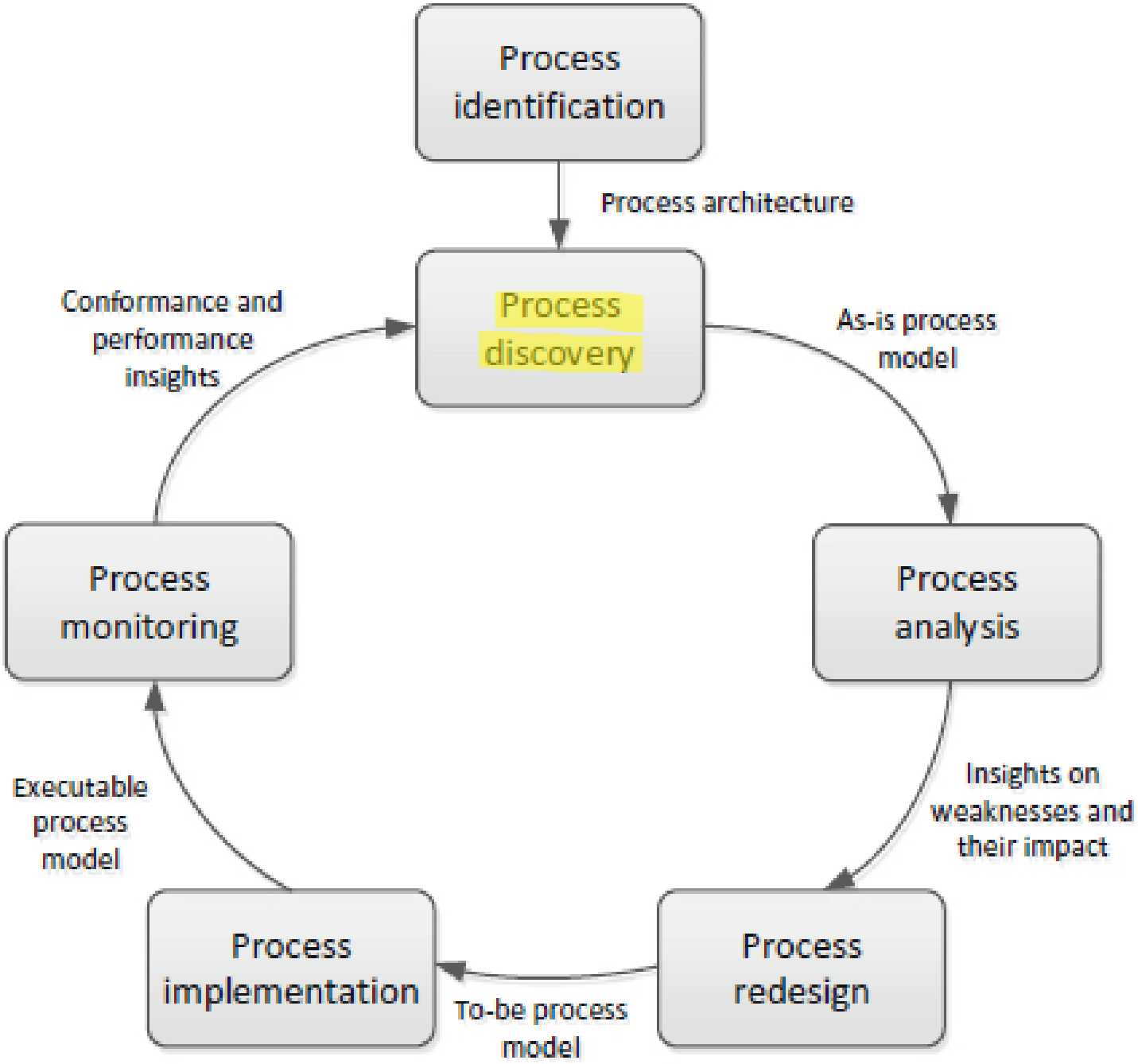


Business Process Management

Process Discovery

All truths are easy to understand once they are discovered: the point is to discover them.

Chapter 05 – Process Discovery (Fundamentals of Business Process Management)


Process identification
Process architecture
Process discovery
As-is process model
Process analysis
Insights on weaknesses and their impact
Process redesign
To-be process model
Process implementation
Executable process model
Process monitoring
Conformance and performance insights

Setting of Process Discovery

- The act of gathering information about the existing process and organization it in terms of an as-is process model.
- All about gathering and organizing information. A much broader activity than process modeling
- The modeling can only be started once the enough information is gathered.
- We need to first define a setting which information can be gathered effectively.

Four tasks to consider

1. Defining setting
2. Gathering information
3. Conducting the modeling task
4. Assuring the process model quality

Process Analyst verses domain experts

- process analysts: responsible for information gathering about a given business process and driving the modeling task, under the leadership of the process owner.
- Domain expert: any individuals who has an intimate knowledge of how a process or specific tasks within that process are performed. This individual could be process participant and or a process owner or operational manager who coordinates a team
- External roles such as partners, suppliers, and customers of the process should be also considered.

who would you hire as a process analyst?

- You are the manager of a consulting company and you need to hire a person for the newly signed BPM project with an online bookstore. Consider the following profiles:
- Mike Miller has ten years of work experience with an online retailer. He has worked in different teams involved with the order-to-cash process of the online retailer.
- Sara Smith has five years of experience working as a process analyst in the banking sector. She is familiar with two different process modeling languages and with several modeling tools.

What Makes a Good Process Analyst

- Getting the right people on board
- Formulate and test hypotheses
- Identify patterns
- Pay attention to model aesthetics

- Focus on improving the process from the customer’s point of view.
- Not focused on tools or technology
- Can sell the value of BPM to others by demonstrating cost or time savings
- Works closely with business strategy or enterprise architecture teams
- Great at running workshops
- Always learning!

What are the skills of a process analyst?

- Personality - Five factors
    - 1. openness (appreciating art, emotion, planning)
    - 2. conscientiousness (tendency to self discipline, achievement, planning)
    - 3. extraversion (being positive, energetic, and seeking company)
    - 4. agreeableness (being compassionate, cooperative)
    - 5. neuroticism (being, anxious, depressed, vulnerable)

Challenges in Process Discovery

- Fragmented process Knowledge: Iterative validation process
- Thinking the process on a case level
- Not familiar with business process modeling language

Process Discovery Methods

1. Evidence-Based Discovery
   1. Document analysis
   2. Observation
   3. Automated process discovery
2. Interview-Based Discovery
3. Workshop-Based Discovery

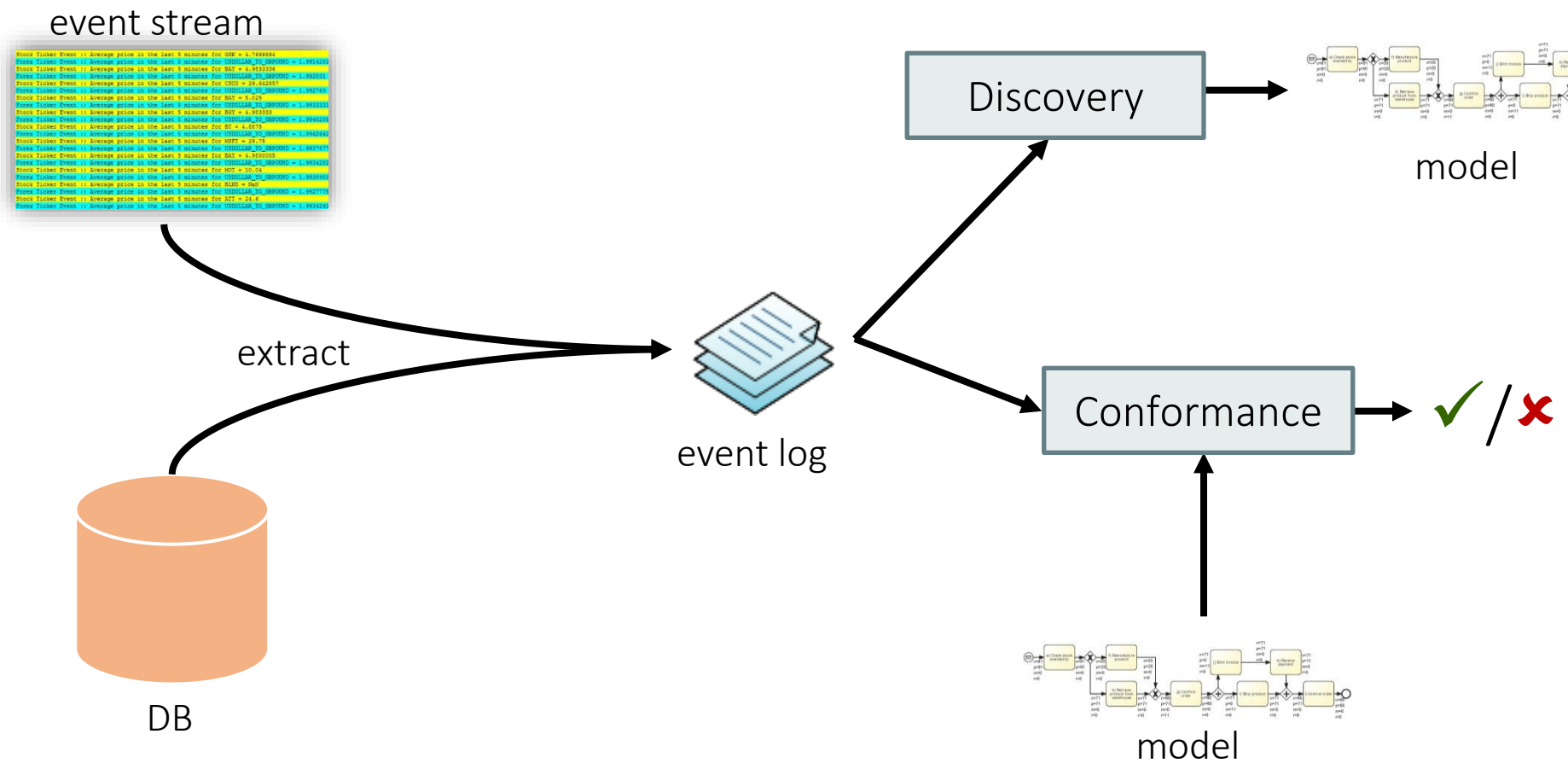
Evidence-Based Discovery – Document Analysis

- Documents point to existing roles, activities and business objects:
    - Process descriptions (ideal scenario)
    - Internal policies
    - Organization charts
    - Employment plans
    - Quality certificate reports
    - Glossaries and handbooks
    - Forms
    - Work instructions
- May not be process-oriented and trustworthy.
- Could be used to gather information before approaching domain experts.

Evidence-Based Discovery – Observation

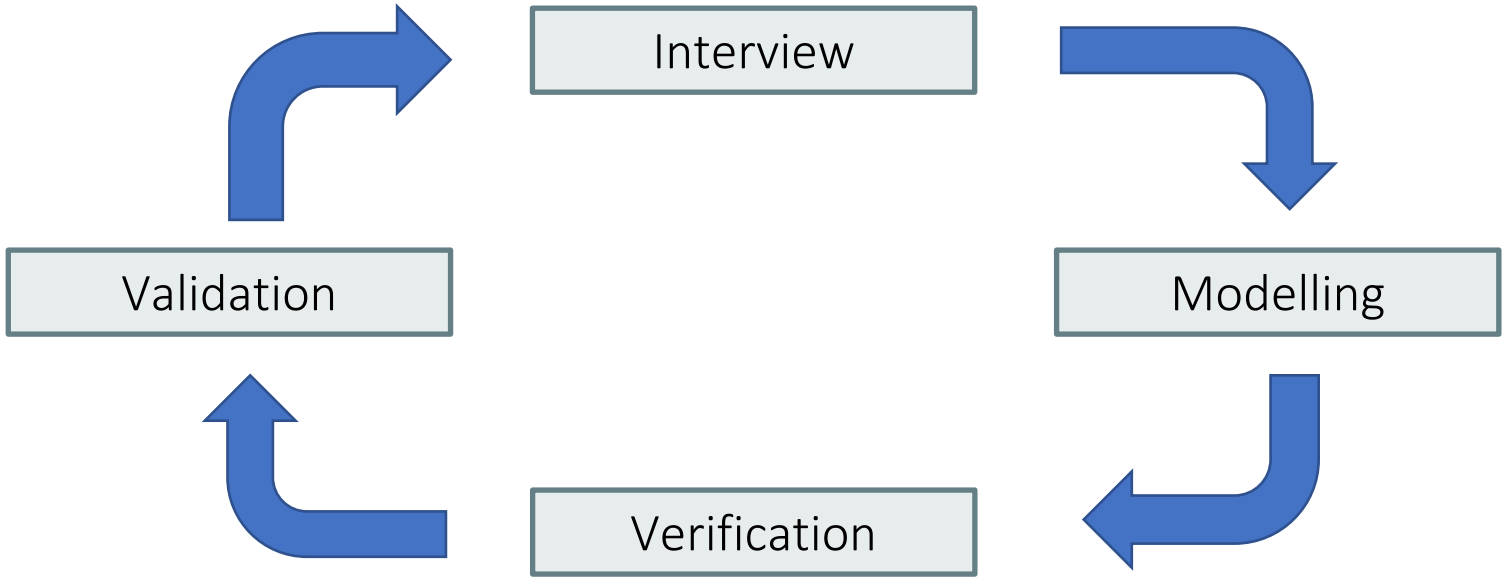
- Follow directly the processing of individual cases:
  - Active role: play a specific role, e.g. customer
  - Passive role: observe participants and their environment
- Trace business objects in the course of their lifecycle
- Active role: no big picture
- Passive role: participants' bias

Evidence-Based Discovery - Automated discovery via Process Mining



case id	event id	properties				
		timestamp	activity	resource	cost	...
1	35654423	30-12-2010:11.02	register request	Pete	50	...
	35654424	31-12-2010:10.06	examine thoroughly	Sue	400	...
	35654425	05-01-2011:15.12	check ticket	Mike	100	...
	35654426	06-01-2011:11.18	decide	Sara	200	...
	35654427	07-01-2011:14.24	reject request	Pete	200	...
2	35654483	30-12-2010:11.32	register request	Mike	50	...
	35654485	30-12-2010:12.12	check ticket	Mike	100	...
	35654487	30-12-2010:14.16	examine casually	Pete	400	...
	35654488	05-01-2011:11.22	decide	Sara	200	...
	35654489	08-01-2011:12.05	pay compensation	Ellen	200	...

Interviews



- Forward vs. backward
- Structured vs. unstructured
- Assumption: analyst and stakeholder share terminology
- Pitfall: exceptional behavior neglected
  > use questions that aim to identify such behavior

Workshops – Brown paper mapping

- Gather all key stakeholders together
- Participants interact to create shared understanding
- Typically one process analyst (facilitator), multiple domain experts, process owner may also attend
- May be software-supported, a model is directly created during the workshop (typically a separate role – tool operator)
- Model is used as reference point for discussions
- Alternative: brown-paper workshops
- Usually 3 to 5 half-day sessions



Discovery and Culture

Before starting with process discovery, it is important to understand the **culture** and the sentiment of an organization.

- There are companies that preach and practice an open culture in which all employees are encouraged to utter their ideas and their criticism. Such organizations can benefit a lot from workshops as participants are likely to present their ideas freely.
- In strictly hierarchical organizations, it is necessary to take special care that every participant gets an equal share of parole in a workshop and that ideas and critique are not held back.

It might be the case that the young dynamic company has a more open culture than the company with extensive health and security regulations. This has to be taken into account when organizing a workshop.

Strengths and Weaknesses

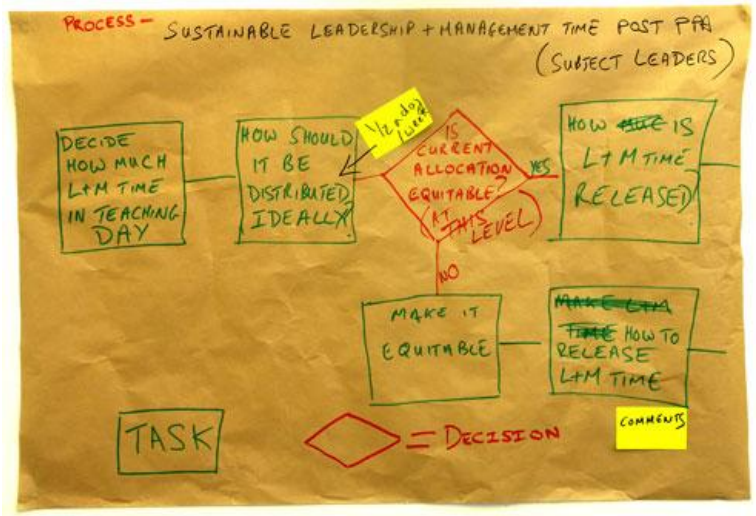
Aspect	Evidence-based	Interviews	Workshops
Objectivity	High	Medium-high	Medium-high
Richness	Medium	High	High
Time consumption	Low-medium	Medium	Medium
Immediacy of feedback	Low	High	High

Strengths and Weaknesses

Technique	Strength	Weakness
Document Analysis	• Structured information • Independent from availability of stakeholders	• Outdated material • Wrong level of abstraction
Observation	• Context-rich insight into process	• Potentially intrusive • Stakeholders likely to behave differently • Only few cases
Automatic Discovery	• Extensive set of cases • Objective data	• Potential issue with data quality
Interview	• Detailed inquiry into process	• Requires sparse time of process stakeholders • Several iterations required before sign-off
Workshop	• Direct resolution of conflicting views	• Requires availability of several stakeholders at the same time

Organizing the Gathered Material

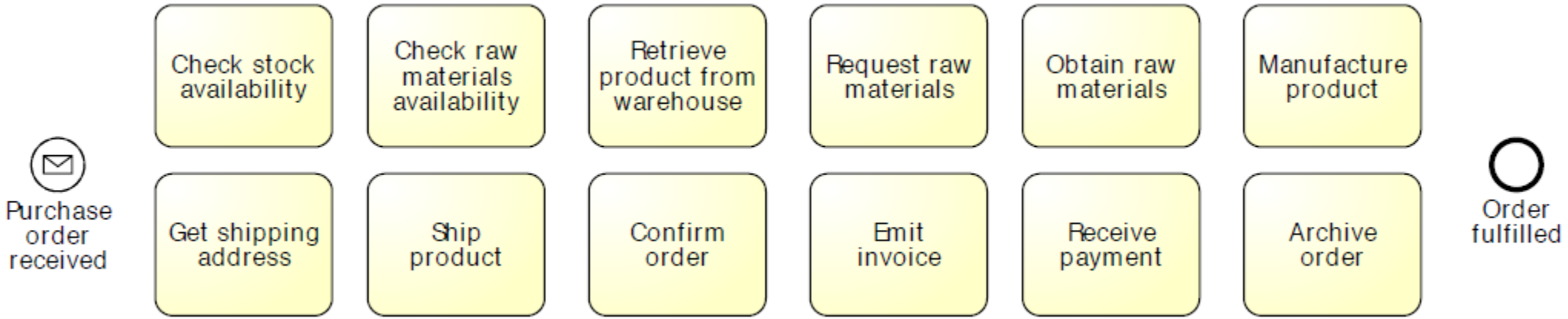
1. Identify the process boundaries
2. Identify activities and events
3. Identify resources and their handovers
4. Identify the control flow
5. Identify additional elements



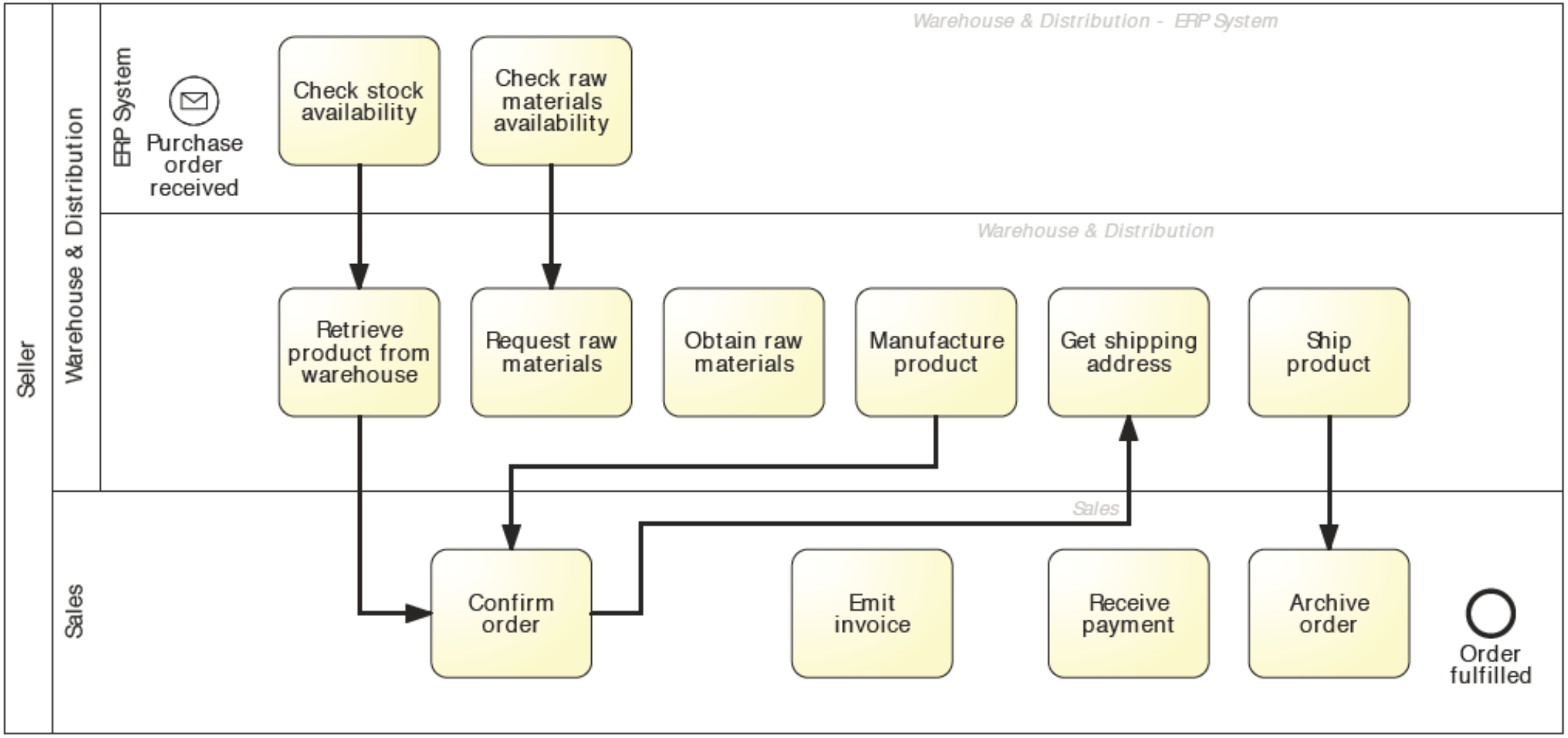
Process Boundaries

- Under which condition does the process start?
- With which result does it end?
- Which perspective do we assume?
- What data are required as input and output to the process?

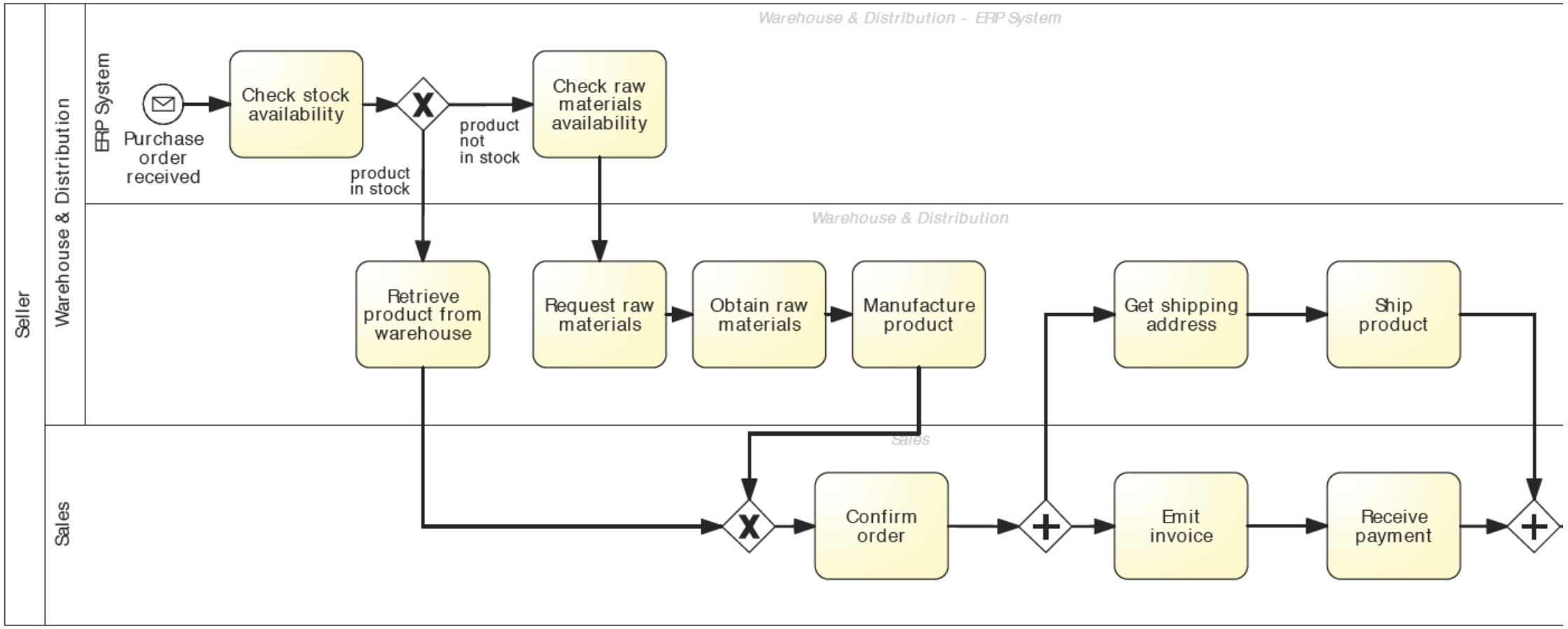
Identify Activities and Events



Identify Resources and Handovers



Identify Control Flow

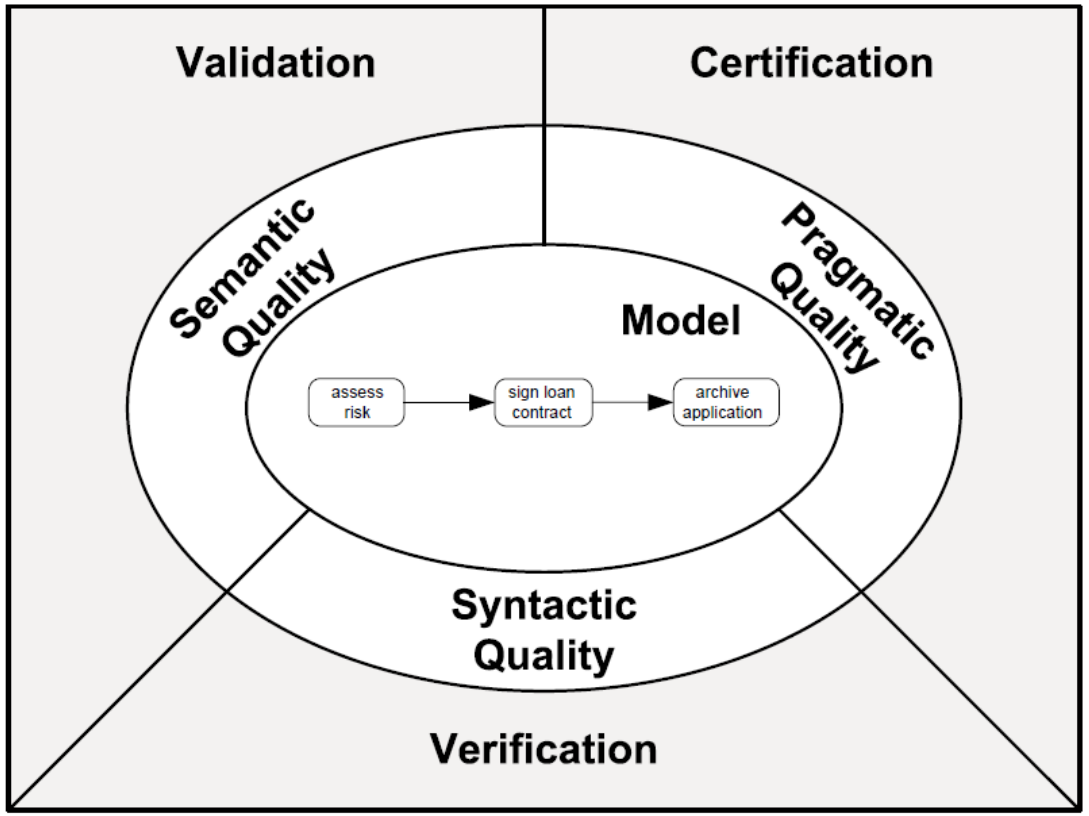


Identify additional elements

- Exceptional Handling (Not only sunny days but also the rainy days)
- Other resources like data

Process Modelling Quality Assurance

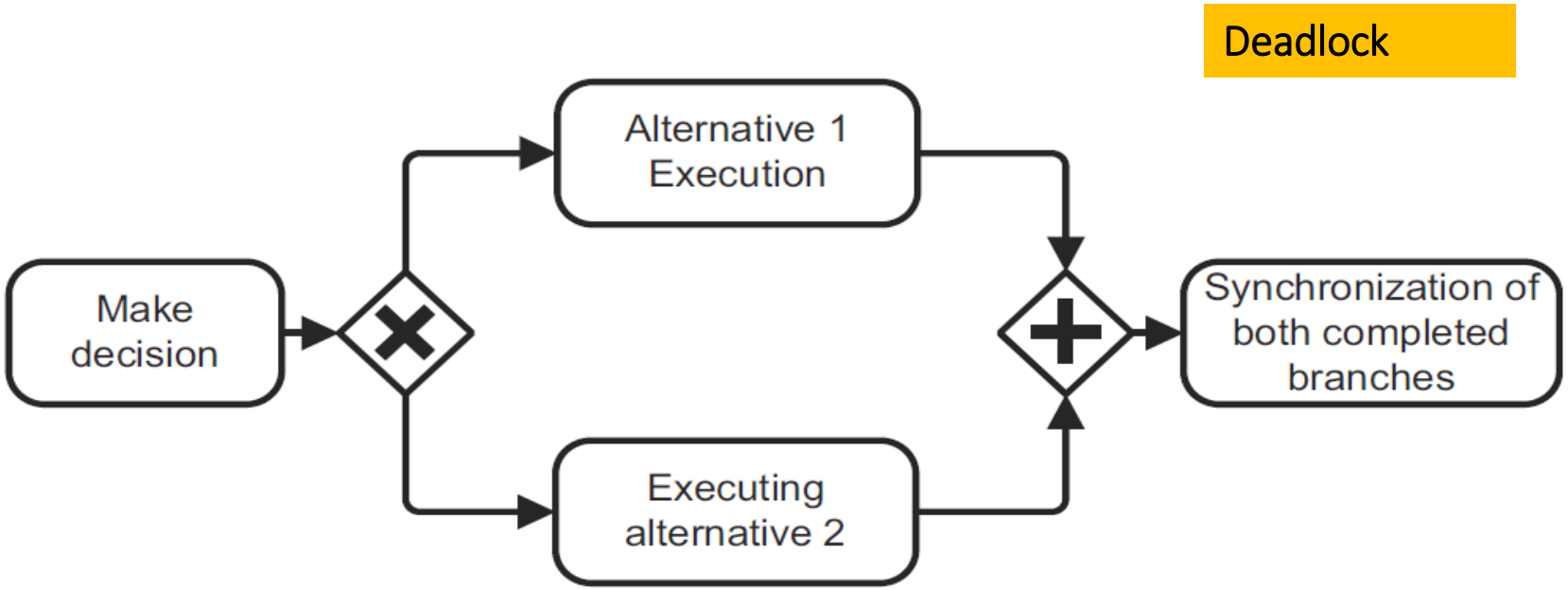
- Validity
- Completeness



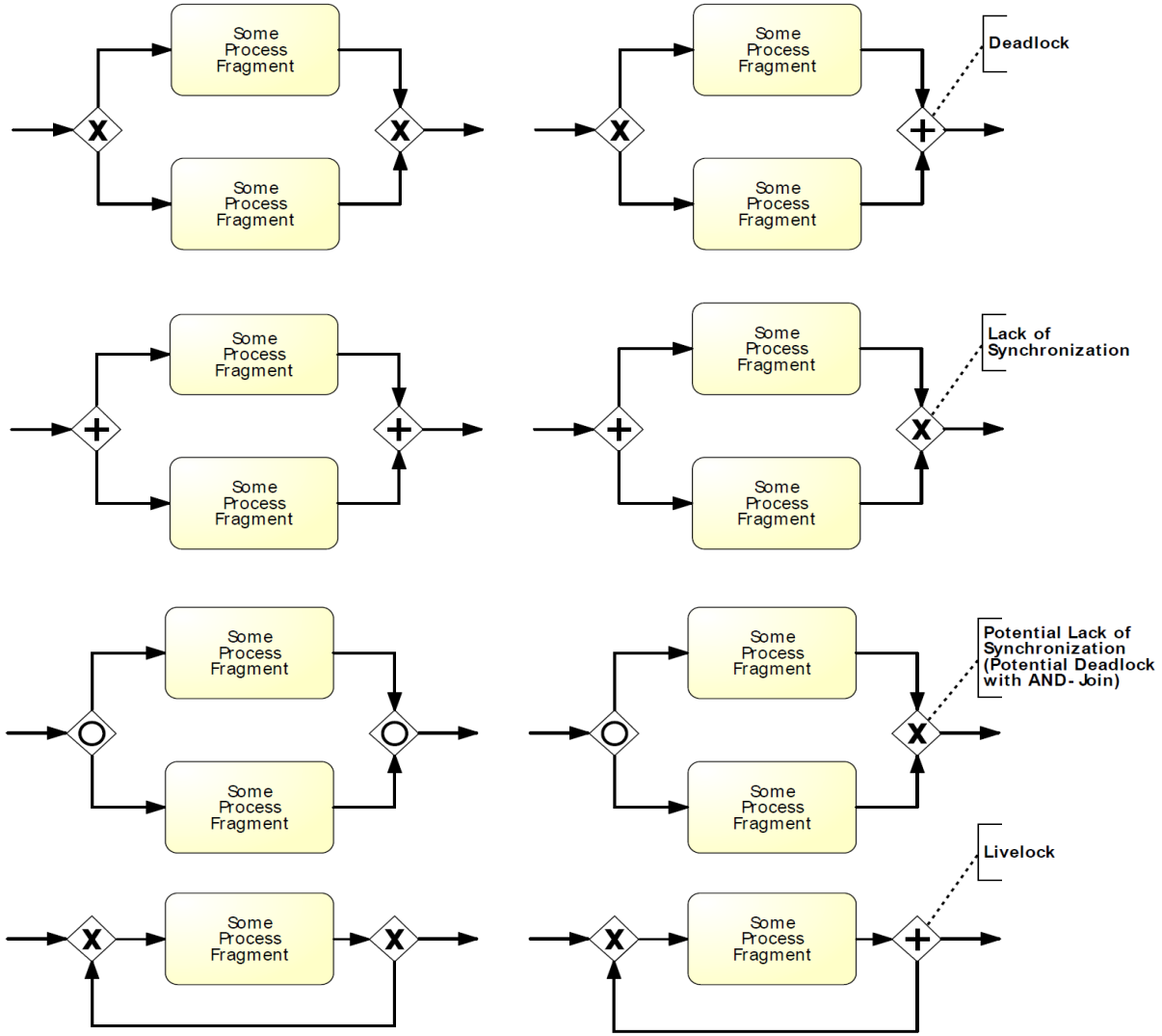
- Understandibility
- Mantainability

- Structural correctness
- Behavioral correctness

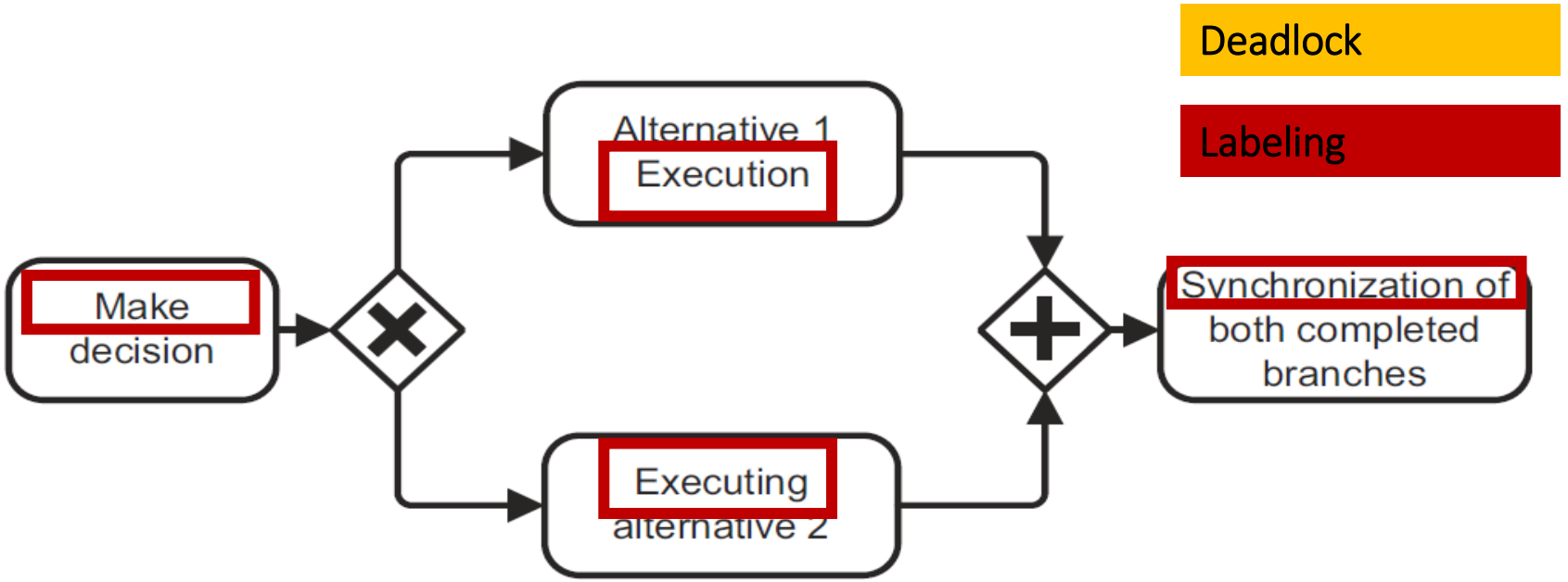
Is this process model of good quality?



Syntactic Quality: Verification



Is this process model of good quality?



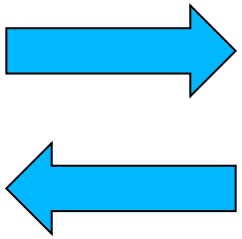
Formulate Labels Adequately

- Activities as Verb-Object
- Events as Object-Past-Participle Verb
- Conditions with reference to Object

Semantic Quality: Validation

- Validity
- Completeness



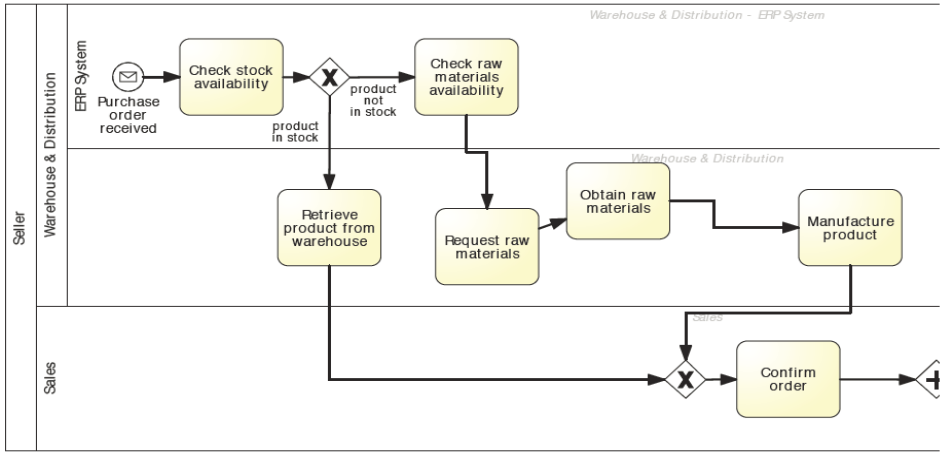




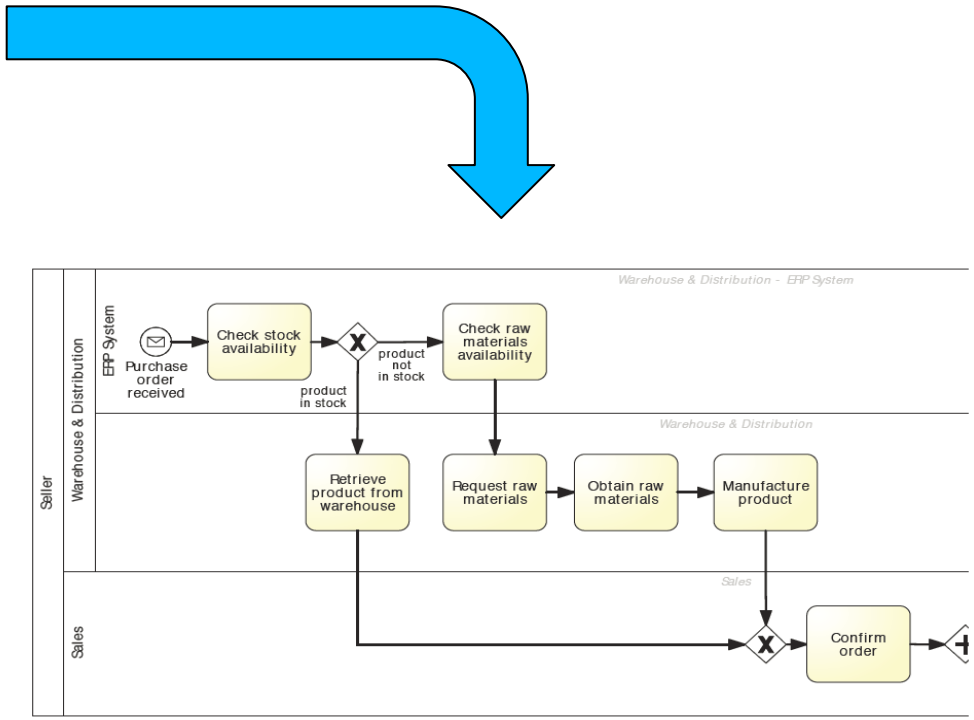
Domain Expert

Process Analyst

Pragmatic Quality – Example: Layout



Better layout helps understanding



Seven Process Modeling Guidelines (7PMG)

G1 Use as few elements in the model as possible

G2 Minimize the routing paths per element

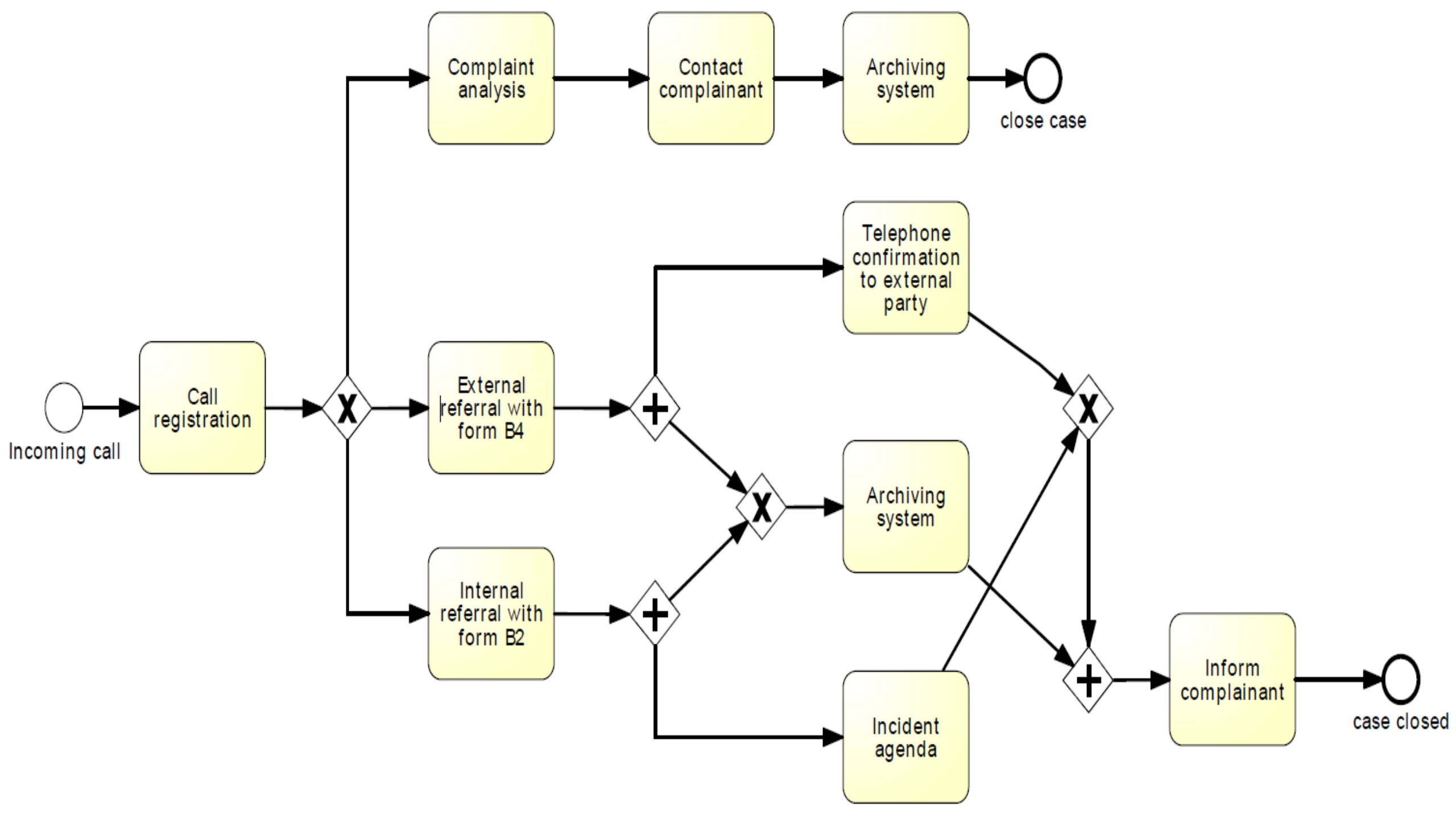
G3 Use one start and one end event

G4 Model as structured as possible

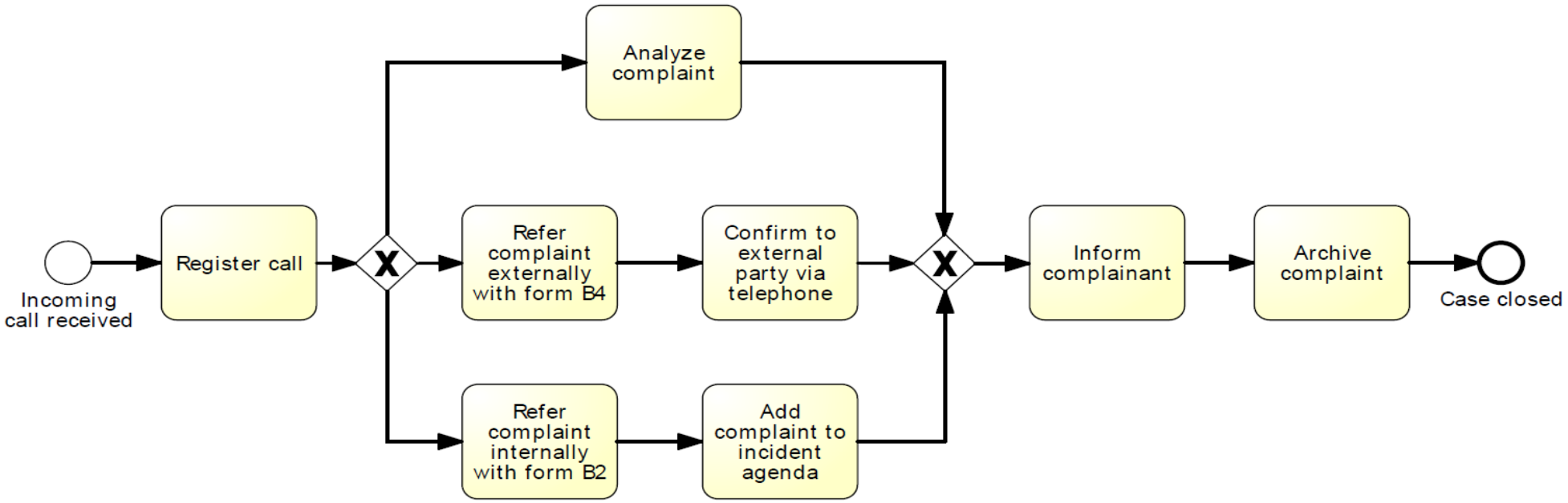
G5 Avoid, where possible, OR routing elements

G6 Use verb-object activity labels

G7 Decompose a model with more than 30 elements


Complaint analysis
Contact complainant
Archiving system
close case
Telephone confirmation to external party
Call registration
External referral with form B4
Incoming call
Archiving system
Internal referral with form B2
Incident agenda
Inform complainant
case closed

Solution


Analyze complaint
Incoming call received
Register call
X
Refer complaint externally with form B4
Confirm to external party via telephone
X
Inform complainant
Archive complaint
Case closed
Refer complaint internally with form B2
Add complaint to incident agenda

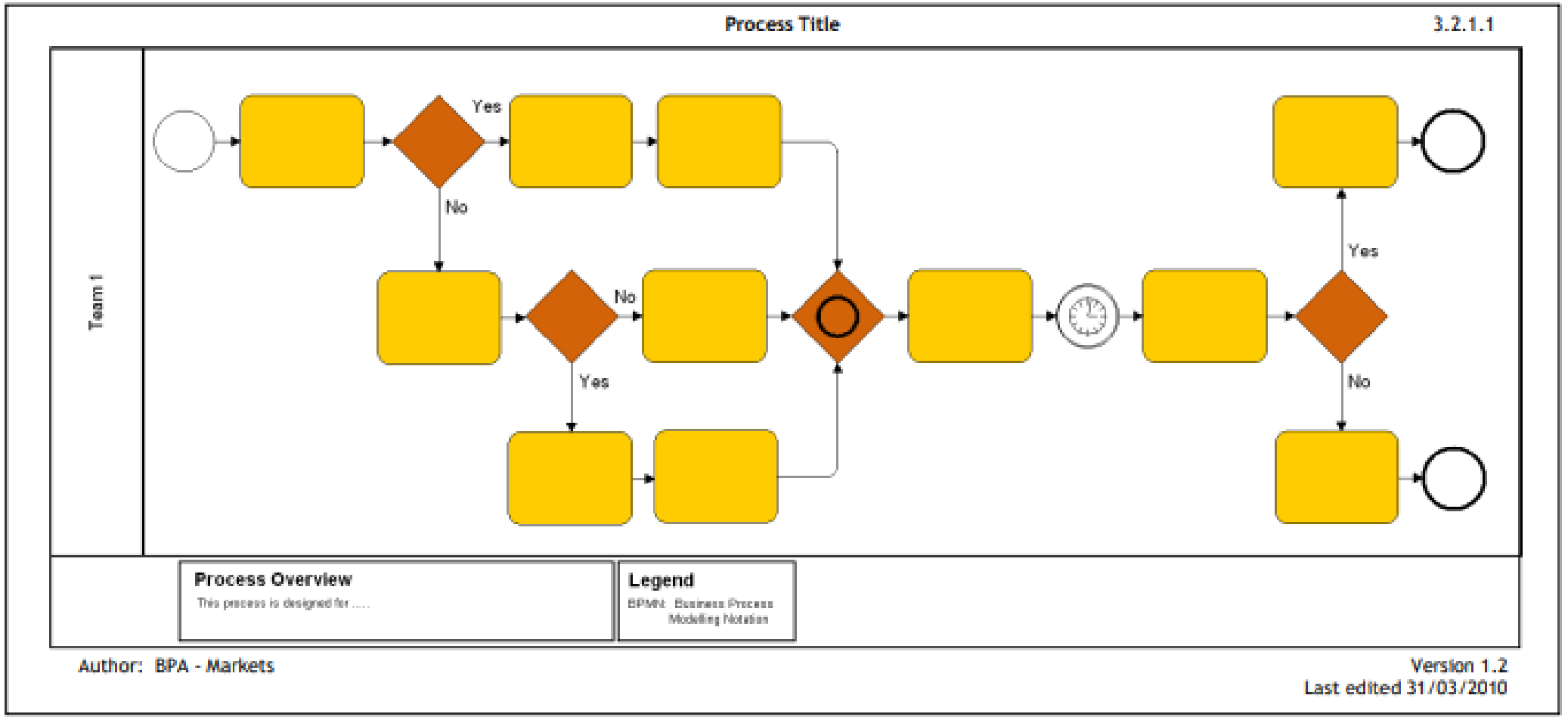
Process Discovery - Summary

- Domain expert and process analyst have different strengths and limitations in process discovery
- There are various discovery methods
- Quality Assurance is important

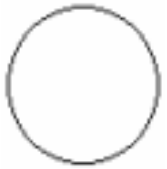


Business Process Management

More about BPMN

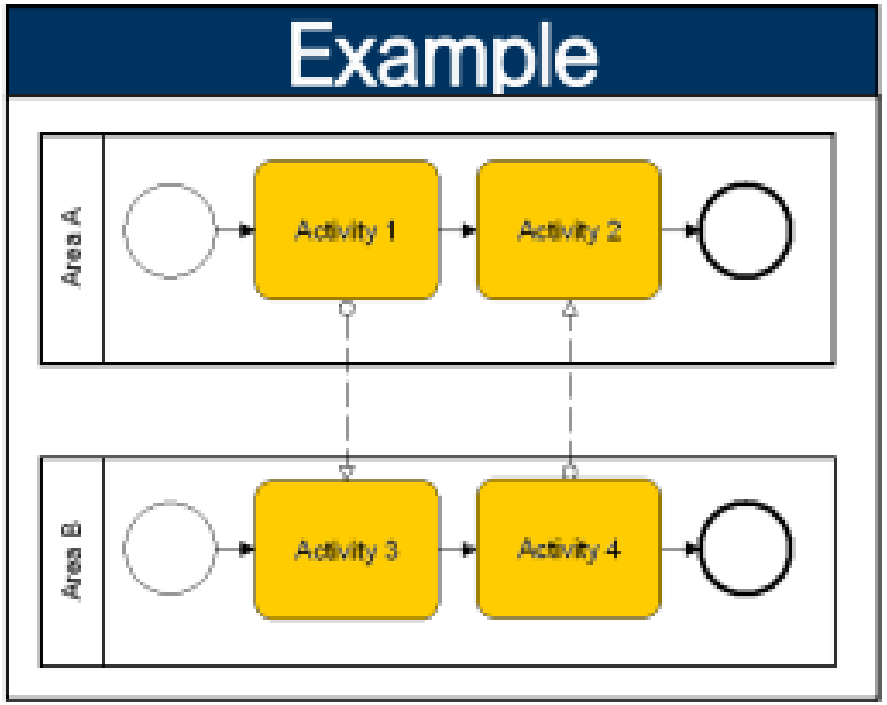

Process Title
3.2.1.1
Team 1
Yes
No
No
Yes
Yes
No
Process Overview
This process is designed for .....
Legend
BPMN: Business Process Modelling Notation
Author: BPA - Markets
Version 1.2
Last edited 31/03/2010

several events types

Trigger	Start	End	Description
General			The start event indicates where a particular process will start. The end event will indicate where the process will end.
Message			A message arrives from a participant and triggers the event. This causes the process to start or end.
Timer			A timer indicates a time based event that will trigger the start or continue the process. The timer can be used to model time based delays.
Link			A link is a mechanism for connecting the end of one process to the start of another

Message flows

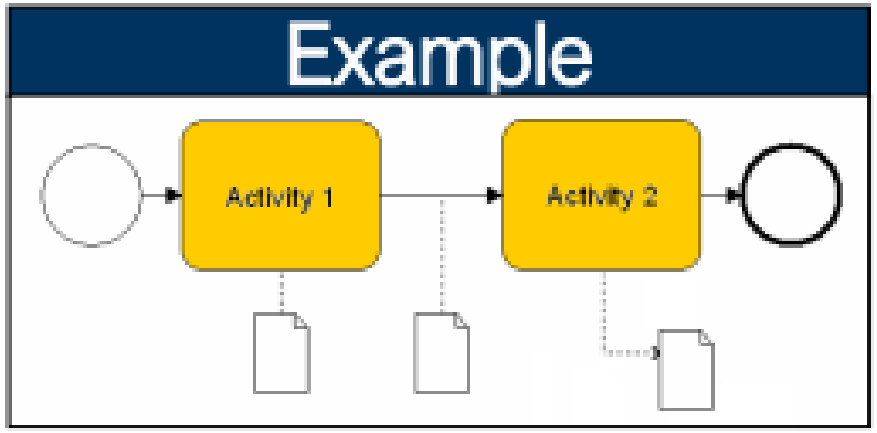
- A message flow is a dashed connector representing a signal (message) send between two pools. It may not be used to connect two nodes within a pool.

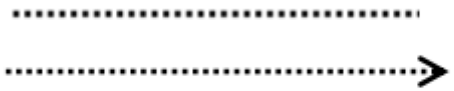




Associations indicate that a message or object (such as a document) is associated with that process

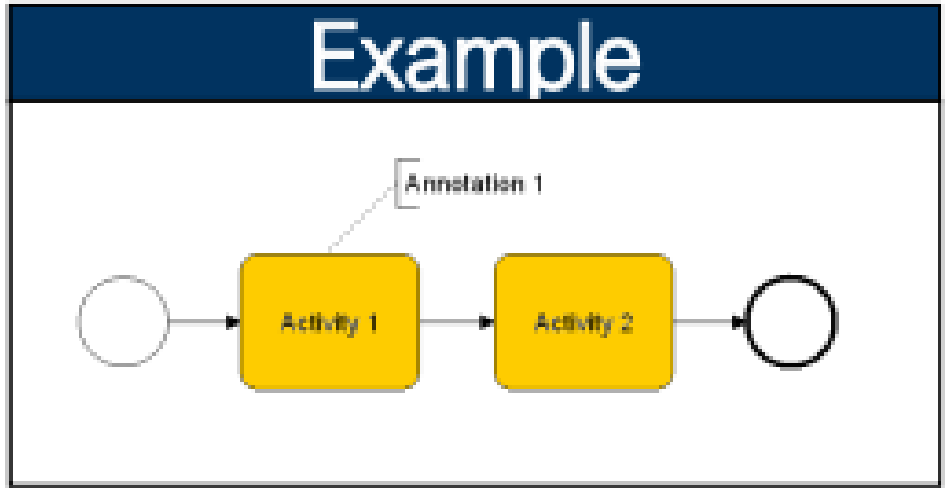
- In BPMN, a data object is linked either to a sequence flow, message flow, or an activity using a dotted line connector called an association.
- When lined to a sequence flow or message flow, the association is directional, with an arrow to or from the activity or event.

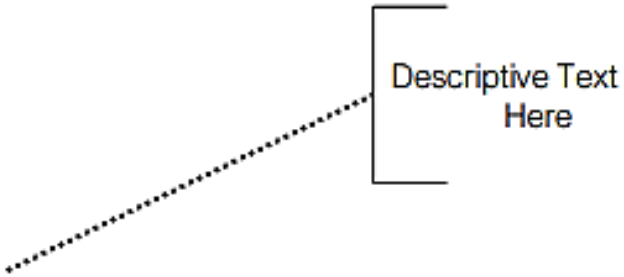




Text annotations

- Arbitrary text may be inserted in the diagram and linked to another object with a text annotation artifact and association connector.
- The text annotation is an enlarged left square bracket symbol





All processes must follow a verb-noun convention (i.echeck report)

Acquire	Borrow, buy, construct, develop, draw, lease, receive, rent
Activate	Authorise, enable, enact, establish, generate, guarantee, implement, initiate, install, integrate, open, reactivate, request
Administer	Admonish, apply, command, consult, direct, enhance, enquire, maintain, measure, monitor, operate, review, safeguard, supervise, waive
Decide	Accept, agree, allocate, approve, confirm, determine, identify, reject, select
Manipulate	Access, accumulate, add, archive, balance, calculate, compare, convert, create, delete, derive, exchange, file, modify, move, read, retrieve, sequence, store
Prepare	Analyse, define, draft, evaluate, forecast, gather, justify, negotiate, plan, propose, schedule, test, validate, verify
Provide	Acknowledge, announce, deliver, deposit, display, issue, lend, notify, offer, pay, print, report, return, send
Record	Capture, recognise, register
Suspend	Delay, make dormant
Terminate	Abandon, accomplish, cancel, close, complete, dispose of, sell, settle, withdraw

ster Class

Activity 1

A typical army recruitment process starts by shortlisting all candidates' applications. Those shortlisted are then called to take the following tests: drug and alcohol, eye, color vision, hearing, blood, urine, weight, fingerprinting, and doctor examination. The color vision can only be done after the eye test, while the doctor examination can only be done after color vision, hearing, blood, urine, and weight have been tested. Moreover, it may be required for some candidates to repeat some of these tests multiple times in order to get a correct assessment, e.g., the blood test may need to be repeated if the candidate has taken too much sugar in the previous 24 h. The candidates who pass all tests are asked to take a mental exam and a physical exam, followed by an interview. Only those who pass all these exams and perform well in the interview can be recruited in the army.

Activity 02 - Model the billing process of an Internet Service Provider (ISP).

The ISP sends an invoice by email to the customer on the first working day of each month (Day 1). On Day 7, the customer has the full outstanding amount automatically debited from its bank account. If an automatic transaction fails for any reason, the customer is notified on Day 8. On Day 9, the transaction that failed on Day 7 is re-attempted. If it fails again, on Day 10 a late fee is charged to the customer's bank account. At this stage, the automatic payment is no longer attempted. On Day 14, the Internet service is suspended until payment is received. If the payment is still outstanding on Day 30, the account is closed and a disconnection fee is applied. A debt-recovery procedure is then started.



Business Process Management

Process Analysis

Analyzing

- Analyze the as-is process to identify opportunities for improvement.
- Activities to be done: Use root cause analysis, value stream mapping, or other techniques to assess inefficiencies.
- Evaluate the process against performance metrics and business objectives.
- Identify areas for automation, streamlining, or redesign.

Process Analysis Techniques

- Value-Added & Waste Analysis
- Issue Register
- Root-Cause Analysis
- Flow analysis
- Queuing analysis
- Simulation

Value Added activities

- This is a step that produces value or satisfaction to the customer.
- When determining whether or not a step is VA, it may help to ask the following questions:
  - Would the customer be willing to pay for this step?
  - Does the customer value this step enough to keep conducting business with us?
  - If we remove this step, would the customer perceive that the outcome of the process is less valuable?

Business value-added activities

- The step is necessary or useful for the business to run smoothly, to collect revenue, or it is required due to the regulatory environment of the business. When determining whether or not a step is BVA, it may help to ask the following questions:
    - Is this step required in order to collect revenue, to improve or grow the business?
    - Would the business (potentially) suffer in the long term if this step were removed?
    - Does it reduce risk of business losses?
    - Is this step required in order to comply with regulatory requirements?

Waste Analysis

- Waste analysis can be seen as the reverse of value added analysis
- *Move*: Wastes that are related to movement. This category includes two types of waste: *transportation* and *motion*.
- *Hold*: Wastes arising from holding something. Again, this category includes two types of waste: *inventory* and *waiting*.
- *Overdo*: Wastes arising from doing more than is necessary in order to deliver value to the customer or the business. This category encompasses three types of waste: *defects*, *overprocessing*, and *overproduction*.

Stakeholder Analysis and Issue Documentation

1. Stakeholder analysis, which allows us to collect issues from complementary perspectives.

2. Issue register, which allows us to document issues in a structured manner.

3. Pareto analysis and PICK charts, which allow us to select a subset of issues for further analysis and redesign.

Stakeholder

- five categories of stakeholders
  - The customer(s) of the process.
  - The process participants.
  - The external parties (e.g., suppliers, sub-contractors) involved in the process.
  - The process owner and the operational managers who supervise the process participants.
  - The sponsor of the process improvement effort and other executive managers who have a stake in the performance of the process.

Issues and *factors:* equipment rental process

- Clerk misunderstood the site engineer's requirements for an equipment.
- Clerk did not select the correct equipment from the supplier's catalog due to inattention.
- Clerk indicated an incorrect delivery date in the PO.
- Supplier did not deliver the equipment that had been ordered.
- Delivered equipment is faulty or is not ready-for-use.
- Supplier delivered the equipment to the wrong construction site or at the wrong time.
- The equipment arrived five working days after the site engineer had requested it, but the site engineer needed the equipment earlier.
    - "Equipment is rejected by the site engineer".

Issue 1: Equipment kept longer than needed

Priority: 1

Description: Site engineers keep the equipment longer than needed

Data and assumptions: BuildIT rents 3,000 pieces of equipment per year. In 10% of cases, site engineers keep the equipment 2 days longer than needed. On average, rented equipment costs € 100 per day

Qualitative impact: Not applicable

Quantitative impact: $0.1 \times 3,000 \times 2 \times 100 =$ € 60,000 in additional rental expenses per year

---

Issue 2: Rejected equipment

Priority: 2

Description: Site engineers sometimes reject the delivered equipment due to non-conformance to their specifications

Data and assumptions: BuildIT rents 3,000 pieces of equipment per year. Each time an equipment is rejected due to a mistake on BuildIT's side, BuildIT is billed the cost of 1 day of rental, that is € 100. 5% of them are rejected due to an internal mistake within BuildIT (as opposed to a supplier mistake)

Qualitative impact: These events disrupt the construction schedules and create frustration and internal conflicts

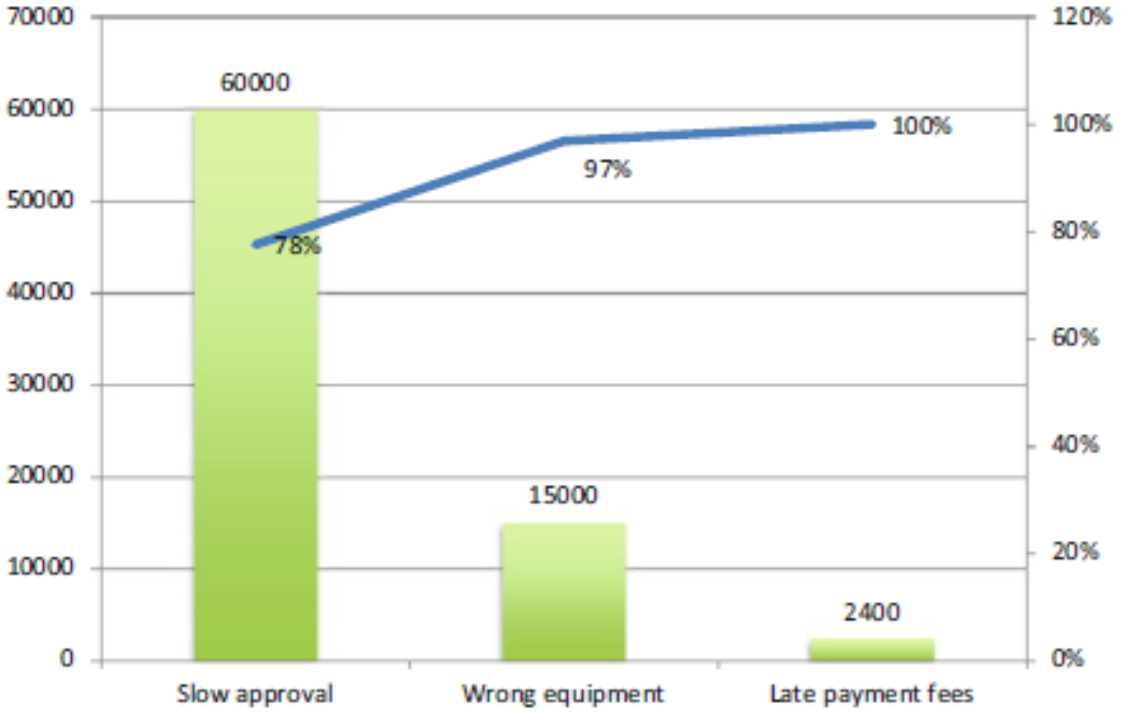
Quantitative impact: $3,000 \times 0.05 \times 100 =$ € 15,000 per year

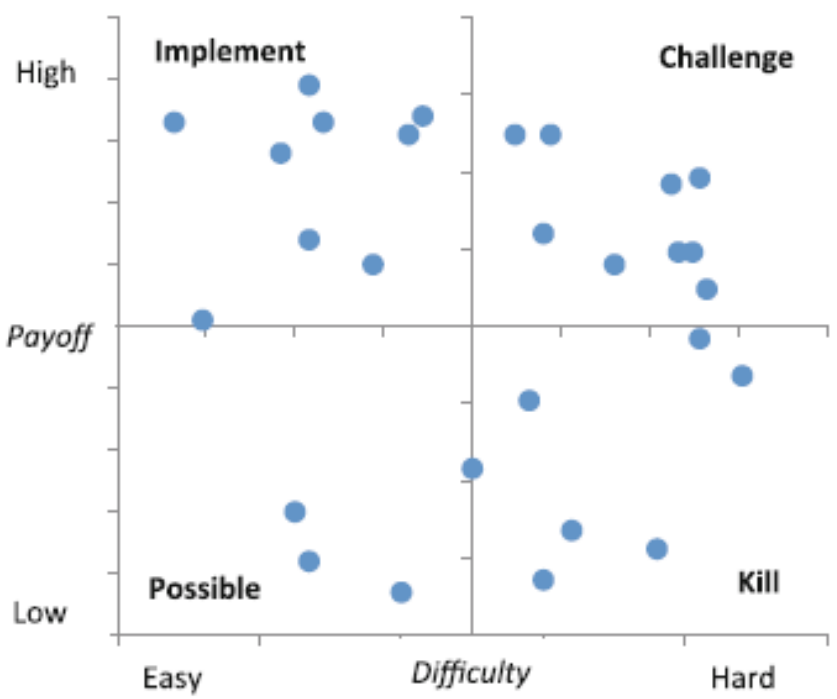
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Issue Register

- Name of the issue. This name should be kept short, typically 2–5 words, and should be understandable by all stakeholders in the process.
- Description. A short description of the issue (e.g., 1–3 sentences) focused on the issue itself as opposed to its consequences or impact, which are described separately.
- Priority. A number (1, 2, 3, . . .) stating how important this issue is relative to other issues. Note that multiple issues can have the same priority number.
- Data and assumptions. Any data used or assumptions made in the estimation of the impact of the issue, such as for example number of times a given negative outcome occurs, or estimated loss per occurrence of a negative outcome. In the early phases of the development of the issue register, the numbers in this column will be mainly assumptions or ballpark estimates. Over time, these assumptions and rough estimates will be replaced with more reliable numbers derived from actual data about the execution of the process.
- Qualitative impact. A description of the impact of the issue in qualitative terms, such as impact of the issue on customer satisfaction, employee satisfaction, longterm supplier relationships, company's reputation, or other intangible impact thatis difficult to quantify.
- Quantitative impact. An estimate of the impact of the issue in quantitative terms, such as time loss, revenue loss, or avoidable costs. This field in the issue register provides a link between qualitative and quantitative analysis techniques.

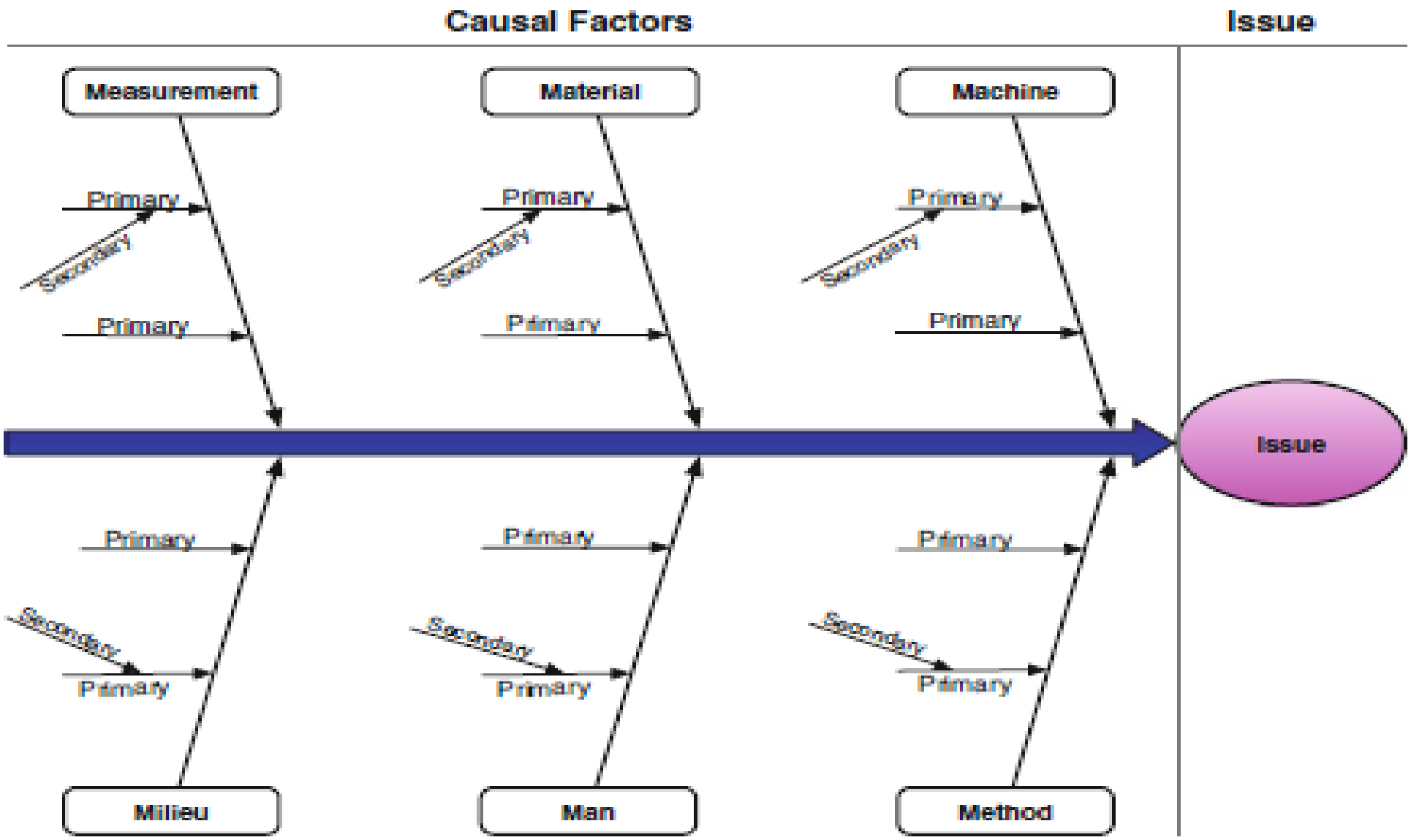
Data analysis





Root Cause Analysis – Cause effective analysis-6M’s

1. Machine
2. Method
3. Material
4. Man
5. Measurement
6. Milieu


Causal Factors
Issue
Measurement
Material
Machine
Primary
Secondary
Primary
Primary
Primary
Primary
Primary
Primary
Primary
Secondary
Primary
Secondary
Primary
Secondary
Primary
Issue
Milieu
Man
Method

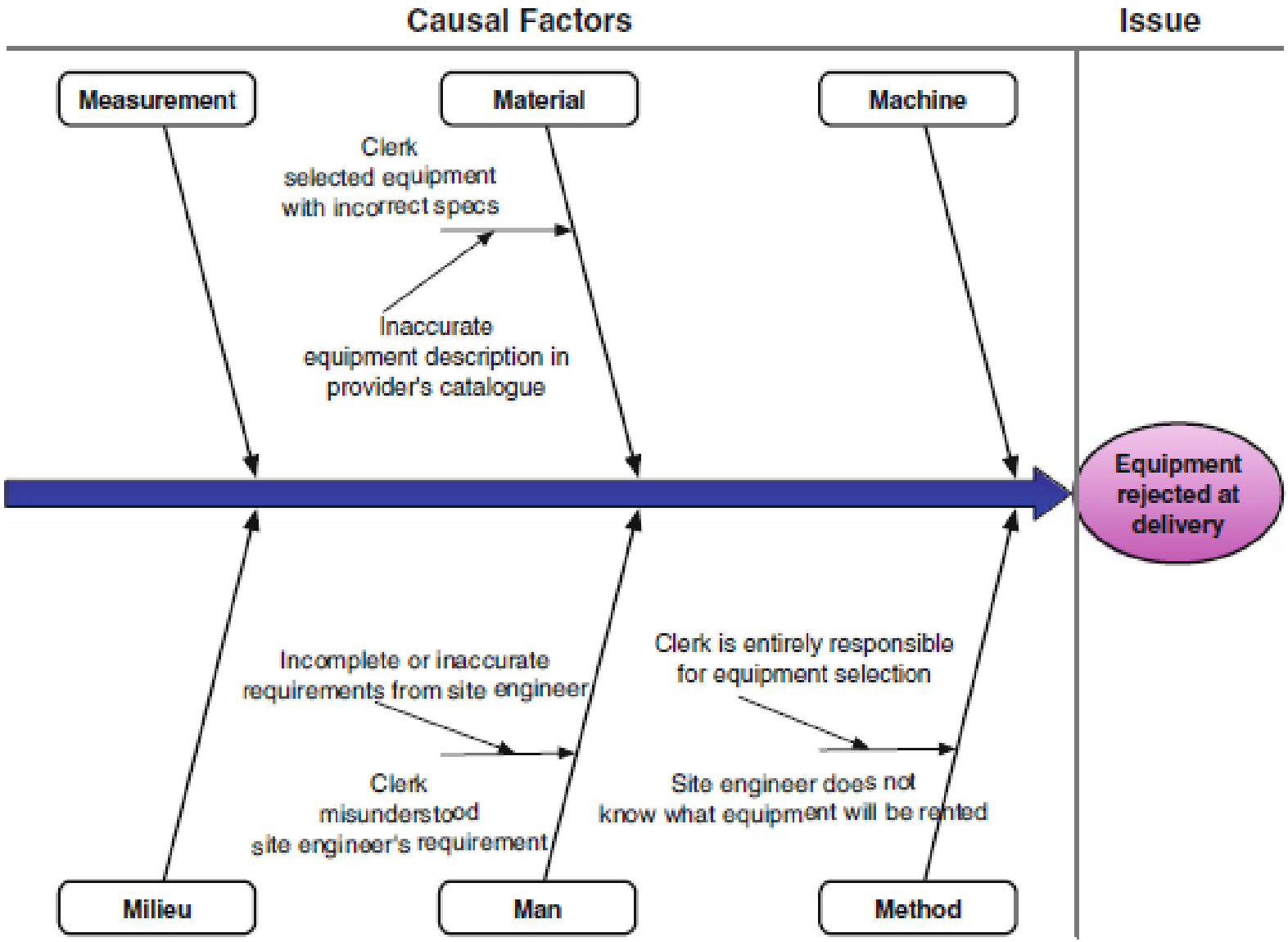
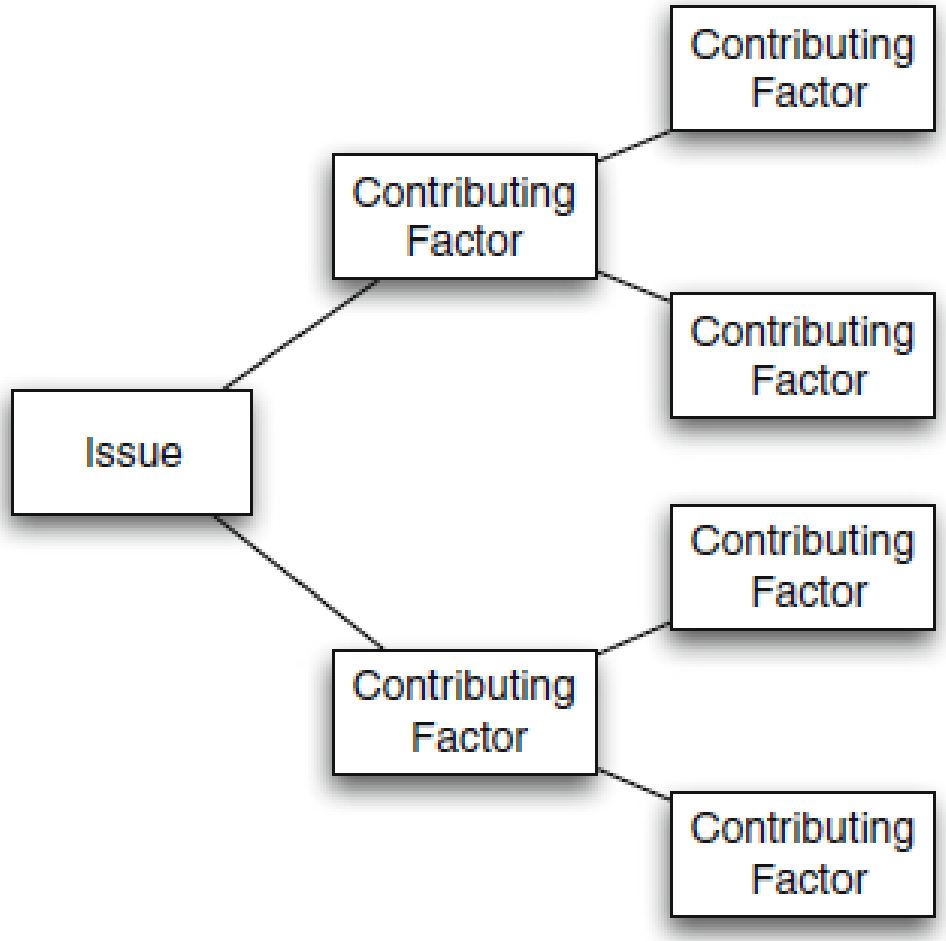


Fig. 6.6 Cause-effect diagram for issue "Equipment rejected at delivery"

Root Cause Analysis – Why-Why Diagrams



Root Cause Analysis – Why-Why Diagrams

- Asking why questions
  - Issue 1 Site engineers sometimes reject delivered equipment, **why?**
    - wrong equipment is delivered, why?
      - miscommunication between site engineer and clerk, why?
        - site engineer provides an incomplete or inaccurate description of what they want.
      - site engineer does not always see the supplier catalogs when making a request and does not communicate with the supplier, why?
        - site engineer generally does not have Internet connectivity.
    - site engineer does not check the choice of equipment made by the clerk.
    - equipment descriptions in supplier's catalog not accurate.

Questions

- Explain the value added analysis
- What are the seven factors in waste analysis?
- Name five categories of stackholders
- Describe root cause with 6M’s
- What is Why –why diagram?

Thank you

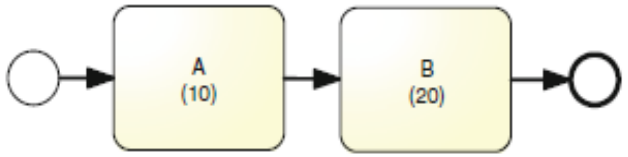


Business Process Management

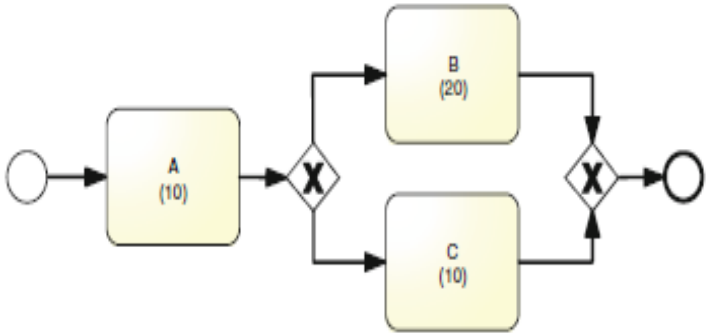
Process Analysis

Quantitative Process Analysis

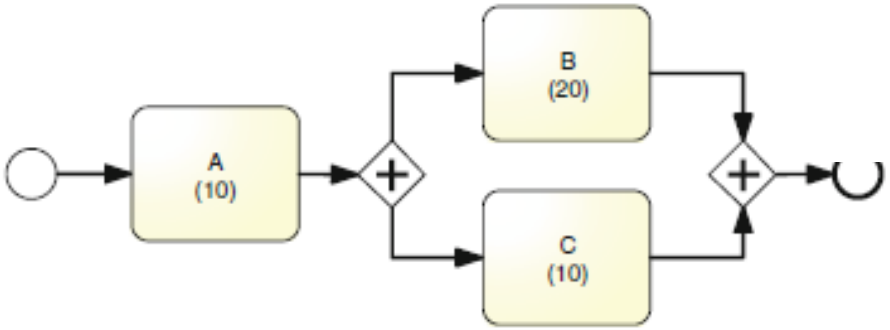
Flow Analysis - Calculating Cycle Time Using Flow Analysis



$$CT = \sum_{i=1}^{n} T_i$$

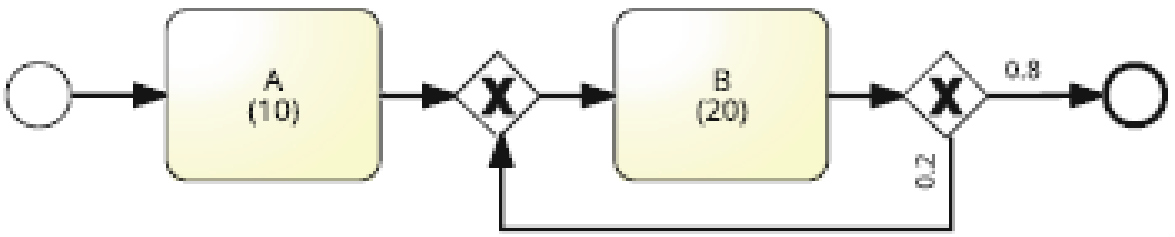


$$CT = \sum_{i=1}^{n} p_i \times T_i$$



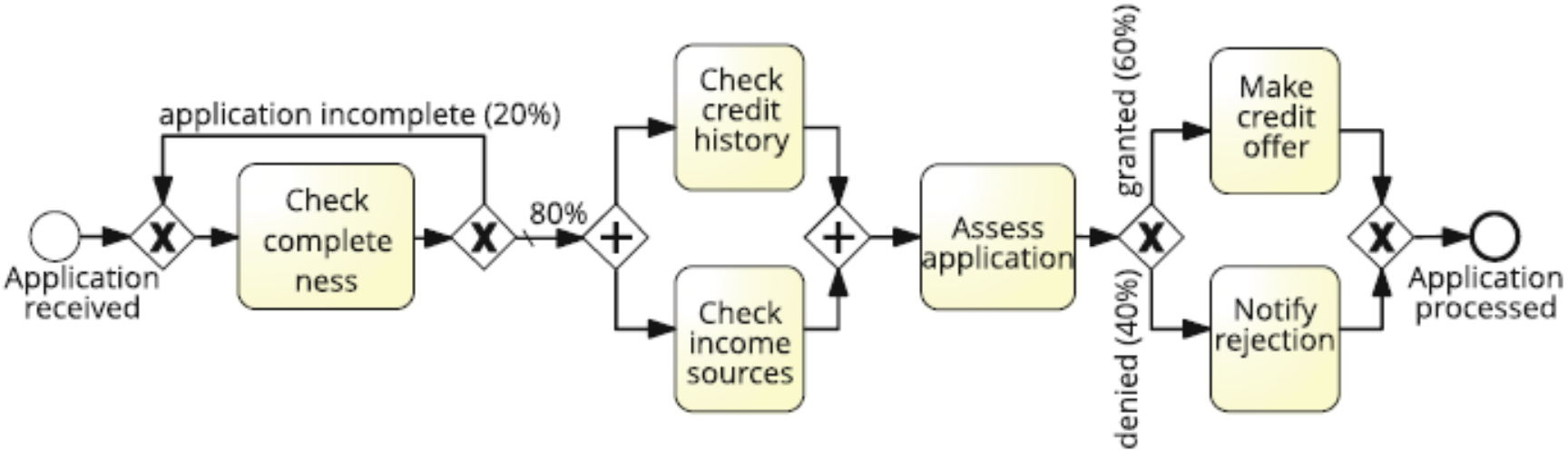
$$CT = Max(T_1, T_2, \ldots, T_n)$$

What is the average cycle time for the following scenario



$$CT = \frac{T}{1-r}.$$

Example – Task 01 calculate the CT



Task Cycle	time
Check completeness	1 day
Check credit history	1 day
Check income sources	3 days
Assess application	3 days
Make credit offer	1 day
Notify rejection	2 days

Queue

- Queueing theory is a collection of mathematical techniques to analyze systems that have resource contention.
- Queueing theory gives us techniques to analyze important parameters of a queue such as the expected length of the queue or the expected waiting time of an individual case in a queue.

Simulation

- The essential idea underpinning process simulation is to use the process simulator for generating a large number of hypothetical instances of a process, executing these instances step-by-step, and recording each step in this execution.
- The output of a simulator then includes the logs of the simulation as well as statistics of cycle times, average waiting times, and average resource utilization.

Question 02

- As a BPM analysis what sort of skills you need to improve?

Question 03

- Why is BPM important for organizations?

Question 04

- What are the key elements of a business process?

Question 05

- How would you prioritize which processes to improve first in a company?

Good Luck